Abridged: On the equation $n'' = n$ and a problem of Erdős and Barbeau on products of prime–reciprocal sums

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Abstract

Erdős Problem #307 (posed by Barbeau, 1976; recorded in Erdős–Graham, 1980) asks whether there exist two finite sets P, Q of primes with $\left(\sum_{p\in P}\frac{1}{p}\right)\left(\sum_{q\in Q}\frac{1}{q}\right) = 1$. Writing $a = \prod P$, $b = \prod Q$, a solution exists if and only if the *arithmetic derivative* has a two–cycle $a' = b$, $b' = a$ with $a \neq b$; equivalently iff $n'' = n$ has a solution that is not a fixed point p^p. That reformulation, and the equivalence with the derivative, are due to Ufnarovski–Åhlander [7], whose Conjecture 4 asserts in exactly these terms that $\left(\sum 1/p_i\right)\left(\sum 1/q_j\right) = 1$ has no solution in distinct primes. Our starting point is that #307 *is* that conjecture — an identification neither literature records — which ties the problem to the Ufnarovski–Åhlander dynamics: their conjecture that every $n'' = n$–cycle has period one would force #307 to have answer NO, while a YES would refute it. Exploiting the equivalence we prove a *rigidity lemma* (each prime–reciprocal sum is automatically in lowest terms, so $N_P = D_Q$, $N_Q = D_P$, and $P \cap Q = \varnothing$) and a quantitative *barrier*: building on Rosen's bound $|P \cup Q| \geq 59$, any solution has $\min(\prod P, \prod Q) \geq 2.09 \times 10^{56}$, for *both* members. This is the first valid unconditional bound of its kind. The prior claims are Kovič's: his Proposition 17 is sound but conditional on the smaller member being odd, while Proposition 18 — the only unconditional cardinality and size bound in the literature, giving $r + s \geq 34$ and $\max\{m, n\} \geq \prod_{i\leq 17} p_i$ — rests on an invalid step, deriving $p_1 q_s < rs$ by multiplying an upper bound against a lower one, where the available products give only $p_1 q_1 < rs < p_r q_s$ and no cardinality bound at all; we give an explicit model and machine–check both points. Against the strongest surviving unconditional bound, that paper's computer search $x > 10^4$, the gain is 52 orders of magnitude. A finite verification over the complete list of 49,961 admissible 59–prime supports then closes the 59–prime level: any solution in fact has $|P \cup Q| \geq 60$ and $\min(\prod P, \prod Q) > 3.50 \times 10^{57}$; a quadratic–reciprocity certificate, requiring no factorisation, then proves the canonical tail family (the first 59 primes plus any q) empty, and together with a factoring pass 46,483 of the 49,961 one–new–prime families at level 60 (93%; the residual is factoring–hard). We add a parity dichotomy (a hypothetical all–odd cycle would need ≥ 1412 primes and have $\min(\prod P, \prod Q) > 10^{2519}$), a proof that longer cycles are strictly harder ($k = 3$ already forces 361,139 primes), and a function–field theorem: over $\mathbb{F}_q[t]$ the derivative has *no* cycles, so the obstruction over $\mathbb{Z}$ is purely archimedean. A local congruence anatomy of the cycle equation, the Erdős–Wintner fact that the abundance density 0.0420 is a genuine constant, and a Thābit/Euler–type constructive calculus (the *Frame Rule* and a divisor–trick) that faithfully parametrises solutions but, by an exact identity $\Delta_0 = M_0 N(1 - s_{M_0} s_N)$, provably does *not* reduce the difficulty below #307 itself (the natural "make Δ_0 small" objective is #307 in disguise). No fixed–shape family can help either: for families of *any* growth rate, not merely polynomial ones, a product identity holding infinitely often forces every member constant, which closes the Thābit/Euler programme (primes $3 \cdot 2^n - 1$ and relatives) that is the

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classical technique for problems of this kind. Recast as a bilinear breeder it yields a conditional reduction (H) $\Rightarrow$YES to an explicit prime–density hypothesis, which a quantitative ledger shows is not a feasible finite search. On the counting side we prove that both Pythagorean layers have density zero unconditionally *on the squarefree stratum*, which is where every tool in the proof lives and where a #307 solution lies by rigidity; the unrestricted statement is not proved. Both Pythagorean layers have density zero on the squarefree stratum by a square–sieve argument; that argument, its conditional rate and the three obstructions to the sharp exponent $N^{1/2}$ are given in the full version and are not reproduced here. The 34 reciprocity-immune families at level 60 are upgraded from probable to proved by Atkin–Morain certificates, archived for independent verification. The rigidity result, the barrier, and the arithmetic core of the counting argument are formalised in Lean 4 (mathlib), extremality proved rather than assumed; the only non–logical input is a verified evaluation of the first 59 primes. The problem remains open. The heuristic case for existence is weaker than it appears: the constant in the expected-count heuristic cannot be measured even in principle, since the barrier places the entire relevant region above $10^{112.9}$ while the largest value of $\sigma(a)\sigma(a')$ observed below 10^7 is 0.5535.

Abridged version. This note omits the deviation ladder, the plus–layer analysis and the square sieve, together with their proofs, and cites their statements by the numbering of the full version, which is archived with the Lean development and the computational scripts. Nothing proved here depends on a proof appearing only there.

1 Introduction

Let P, Q be finite, nonempty sets of (distinct) primes. Erdős Problem #307 [10], attributed to Barbeau's 1976 "Computer challenge corner" problem #477 [2] and listed in Erdős–Graham [3], asks:

$$\text{do there exist finite sets of primes } P, Q \text{ with } \Big(\sum_{p\in P}\frac{1}{p}\Big)\Big(\sum_{q\in Q}\frac{1}{q}\Big) = 1\,?$$

The problem is verifiable in the sense that a single example would settle it. No example is known. The published partial facts are a remark of J. Rosen on MathOverflow [9], that any solution needs at least 59 primes in $P \cup Q$, and the problem page [10], which asserts $|P \cup Q| \geq 60$.

The relation between the two is worth stating precisely, since this paper proves the second. The page's justification reads, in full, that P and Q are disjoint and $\sum_{p\in P\cup Q} 1/p \geq 2$, "whence $|P \cup Q| \geq 60$". That implication does not hold as displayed: the reciprocals of the first 59 primes already sum to $2.00235 > 2$, so a mass of 2 is attainable on 59 primes and the stated hypotheses yield $|P \cup Q| \geq 59$, which is Rosen's bound. Reaching 60 requires excluding the 49,961 admissible 59–element supports one at a time, which is what Proposition 4.5 does and what `erdos307_sixty` machine-checks. We therefore claim no priority for the *statement* $|P \cup Q| \geq 60$, which predates this work on the problem page, and we do claim the proof. The known "solutions" of Cambie and others solve only the *weakened* version in which 1 is permitted as an element, e.g. $(1+\frac{1}{5})(\frac{1}{2}+\frac{1}{3}) = 1$; these do not bear on the prime version.

Expanding the product exhibits #307 as a rigidly shaped Egyptian fraction for 1: $1 = \sum_{p\in P}\sum_{q\in Q} 1/(pq)$, whose denominators are the *rectangular grid* $P \times Q$ of semiprimes; a *rank–one*, multiplicatively separable representation. The unrestricted companion (representing 1 by reciprocals of distinct two–prime–factor numbers, with no product structure imposed) is by contrast *solved*: Johnson (1978) gave a 48–term decomposition, Watanabe [5] 47–term ones, and Graham's survey [4] records exactly this contrast, noting after Barbeau that the product form "is not known." Thus #307 is precisely the rank–one restriction of a solved problem. Nor is the density–zero nature of the prime denominators the obstruction in itself: Bloom [6] represents every positive rational by distinct unit fractions drawn from the (equally sparse) shifted primes $\{p-1\}$. What resists is the rank–one rigidity (the demand that the representation *factor* as a product of two prime–reciprocal sums) which the barrier (Theorem 4.3) shows costs at least 59 primes, hence of order 870 semiprime terms against Johnson's 47.

Our starting point is an identification of #307 with a question about the *arithmetic derivative* $n \mapsto n'$, the unique map with $0' = 1' = 0$, $p' = 1$ for primes p, and the Leibniz rule $(ab)' = a'b + ab'$. (The map goes back to J. Mingot Shelly (1911) and appeared as a 1950 Putnam problem; Barbeau [1] gave the first systematic study, with the modern account in Ufnarovski–Åhlander [7] and OEIS [11].) For $n = \prod_i p_i^{e_i}$ one has $n' = n \sum_i e_i/p_i$. We show (Section 3) that #307 is *equivalent* to the existence of a two–cycle of $n \mapsto n'$ that is not a fixed point. The two–cycle reformulation itself is not new, and it is older than is generally recorded. Ufnarovski–Åhlander [7] prove that a period–two point of $n \mapsto n'$ has squarefree members with disjoint supports, and deduce that the period–two case of their Conjecture 3 is equivalent to their Conjecture 4: $\big(\sum_{i \le k} 1/p_i\big)\big(\sum_{j \le l} 1/q_j\big) = 1$ has no solution in distinct primes. That is #307, in the derivative literature, in 2003. It was restated "in disguise" by Seva [19] in 2019 (via $Q \mapsto Q \sum_{p|Q} 1/p$, which agrees with the arithmetic derivative on squarefree integers, where the question lives), who is the first to name it as #307, and obtained independently by Bado [20]; see Section 12. What is new here is the identification of the two problems — #307 is Ufnarovski–Åhlander's Conjecture 4 — and the resulting two–way bridge, importing their period–one conjecture and exporting our bounds. Remarkably, Barbeau authored foundational papers on both sides (1961 and 1976 [1, 2]).

The bridge is productive in both directions.

- *Importing* from the derivative literature: the Ufnarovski–Åhlander theorem that any two–cycle has squarefree members with disjoint supports [7] makes the "finite sets of distinct primes" hypothesis of #307 automatic, and links #307 to the Ufnarovski–Åhlander conjecture that all derivative cycles have period one.

- *Exporting* to that literature: our barrier (Theorem 4.3) is, we believe, the first valid unconditional bound of its kind. Kovič [8] states three bounds on an $n'' = n$ solution. A computer search gives $x > 10^4$, unconditionally. Proposition 17 gives $> 3 \cdot 5 \cdots 29 \approx 3.23 \times 10^9$, but only when the *smaller* member is odd; Kovič notes explicitly that no estimate follows otherwise. Proposition 18 claims the unconditional $r + s \ge 34$ and $\max\{m, n\} \ge \prod_{i=1}^{17} p_i \approx 1.92 \times 10^{21}$; its proof does not establish them, for the reason set out in Section 12. Against the strongest *valid unconditional* bound, 10^4, our $\min(\prod P, \prod Q) \ge 2.09 \times 10^{56}$ is a gain of 52 orders of magnitude; against the conditional Proposition 17 it is 47, in the case where that proposition applies. On cardinality there is no prior bound to improve: $|U| \ge 60$ appears to be the first, and it re-derives Kovič's claimed $r + s \ge 34$ as a corollary by an unrelated route.

Results and organisation. Section 2 proves the Rigidity Lemma (Theorem 2.1): the fraction $\sum_{p \in S} 1/p$ is automatically in lowest terms, which forces, for any solution, the exact identities $N_P = D_Q$, $N_Q = D_P$, disjointness $P \cap Q = \varnothing$, and $D_Q = s\, D_P$ where $s = \sum_{p \in P} 1/p$. The disjointness core was anticipated by R. Israel [9]; the full rigidity is what powers everything later. Section 3 establishes the bridge. Section 4 proves the barrier, the parity dichotomy, and that longer cycles are strictly harder (so two–cycles are extremal). Section 5 gives a local congruence anatomy, a heuristic model, and a proof (via Erdős–Wintner) that the relevant abundance density is a genuine constant. Section 6 adds a function–field lens: over $\mathbb{F}_q[t]$ the derivative has *no* cycles, so the obstruction in $\mathbb{Z}$ is purely archimedean. Section 7 develops the Frame Rule and divisor–trick, and, crucially, records the exact identity that renders the natural constructive objective circular.

Section 8 recasts the rule as a bilinear breeder, records a conditional reduction (H) $\Rightarrow$ YES to an explicit prime–density hypothesis, and shows quantitatively why (H) does not amount to a feasible finite search. Section 9 documents the computations, and Section 10 the Lean 4 formalisation. We are explicit throughout about what is proved, what is heuristic, and what is borrowed; the problem remains open.

Notation. For a finite set S of distinct primes put

$$D_S := \prod_{p\in S} p, \qquad N_S := \sum_{p\in S} \frac{D_S}{p}, \qquad \text{so} \qquad \sum_{p\in S} \frac{1}{p} = \frac{N_S}{D_S}.$$

We call N_S the *cofactor sum.* For a solution we write $a = D_P$, $b = D_Q$, $s = N_P/D_P = \sum_{p\in P} 1/p$ and $t = N_Q/D_Q = \sum_{q\in Q} 1/q$, so $st = 1$, and (without loss of generality) $s > 1 > t$. We write $\omega(n)$ for the number of distinct primes dividing n.

2 The Rigidity Lemma

Lemma 2.1 (Rigidity / lowest terms)**.** *For a finite set S of distinct primes,* $\gcd(N_S, D_S) = 1$*. Equivalently, $\sum_{p\in S} 1/p$ is already in lowest terms, and $v_p\big(\sum_{r\in S} 1/r\big) = -1$ for every $p \in S$.*

Proof. The prime divisors of $D_S = \prod_{r\in S} r$ are exactly the elements of S, so it suffices to show $p \nmid N_S$ for each $p \in S$. Fix $p \in S$ and reduce $N_S = \sum_{r\in S} D_S/r$ modulo p. For $r \neq p$ the term $D_S/r = \prod_{u\in S\setminus\{r\}} u$ still contains the factor p, so $p \mid D_S/r$. The single surviving term is $D_S/p = \prod_{u\in S\setminus\{p\}} u$, whence

$$N_S \equiv \prod_{u\in S\setminus\{p\}} u \pmod{p}.$$

Each u in this product is a prime distinct from p, so $p \nmid u$; as p is prime it does not divide the product. Thus $N_S \not\equiv 0 \pmod{p}$, i.e. $p \nmid N_S$. As p ranged over all prime divisors of D_S, $\gcd(N_S, D_S) = 1$. The valuation statement is the same computation: $v_p(N_S/D_S) = v_p(N_S) - v_p(D_S) = 0 - 1 = -1$. $\square$

Remark 2.2. Distinctness and primality are both essential: v_p of the cofactor of a repeated prime need not be -1, and the final step uses Euclid's lemma. The case $|S| = 1$ is consistent ($N_{\{p\}} = 1$).

Theorem 2.3 (Solution structure)**.** *Suppose P, Q are finite sets of distinct primes with $\dfrac{N_P}{D_P}\cdot\dfrac{N_Q}{D_Q} = 1$. Then*

$$N_P = D_Q, \qquad N_Q = D_P, \qquad P \cap Q = \varnothing,$$

N_P is squarefree with prime support exactly Q (and N_Q squarefree with support P), and $D_Q = s\,D_P$ where $s = N_P/D_P$.

Proof. Cross-multiplying, $N_P N_Q = D_P D_Q$. By Lemma 2.1 both N_P/D_P and N_Q/D_Q are in lowest terms. Since $N_Q/D_Q = D_P/N_P$ and the lowest–terms representative of a positive rational is unique, comparing N_Q/D_Q with D_P/N_P gives $N_Q = D_P$ and $D_Q = N_P$.

Disjointness (the Israel mechanism [9]). If $r \in P \cap Q$ then $r \mid D_P$ and, by $N_P = D_Q = \prod Q$, also $r \mid N_P$, but $\gcd(N_P, D_P) = 1$ forces $r \nmid N_P$, a contradiction. (Equivalently $v_r(s) = v_r(t) = -1$, so $v_r(st) = -2 \neq 0 = v_r(1)$.) Hence $P \cap Q = \varnothing$.

Finally $N_P = D_Q = \prod_{q\in Q} q$ is a product of distinct primes, hence squarefree with prime support Q; symmetrically for N_Q. And $s\,D_P = (N_P/D_P)D_P = N_P = D_Q$. $\square$

The disjointness in Theorem 2.3 means a solution is genuinely a partition of $U := P \cup Q$, and $D_P D_Q = \prod_{p\in U} p$. The valuation form $v_p(\sum 1/p_i) = -1$ underlying disjointness is due to R. Israel [9], and appears also in Seva [19]; the cross–matching form ($N_P = D_Q$, $N_Q = D_P$) was obtained independently and concurrently by Bado [20, Thm. A] (Section 12).

3 The bridge to the arithmetic derivative

Lemma 3.1. *For squarefree m, the arithmetic derivative equals the cofactor sum: $m' = \sum_{p|m} m/p = N_{\{p:\, p|m\}}$. Moreover* $\gcd(m, m') = 1$.

Proof. For $m = \prod_{p|m} p$ squarefree, $m' = m \sum_{p|m} 1/p = \sum_{p|m} m/p$, which is exactly the cofactor sum of the prime set of m. Coprimality is Lemma 2.1. ☐

Theorem 3.2 (Bridge)**.** *The following are equivalent.*

(1) *Erdős #307 has a solution: finite prime sets P, Q with $\left(\sum_{p\in P} 1/p\right)\left(\sum_{q\in Q} 1/q\right) = 1$.*

(2) *The arithmetic derivative has a two–cycle: integers $a \neq b$ with $a' = b$ and $b' = a$.*

(3) *The equation $n'' = n$ has a solution n that is not a fixed point (i.e. $n' \neq n$, so n is not of the form p^p).*

(*The implication* (2) ⇒ (1) *uses the Ufnarovski–Åhlander squarefreeness theorem for cycles; all other implications, and everything used in the barrier of Section* 4*, are unconditional; see Remark* 3.3.)

Proof. (1) ⇒ (2). Let $a = D_P, b = D_Q$. By Lemma 3.1, $a' = N_P$ and $b' = N_Q$, and by Theorem 2.3, $N_P = D_Q = b$ and $N_Q = D_P = a$. Thus $a' = b$, $b' = a$, and $a \neq b$ since $s \neq 1$ (a prime–reciprocal sum is never 1, being N/D in lowest terms with $D > 1$).

(2) ⇒ (1). Let $a' = b, b' = a$ with $a \neq b$. By the Ufnarovski–Åhlander analysis of two–cycles (cf. [7, 8]), both a and b are squarefree; their prime supports P, Q are then disjoint since $\gcd(a, b) = \gcd(a, a') = 1$. By Lemma 3.1, $a' = N_P$ and $b' = N_Q$; the cycle gives $N_P = b = D_Q$ and $N_Q = a = D_P$, so $\left(\sum_{p\in P} 1/p\right)\left(\sum_{q\in Q} 1/q\right) = \frac{N_P}{D_P}\frac{N_Q}{D_Q} = \frac{D_Q}{D_P}\frac{D_P}{D_Q} = 1$.

(2) ⇔ (3). A two–cycle $a \neq b$ gives $a'' = a$ with $a' = b \neq a$, so a is a non–fixed solution of $n'' = n$. Conversely a solution of $n'' = n$ with $n' \neq n$ yields the two–cycle (n, n'). The fixed points of $n \mapsto n'$ are exactly the numbers p^p [1, 7]. ☐

Remark 3.3 (Status of the cited input)**.** Part (2) ⇒ (1) uses that a derivative two–cycle has squarefree, support–disjoint members. Squarefreeness is the content of the Ufnarovski–Åhlander analysis of cycles (a non–squarefree n has $p \mid \gcd(n, n')$ for some p, and along $n \mapsto n'$ such common factors propagate, precluding return); disjointness then follows from $\gcd(a, a') = 1$. We use it as a cited theorem. Even *without* it, the implication (1) ⇒ (2) and the entire barrier below are unconditional, because Rigidity already delivers squarefree, disjoint P, Q from a #307 solution.

Formalised. `Erdos307.bridge_forward` is (1) ⇒ (2) and cites nothing; the distinctness $a \neq b$ is *derived* there, not assumed (`dprod_ne_dprod`). Support–disjointness is available separately as `disjoint_primeFactors_of_cycle`, also from Rigidity; it is not consumed by `bridge_forward`, which does not need it. `bridge_backward` is (2) ⇒ (1) with the Ufnarovski–Åhlander squarefreeness appearing as an explicit hypothesis, and `two_cycle_iff_double_fixed` is (2) ⇔ (3) for an arbitrary self–map, on *no* axioms at all. 0 `sorry` (`lean/Erdos307/Bridge.lean`).

Corollary 3.4. *If the Ufnarovski–Åhlander conjecture holds in the weak form "every cycle of $n \mapsto n'$ has period one," then Erdős #307 has answer* NO. *Conversely, a* YES *to #307 exhibits a period–two cycle and refutes that statement.*

4 The barrier

Lemma 4.1 (Strict AM–GM). *If $s, t > 0$ with $st = 1$ and $s \neq 1$, then $s + t > 2$.*

Proof. $s + t - 2 = s + 1/s - 2 = (s-1)^2/s > 0$ since $s \neq 1$. □

Lemma 4.2 (The number 59, after Rosen [9]). *Any finite set U of distinct primes with $\sum_{p \in U} 1/p > 2$ has $|U| \geq 59$. Moreover, among k–element prime sets, $\sum 1/p$ is maximised by the k smallest primes.*

Proof. Replacing any prime of U by a smaller prime not in U increases $\sum 1/p$, so the maximum over k–element sets is at the first k primes. The reciprocal sum of the first 58 primes is $1.99874\ldots < 2$ and of the first 59 primes is $2.00235\ldots > 2$, so a sum exceeding 2 needs at least 59 primes. □

Theorem 4.3 (Barrier). *For any solution (P, Q) of #307, with $U = P \cup Q$:*

(a) $|U| \geq 59$ *and* $\displaystyle\prod_{p \in U} p \geq \Pi_{59} := \prod_{i=1}^{59} p_i \approx 8.77 \times 10^{112}$*;*

(b) $\min(\prod P, \prod Q) \;\geq\; \sqrt{\Pi_{59}/T_{59}} \;=\; 2.0929\ldots \times 10^{56}$*, where $T_{59} = \sum_{i=1}^{59} 1/p_i = 2.00235\ldots$. In particular both $\prod P$ and $\prod Q$ exceed* 2.09×10^{56}.

Consequently every $n'' = n$ solution has both members $> 2.09 \times 10^{56}$. The strongest valid unconditional bound previously available is the computer search $x > 10^4$ of [8], so this is a gain of 52 *orders of magnitude; against the conditional Proposition 17 of that paper (*3.23×10^9*, when the smaller member is odd) it is* 47*. The unconditional $r + s \geq 34$ and $\max\{m, n\} \geq 1.92 \times 10^{21}$ claimed in Proposition 18 of [8] are not established by their proof (Section 12); the bound above does not rely on them and in fact implies the first.*

Proof. (a) By $st = 1$, $s \neq 1$ and Lemma 4.1, $\sum_{p \in U} 1/p = s + t > 2$, so $|U| \geq 59$ by Lemma 4.2; since P, Q are disjoint (Theorem 2.3), $\prod_{p \in U} p = D_P D_Q$, and a product of ≥ 59 distinct primes is at least the product of the 59 smallest, Π_{59}.

(b) Take $D_P \leq D_Q$ to be the minimal side. By Theorem 2.3, $D_Q = s\, D_P$ with $s > 1$, hence

$$s\, D_P^2 \;=\; D_P\, D_Q \;=\; \prod_{p \in U} p \;=: R, \qquad \text{so} \qquad D_P^2 = \frac{R}{s}.$$

Since $s \leq s + t = T(U) := \sum_{p \in U} 1/p$ (as $t > 0$),

$$D_P^2 = \frac{R}{s} \;\geq\; \frac{R}{T(U)}.$$

It remains to bound $R/T(U)$ below over all admissible U. Among k–element prime sets the quotient $\left(\prod U\right)/\left(\sum_{p \in U} 1/p\right)$ is minimised by the k smallest primes (smallest numerator and, by Lemma 4.2, largest denominator simultaneously), and passing from 59 to more primes only increases it (the product grows multiplicatively while the reciprocal sum grows by a single small term). Hence the global minimum over admissible U is at $U = \{p_1, \ldots, p_{59}\}$, giving $R/T(U) \geq \Pi_{59}/T_{59}$. Therefore

$$D_P^2 \;\geq\; \frac{\Pi_{59}}{T_{59}} \;=\; \frac{\Pi_{59}^2}{N_{59}} \;=\; 4.3806\ldots \times 10^{112}, \qquad D_P \;\geq\; 2.0929\ldots \times 10^{56},$$

where $N_{59} = T_{59}\,\Pi_{59}$ is the integer cofactor sum of the first 59 primes. □

Remark 4.4 (Why the *minimum* is bounded, not only the maximum)**.** The product identity $D_P D_Q \geq \Pi_{59}$ alone gives only $\max(\prod P, \prod Q) \geq \sqrt{\Pi_{59}} \approx 2.96 \times 10^{56}$. The stronger statement (b), a lower bound on the *smaller* side, genuinely uses rigidity: it is the identity $D_Q = s\,D_P$ (equivalently $D_P^2 = R/s$) together with $s \leq T(U)$ that converts the product bound into a bound on D_P. We verified numerically that $\min_U R(U)/T(U)$ is attained exactly at the first 59 primes (the minimising configuration is unique), so the constant 2.09×10^{56} is sharp for this method.

Proposition 4.5 (Closing the 59–prime level)**.** *No solution of #307 has* $|P \cup Q| = 59$*. Hence* $|P \cup Q| \geq 60$*,* $\prod_{p \in U} p \;\geq\; \Pi_{60} > 2.46 \times 10^{115}$*, and*

$$\min\bigl(\textstyle\prod P, \prod Q\bigr) \;\geq\; \sqrt{\Pi_{60}/T_{60}} \;=\; 3.5053\ldots \times 10^{57}.$$

Proof (finite verification). Let $U = P \cup Q$ with $|U| = 59$; as in Theorem 4.3(a), $T(U) > 2$. The candidate list is finite and complete: (i) every prime $p \leq 167$ lies in U; omitting p caps $T(U)$ at $T_{60} - 1/p < 2$, since the 59 largest prime reciprocals avoiding p are those of the first 60 primes less p, and $1/p > T_{60} - 2 = 0.00590\ldots$ exactly for $p \leq 167$; (ii) every element $p' \geq 281$ satisfies $T_{58} + 1/p' > 2$ (the other 58 reciprocals sum to at most T_{58}), so $p' \leq 787$. Thus U consists of the 39 primes up to 167 plus twenty primes from $(167, 787]$, and exactly 49,961 such sets have $T(U) > 2$. For each, a solution supported on U forces both $N' + 2N = (a+b)^2$ and $N' - 2N = (a-b)^2$ to be perfect squares, where $N = \prod U$ and $N' = a^2 + b^2$ (rigidity; the $k = 0$ case of Proposition 12.2); one integer test that simultaneously excludes all 2^{58} splits of U. Exact–integer verification shows $N' + 2N$ is a non–square for *every* one of the 49,961 candidates, by two independent implementations with different enumeration orders and prunings (`code/close59.py` and `hunt/close59.rs` in the repository). Hence no 59–element support closes, and $|U| \geq 60$. The bound on the minimal side repeats the proof of Theorem 4.3(b) with the minimum of R/T now over $|U| \geq 60$, attained at the first 60 primes: $D_P^2 \geq \Pi_{60}/T_{60} = \Pi_{60}^2/N_{60}$, where N_{60} is the cofactor sum of the first 60 primes. □

Corollary 4.6 (The coprime variant inherits the closed level)**.** *Consider the weakened variant in which the elements of P and Q are merely pairwise coprime integers* ≥ 2 (*the open form of the variant,* $1 \notin P \cup Q$)*. Any solution has* $|P \cup Q| \geq 60$ *elements, and* $\min(\prod P, \prod Q) > 3.50 \times 10^{57}$*.*

Proof. Rigidity holds verbatim for pairwise–coprime families (the cofactor–sum argument uses only coprimality), so again $T(U) = s + t > 2$. The least prime factors $\ell(e)$ of the elements are distinct, and $\sum_e 1/e \;\leq\; \sum_e 1/\ell(e)$. Suppose $|U| = 59$ and some element e is not prime, so $e \geq \ell(e)^2$. (Formalised: `Erdos307.coprime_loss` is the per–element bound, `loss_antitone` its monotonicity, and `loss_at_277` the numeric consequence (`lean/Erdos307/Coprime60.lean`); the slack itself is the level–59 census.) If $\ell(e) \leq 277$, the loss is $1/\ell - 1/e \geq (\ell - 1)/\ell^2 \geq 276/277^2 = 0.00359\ldots$, exceeding the maximal available slack $T_{59} - 2 = 0.00235\ldots$; if $\ell(e) \geq 281$, then $e \geq 281^2$ while the other 58 reciprocals sum to at most T_{58}, giving $T(U) \leq T_{58} + 281^{-2} = 1.99875\ldots < 2$. Either way $T(U) < 2$, a contradiction: at 59 elements every element is prime, and Proposition 4.5 applies verbatim. The product bound transfers because $\prod_e e \;\geq\; \prod_e \ell(e)$ and $T(U) \;\leq\; \sum_e 1/\ell(e)$, so the extremal ratio bound at the first 60 primes still applies. □

Remark 4.7 (Verification tier, and why the ladder stops here)**.** Theorem 4.3 is machine–checked in Lean 4, and so is Proposition 4.5: the theorem `erdos307_sixty` formalises the entire argument; the forced–primes and element bounds, the Pythagorean plus–square, and a pruned depth–first search over the pool whose *completeness is proved, not assumed* (`dfs_sound`), the search itself running under `native_decide`; on top of the two independent unverified programs. The level cannot be climbed further by finite means: already at $|U| = 60$ the family $U = \{p_1, \dots, p_{59}\} \cup \{q\}$ qualifies for *every* prime $q > 277$, and on it the two square conditions read $x^2 = Aq + \Pi_{59}$, $y^2 = Bq + \Pi_{59}$ with $A, B = N_{59} \pm 2\Pi_{59}$, i.e. the Pell–type system $Bx^2 - Ay^2 = -4\Pi_{59}^2$ with 113–digit coefficients. We verified that no accessible modulus obstructs it (mod 32, and all odd $\ell < 4000$: the conditions pin $q \equiv 5 \pmod 8$ and thin each $\ell \mid \Pi_{59}$ to about half of its residue classes and each $\ell \nmid \Pi_{59}$ to about a quarter, but never to zero), consistent with the local–solubility theme of Section 5. The obstruction, however, lives at the primes *of* A and B, and it is detectable *without factoring*: Proposition 4.8 below closes this family for *every* q by a single Jacobi–symbol computation. (An exact sweep to $q \le 10^{12}$, `hunt/tail_sweep.rs`, 1.25×10^{11} candidates, none with $Aq + \Pi_{59}$ even a single square, which corroborates it independently.) The tail family also exposes the self–similarity of the ladder: writing $a = qm$, so that $m\,b = \Pi_{59}$ partitions the first 59 primes, its two closing conditions collapse to the exact equation

$$D(m)\,D(b) + m^2 \;=\; \Pi_{59}$$

together with the divisibility $m \mid D(b)$ (then $q = D(b)/m$, required prime); verified exhaustively on a 12–prime toy universe. Equivalently, the deficit $1 - \sigma(m)\sigma(b)$ must equal m^2/Π_{59} *exactly*: the circularity $\Delta_0 = M_0 N(1 - s_{M_0} s_N)$ of Section 7 in its purest form, with the deviation forced to be the perfect square m^2. Closing even this one family is thus a 2^{58}–fold exact–coincidence search of the same type as #307 itself (expected count $\sim 2^{58}/\Pi_{59} \approx 10^{-95}$): each rung of the ladder past 60 contains the whole problem. The one case with *no* free large prime (a 60–set drawn entirely from small primes) is instead a finite search, and empty in the minimal range: a direct both–squares enumeration of all 60–prime sets with $T > 2$ supported in the first 70 primes (4,011,289 sets; `hunt/close60.rs`) finds none passing even the plus–square, so no level–60 solution has all its primes ≤ 349. (The first 68 primes give 1,258,448 sets, also empty, and reproduce independently.) By Corollary 13.5 of the full version this verdict is decisive rather than provisional: at these masses a support passing the plus– and minus–squares, with the parity of Proposition 13.1(2) of the full version, *is* a two–cycle, so a survivor would have been a solution and not a candidate. The enumeration extends with compute.

Proposition 4.8 (The canonical tail family is empty: a reciprocity kill)**.** *No solution of #307 has support $U = \{p_1, \dots, p_{59}\} \cup \{q\}$, for any prime q.*

Proof. A solution supported on U forces $x^2 = Aq + \Pi_{59}$ with $x = a + b$ and $A = N_{59} + 2\Pi_{59}$ (Remark 4.7). Every prime $\ell \mid A$ exceeds 277: for $\ell \le 277$ one has $\ell \mid \Pi_{59}$, so $\ell \mid A$ would force $\ell \mid N_{59}$, contradicting $\gcd(N_{59}, \Pi_{59}) = 1$ (rigidity). Hence $\ell \nmid \Pi_{59}$, and reduction mod ℓ gives $x^2 \equiv \Pi_{59} \pmod{\ell}$, requiring the Legendre symbol $(\Pi_{59} \mid \ell) \ne -1$. But the Jacobi symbol satisfies

$$\left(\tfrac{\Pi_{59}}{A}\right) \;=\; -1$$

(a finite gcd–speed computation, script `code/tailkill.py`; no factorisation of the 114–digit A is needed), so some $\ell \mid A$ of odd multiplicity has $(\Pi_{59} \mid \ell) = -1$. The family is empty, for every q.

The companion symbol at $B = N_{59} - 2\Pi_{59}$ equals -1 as well; necessarily: for any admissible base S (writing $D = D_S$, $N = N_S$, $A_S, B_S = N \pm 2D$) one has $A_S \equiv B_S \pmod 8$ (as $4D \equiv 8 \bmod 16$) and $A_S \equiv B_S \equiv N \pmod p$ for every odd $p \mid D$, while the quadratic–reciprocity sign difference carries the exponent $\frac{p-1}{2} \cdot \frac{A_S - B_S}{2}$ with $\frac{A_S - B_S}{2} = 2D$ even; hence $(D \mid A_S) = (D \mid B_S)$ always; one binary certificate per tail family. Running it over all 49,961 admissible bases of Proposition 4.5 proves 31,219 of the one–new–prime families at level 60 empty (62.5%, curiously near 5/8); the remaining 18,742 are Jacobi–inconclusive: their -1–Legendre primes, if any, occur to an even count, visible only through a factorisation of A_S. A factoring attack on those (trial division and Brent–Pollard ρ over a Montgomery core, `hunt/survivor_kill.rs`; the found factors are kept as a pairwise–coprime basis, so a kill certificate (an odd–multiplicity basis piece P with $(D_S \mid P) = -1$) needs no primality testing at all), pushed to ρ–budget 10^6 with a Pollard $p-1$ stage, closes a further 15,264: in total 46,483 of the 49,961 one–new–prime families at level 60 are proven empty (93.0%). The 3,478 still open resist all of this for a structural reason, not a want of effort: each is Baillie–PSW *composite* yet has no prime factor below $\sim 10^{12}$, so its A_S (or B_S) is a *balanced* semiprime of two $\sim$57–digit primes, splitting one is factoring at cryptographic scale. Two cheap tests confirm no shortcut survives: a product–tree batch–GCD over all 37,484 values A_S, B_S finds only shared primes below 10^9 (already found by ρ, so *no* free kills), and Baillie–PSW leaves exactly 34 families in which A_S and B_S are both *probable* primes; these are immune *conditionally on that primality* (a prime A_S with $(D_S \mid A_S) = +1$ has no inert factor, hence no congruence obstruction for any q). The conditionality is not pedantry and not removable by more of the same computation: Baillie–PSW is rigorous only in its *composite* verdicts, so the 46,483 kills are unconditional while the 34 immunities are not. Upgrading them would require primality *certificates* for thirty-four 114–digit integers (ECPP or a Pocklington factorisation of $A_S - 1$). The elementary route does not reach: immunity needs *both* A_S and B_S prime, and partial factorisation of $n-1$ (trial division to 3×10^6 together with the fully split part) leaves $\log F/\log n$ below 1/3 on at least one side of every one of the 34, so neither Pocklington ($F > \sqrt{n}$) nor Brillhart–Lehmer–Selfridge ($F > n^{1/3}$) applies to any family: 0 of 34 are certifiable this way (`code/immune_certify.py`). This is now settled. Running the Adleman–Pomerance–Rumely –Cohen–Lenstra test on all 68 integers proves every one of them prime in 8 seconds of wall clock, so the 34 immunities upgrade from [C] to [P] and are *unconditional* (`code/immune_prove.gp`, PARI/GP 2.17.4, whose `isprime` is a proof and not a probable–prime test). The run reproduces the whole chain independently: 49,961 admissible bases, 31,219 Jacobi kills, and, among the 81 bases whose A_S and B_S are both prime, exactly the 34 with $(D_S \mid A_S) = +1$; the other 47 carry $(D_S \mid A_S) = -1$ and were already killed. The primality is moreover backed by *independently checkable* evidence rather than a verdict: `code/immune_certify.gp` emits Atkin–Morain ECPP certificates for all 68 integers, archived at `certs/immune_ecpp.txt` (512 KB) so a third party can check them without trusting the prover. That file is a `primecertexport` dump and is not machine–readable: PARI's `read` parses its header as a polynomial, so `primecertisvalid` cannot consume it, and earlier versions of this paper and of the repository documented that command in error. The working check is `code/immune_cert_verify.gp`, which extracts the 68 asserted subjects and re–proves each by ECPP in 1.3 seconds: 68/68 prime, 0 failures. The structural half of the campaign is now formal: the identity $(D_S \mid A_S) = (D_S \mid B_S)$, which is what makes one binary test decide a family rather than two, is proved in Lean 4 as `Erdos307.jacobiSym_A_eq_B` (`lean/Erdos307/Campaign.lean`), on the three standard axioms and with no `native_decide`; it follows from periodicity, $A_S - B_S = 4D_S$ being exactly the period of $J(a \mid \cdot)$. The primality itself is not formalised and cannot presently be: mathlib's only route to a large prime is `lucas_primality`, which needs a complete factorisation of $n-1$, and that is precisely what the 0 of 34 above rules out. Note the two counts answer different questions and both are correct: 81 is the number of prime pairs, 34 the number of *immune* families among them. A complete kill/immune split of the residual is in any case beyond feasible computation, and the honest state of the single–tail sector is 46,483 *empty* [P], 34 *immune and now unconditionally* [P] *by the* APRCL *run above,* 3,478 *factoring–hard.* ∎

Proposition 4.9 (Sector decomposition of level 60)**.** *Let U be a 60–element prime support with $T(U) > 2$. Then exactly one of the following holds:* (i) $T(U) - 2 \geq 1/\max U$*, and* $U = S \cup \{q\}$ *for an admissible base* S *(one of the* 49,961 *of Proposition 4.5); the* single–tail sector*, or* (ii) $T(U) - 2 < 1/\max U$*, and, writing $p > q$ for the two largest elements and W for the rest,* $T(W \cup \{p\}) < 2$ *and* $T(W \cup \{q\}) < 2$*; the* pair sector*, whose members escape every single–tail family.*

Proof. U lies in the tail family of $U \setminus \{r\}$ iff $T(U) - 1/r \geq 2$, which is easiest for $r = \max U$; (i) is that case ($T = 2$ being impossible by rigidity, the base is admissible). In case (ii) no removal reaches 2, and a fortiori the two stated 59–subsets stay below 2. □

Remark 4.10 (The campaign; completeness of the certificate; the arity barrier)**.** The reciprocity certificate captures *every* q–independent congruence obstruction of a tail family: at a prime $\ell \mid A_S B_S$ the system degenerates to the rank–one congruences $x^2 \equiv D_S$ or $y^2 \equiv D_S \pmod{\ell}$, while at $\ell \nmid 2A_S B_S$ the conditions define conics in (x, q) or (y, q) with $\sim \ell$ points, deleting at most half of the residue classes of q and never all (the 2–adic layer pins $q \equiv 5 \bmod 8$ without emptiness); by CRT and Dirichlet, admissible q survive any finite set of such conditions. Hence a tail family admits a congruence kill iff $(D_S \mid \ell) = -1$ for some $\ell \mid A_S B_S$, which the Jacobi symbol detects for 31,219 bases, and partial factorisation (`hunt/survivor_kill.rs`) for a further 15,264: in total 46,483 of the 49,961 single–tail families at level 60 are proven empty (93.0%), 3,478 remaining open (each a balanced ∼57–digit semiprime, factoring–hard). The weapon is intrinsically *arity–one*, and provably fails on the pair sector. With $s = p + q$, $t = pq$, a pair–sector member requires the two simultaneous squares $x^2 = A_W t + D_W s$ and $y^2 = B_W t + D_W s$ (the would–be third condition $s^2 - 4t = \square$ is automatic, being $(p - q)^2$), together with the reciprocity–linked constraints $(D_W p \mid q) = (D_W q \mid p) = 1$ (mod q, a solution with $q \mid a$ has $N_U \equiv D_W p$, forcing $D_W p$ to be a square). At a prime $\ell \mid A_W$ the condition $x^2 \equiv D_W s \pmod{\ell}$ constrains only s, which the free second prime absorbs; the pair system is in fact locally soluble at *every* modulus (a theorem, Proposition 4.11 below, corroborated by a direct scan over all $\ell \leq 600$) so, unlike the single–tail family, the pair sector admits *no* congruence obstruction at all. It is a genuine two–parameter surface per base, #307 in miniature (Remark 4.13); reciprocity cannot touch it, and only a direct both–squares search or a descent argument can.

Proposition 4.11 (The pair sector admits no congruence kill)**.** *Let W be any pair–sector base and $A = A_W$, $B = B_W$, $D = D_W$, so that a pair–sector solution requires $x^2 = A\,\alpha\beta + D(\alpha + \beta)$ and $y^2 = B\,\alpha\beta + D(\alpha + \beta)$ with α, β the two new primes. For every prime power ℓ^j this system has a solution in units α, β and integers x, y at a nonsingular point. Consequently, by CRT and Dirichlet, no finite system of congruence conditions empties any pair–sector family: the arity barrier of Remark 4.10 is a theorem, and the certificate programme provably ends at arity one.*

Formalised, in part. *`Erdos307.no_pinning_isUnit` is the step that makes the tail–kill mechanism unavailable, `collapse_identity` is regime* (2)*'s observation that the two square conditions are one, `regime_one_first` and `regime_one_second` are the explicit unit of regime* (1)*, and `targets_agree_mod_eight` is the mod–8 collapse of regime* (3)*. The lift from ℓ to ℓ^j that regime* (1) *needs is `square_lifts_to_prime_powers`, proved from `hensel_all_powers`: if ℓ is an odd prime not dividing c and c is a square modulo ℓ, it is a square modulo every power of ℓ. Weil's bound for the point count in regime* (2) *is the one remaining input, and it is* cited*, not pending: the quantity $\delta = u^2 - 4m$ is a quartic in y, so the count is Hasse–Weil for $v^2 = \delta(y)$, and Mathlib has elliptic–curve infrastructure but no point–count bound. Formalising Hasse's theorem is a contribution to Mathlib rather than to this problem, so the star here records a citation.* 0 *`sorry`* (`lean/Erdos307/PairLocal.lean`).

Proof. First, no ℓ pins either target to a constant: that needs $\ell \mid A$ (or B) *and* $\ell \mid D$, whence $\ell \mid N$, contradicting $\gcd(N, D) = 1$ (rigidity); so the tail–kill mechanism is unavailable, and solubility is decided in three regimes.

(1) ℓ odd, $\ell \mid D$. This covers every odd $\ell \le 103$: at level 60, omitting a prime p caps $T(U)$ at $T_{61} - 1/p$, and $1/p > T_{61} - 2 = 0.00944\ldots$ exactly for $p \le 103$, so all such primes lie in U, hence in W (the two adjoined primes are the largest). Take $\beta = 1$, $x = 1$, $\alpha = (1 - D)/(A + D) \bmod \ell^j$, a unit since $A + D \equiv N \not\equiv 0 \pmod{\ell}$: the first equation holds by construction, and the second target $(B + D)\alpha + D \equiv N \cdot N^{-1} = 1 \pmod{\ell}$ is a unit congruent to a square, so y exists modulo ℓ^j.

(2) ℓ odd, $\ell \nmid 2D$ (in particular every $\ell \ge 107$ outside W). The two conditions are secretly *one*: setting $m = (x^2 - y^2)/4D$ and $u = (x^2 - Am)/D$, the relation $A - B = 4D$ gives the polynomial identity $Bm + Du = y^2$. So any (x, y) with $xy \ne 0$, $x^2 \ne y^2$ and $\delta := u^2 - 4m$ a square delivers α, β as the roots of $z^2 - uz + m$ (nonzero, as $\alpha\beta = m \ne 0$). For each admissible x, δ is a quartic in y and not a square in $\overline{\mathbb{F}}_\ell[y]$ (a factorisation $(Q - R)(Q + R) = 16D^3(x^2 - y^2)$ would force $\ell \mid 4D$), so Weil's bound gives $\frac{1}{2}\ell^2 + O(\ell^{3/2})$ admissible pairs; positive well below $\ell = 107$. The point is nonsingular ($xy \ne 0$; the Jacobian contains $\operatorname{diag}(2x, 2y)$), so it lifts to every ℓ^j.

(3) $\ell = 2$. Both targets are odd and agree modulo 8 (as $\alpha\beta$ is odd, $D(\alpha + \beta) \equiv 0 \bmod 4$, and $(A - B)\alpha\beta = 4D\alpha\beta \equiv 0 \bmod 8$), and a finite table over ($A \bmod 8$, $D \bmod 8$) shows that for every base some achievable pair $(\alpha\beta, \alpha + \beta) \bmod 8$ makes the common target $\equiv 1 \pmod 8$; a 2–adic square, soluble modulo 2^j for every j.

All three regimes, the collapse identity, the threshold $1/103 > T_{61} - 2 > 1/107$, and the mod–32 layer were verified computationally on three bases and every $\ell \le 600$ (`code/pairlocal.py`; the measured density of admissible (x, y) in regime (2) is 0.497–0.503 against the predicted $\frac{1}{2}$). ☐

Remark 4.12 (Anatomy of the certificate; the empirical 5/8 law)**.** Writing $D = 2D_1$, the reciprocity signs in the certificate cancel ($\frac{A-1}{2} + \frac{N-1}{2} = N - 1 + 2D_1$ is even), leaving the clean product

$$\left(\tfrac{D}{A}\right) = \left(\tfrac{2}{A}\right)\left(\tfrac{D_1}{N}\right), \qquad \left(\tfrac{2}{A}\right) = +1 \iff N \equiv 3, 5 \pmod 8,$$

and one layer down $(N \mid D_1) = \prod_{p \mid D_1} \left(\frac{D/p}{p}\right)$, since $N \equiv D/p \pmod p$: the cofactor structure of rigidity recurs inside its own certificate. This product evaluates in closed form: with $D/p = 2D_1/p$ and quadratic reciprocity,

$$(N \mid D_1) = (2 \mid D_1) \prod_{\{p,q\} \subset D_1} (-1)^{\frac{p-1}{2}\frac{q-1}{2}} = (-1)^{s + \binom{t}{2}}, \quad s = \#\{p \mid D_1 : p \equiv 3, 5\ (8)\},\ t = \#\{p \mid D_1 : p \equiv 3\ (4)\},$$

a Rédei genus character of D_1, *independent of* N. This is forced, not merely observed: $(N \mid D_1) = \prod_{p \mid D_1} (N \mid p)$, and cofactor survival gives $N \equiv D/p = 2D_1/p \pmod p$, so $(N \mid p) = (2 \mid p)\,(D_1/p \mid p)$. The first factors multiply to $(-1)^s$ since $(2 \mid p) = -1$ exactly for $p \equiv 3, 5 \pmod 8$; and, each unordered pair $\{p, q\}$ occurring once in each of the two cofactors,

$$\prod_{p \mid D_1} \left(\frac{D_1/p}{p}\right) = \prod_{\{p,q\} \subset D_1} \left(\frac{q}{p}\right)\left(\frac{p}{q}\right) = \prod_{\{p,q\} \subset D_1} (-1)^{\frac{p-1}{2}\frac{q-1}{2}} = (-1)^{\binom{t}{2}},$$

by quadratic reciprocity, since $\frac{p-1}{2}$ is odd exactly for $p \equiv 3 \pmod 4$. (Checked independently, prime by prime, on 20,000 bases: 0 violations.) The inner symbol is thus not a coin but a genus invariant; its empirical fairness (49.9%, 50.0% over the 49,961 bases) is that character's equidistribution; and $(D_1 \mid N)$ follows by the reciprocity sign $(-1)^{\frac{D_1-1}{2}\frac{N-1}{2}}$, hence depends on N only through $N \bmod 4$. In fact the entire certificate closes into a finite functional. Let $v(S) = (n_1, n_3, n_5, n_7) \bmod 4$ count the odd primes of S in the residue classes $1, 3, 5, 7$ modulo 8. Every factor above is a function of v alone: $s \equiv n_3 + n_5$, $t = n_3 + n_7$ (with $\binom{t}{2}$ needing only $t \bmod 4$), $D_1 \equiv 3^{n_3}5^{n_5}7^{n_7} \pmod 8$, $S(D_1) \equiv (n_1 + n_5) - (n_3 + n_7) \pmod 4$, and (by the mod–8 law of Proposition 13.14 of the full version, $D_1' \equiv D_1 S(D_1) \pmod 8$) also $N = D_1 + 2D_1' \equiv D_1 + 2\big(D_1 S(D_1) \bmod 4\big) \pmod 8$. Hence

$$(D \mid A) \;=\; K\big(v(S)\big):$$

the kill is decided by sixteen residue counts modulo 4. The realized vectors are exactly the coset $n_1 + n_3 + n_5 + n_7 \equiv 58 \equiv 2 \pmod 4$ (all 64 of its classes occur), K is constant on each, and the functional reproduces the direct 113–digit Jacobi computation on every one of the 49,961 bases (`code/kill_classvector.py`). The five–eighths law is then an exact finite statement, and its mechanism is the *reciprocity switch*: for $N \equiv 1 \pmod 4$ the reciprocity sign is inert and the kill reduces to the genus coin; exact conditional rate $12{,}699/24{,}975 = 0.5085$; while for $N \equiv 3 \pmod 4$ the sign couples the parity of t into the product and the rate jumps to $18{,}520/24{,}986 = 0.7412$ (by class modulo 8: 0.509, 0.741, 0.508, 0.741 at $N \equiv 1, 3, 5, 7$; the 3/4 classes are $N \equiv 3 \pmod 4$, not the $(2 \mid A)$ classes). The blend $\frac{1}{2} \cdot \frac{24975}{49961} + \frac{3}{4} \cdot \frac{24986}{49961} = 0.62503$ against the exact $31{,}219/49{,}961 = 0.62487$ is the whole of the "coincidence". The correlation is an ensemble property, a random–subset proxy gives $\approx$0.45/0.55; and the functional yields no second certificate: it recomputes the same symbol.

Remark 4.13 (The kill–program is one–sided)**.** Every kill is a proof that no solution hides in that family, but if #307 has a solution, the solution's own family is unkillable, so no accumulation of kills can overshoot the truth: the program's asymptote *is* the existence question. Together with the self–similarity of Remark 4.7 (each rung past 60 contains the whole problem), this locates the campaign precisely: it corners where a solution could live, one certified region at a time, and can terminate only if #307 has answer NO, which the heuristics disfavour.

Remark 4.14 (The reciprocity kills are the local factors of the expected count)**.** The divergent naive count of Section 5 and the reciprocity sieve of Proposition 4.8 are the *same* computation. Decompose the expected number of solutions over tail families $S \cup \{q\}$: the summand for a base S is $\kappa_S \left(D_S\sqrt{\mathrm{mass}^2 - 4}\right)^{-1} \sum_q q^{-1}$, where $\kappa_S = \prod_\ell$(local solubility density at ℓ) is exactly the reciprocity data. A reciprocity–killed base carries an inert $\ell \mid A_S$ with $(D_S \mid \ell) = -1$, so $N_S + 2D_S \equiv D_S \pmod \ell$ is a non–residue for *every* q, forcing $\kappa_S = 0$; the family drops from the count entirely. An immune base (A_S, B_S both prime, both residues) has no inert prime, so $\kappa_S > 0$ and its summand is a divergent $\sum_q q^{-1}$ (verified on an explicit 114–digit immune base: local density 1 at every tested prime, partial sums growing without bound). Hence the campaign of Remark 4.10 is a Selberg–sieve evaluation of the singular series: the kills remove $\sim$93% of the count's mass, and the residual log–divergence (the entire "weakly favours YES" signal) is carried by the 34 conditionally immune families together with the pair sector. (Conditionally, because their immunity rests on the Baillie–PSW primality of A_S, B_S; the pair sector's contribution, by contrast, is unconditional, being a theorem, Proposition 4.11.) This does not touch the reliability of the count (which ignores the $N(r) \le 1$ rigidity), but it pins down *where* the heuristic's YES lives.

Proposition 4.15 (Local completeness of the reciprocity sieve)**.** *For an admissible base S and an odd prime ℓ, the admissible tail residues $Q_\ell = \{q \bmod \ell : (A_S q + D_S \mid \ell) \neq -1$ and $(B_S q + D_S \mid \ell) \neq -1\}$ fall into exactly three regimes.* (i) *$\ell \mid D_S$, which covers every $\ell \le 167$ uniformly over the* 49,961 *admissible bases, since $1/p \ge (T_{59} - 2) + \frac{1}{281}$ forces every prime $p \le 167$ into every base: then $A_S \equiv B_S \equiv N_S \not\equiv 0 \pmod \ell$, the two conditions collapse to the single condition $(N_S q \mid \ell) \neq -1$, and $|Q_\ell| = (\ell + 1)/2 \ni 0$.* (ii) *$\ell \mid A_S B_S$: one condition is the constant $(D_S \mid \ell)$, so $Q_\ell = \emptyset$ iff $(D_S \mid \ell) = -1$ (exactly the certificate of Proposition 4.8)* *and otherwise* $|Q_\ell| \ge (\ell-1)/2$. (iii) $\ell \nmid A_S B_S D_S$ *(hence* $\ell \ge 173$*): the complete character sums give* $|Q_\ell| \ge (\ell-3)/4 - 2 \ge 40$ *(the linear sums vanish, the quadratic sum is* $-\chi(A_S B_S)$*, boundary terms* ≤ 2*).* (*Regimes* (i) *and* (ii) *are formalised:* `Erdos307.regime_one_collapse`*,* `regime_one_same_argument` *and* `regime_two_constant` (`lean/Erdos307/LocalComplete.lean`)*. Regime* (iii)*'s counting step is* `regime_three_count`*, which takes the two complete character–sum evaluations as hypotheses. Those evaluations,* $\sum_q \chi(aq+b) = 0$ *for* $a \neq 0$ *and* $\sum_q \chi((aq+b)(cq+d)) = -\chi(ac)$ *for* $ad \neq bc$*, are classical and are* cited, *not pending: Mathlib does not carry them, and supplying them is a contribution to Mathlib rather than to this problem. The star on this proposition records a citation, not unfinished work.) Consequently* $Q_\ell \neq \emptyset$ *forces* $|Q_\ell| \ge 2$ *at every modulus, the surviving local density is positive (e.g.* $\prod_{\ell \le 300} |Q_\ell|/\ell \approx 2.2 \times 10^{-19} > 0$ *on the canonical base), and by CRT* no finite system of congruences (no covering) can annihilate an unkilled family*: the binary certificate of Proposition 4.8 is the complete local theory of the tail sector, there is no "second kill layer," and the one–sidedness of Remark 4.13 is structural rather than an artefact of method. (All three regimes verified exhaustively on the canonical base: collapse and* $|Q_\ell| = (\ell+1)/2$ *at* $\ell \in \{3,5,7,101,277\}$*; the Weil floor at every prime* $281 \le \ell \le 600$*;* A_S, B_S *carry no factor below* 10^6*, consistent with the balanced–semiprime wall of Remark 4.10.)*

Proposition 4.16 (The Pythagorean tail prime is bounded)**.** *Let* S *be a finite set of primes,* $D = \prod S$*,* $N = \sum_{p \in S} D/p$*,* $A = N + 2D$*,* $B = N - 2D$*, and suppose* $D < N$*. Let* q *be a prime with* $q \nmid 2D$ *for which* $Aq + D$ *and* $Bq + D$ *are both perfect squares. Then there are integers* m, c *with* $mc = 4D$ *and*

$$q^2 m^2 - 4Nq + c^2 - 4D = 0, \qquad \big(qm^2 - 2N\big)^2 = 4\big(AB + Dm^2\big),$$

so that $q = 2(N \pm k)/m^2$ *with* $k^2 = AB + Dm^2$ *and* $m \mid 4D$*. Moreover* m *is* even*, say* $m = 2\alpha$ *with* $\alpha \mid D$ *and* $\beta = D/\alpha$*, and then* q *is a root of*

$$\alpha^2 q^2 - Nq + (\beta^2 - D) = 0, \qquad \textit{so} \qquad q = \frac{N + \sqrt{N^2 - 4D^2 + 4\alpha^2 D}}{2\alpha^2},$$

a quantity strictly decreasing in α*. Hence the maximum is at* $\alpha = 1$ *and*

$$q \;\le\; \tfrac{1}{2}\Big(N + \sqrt{N^2 - 4D^2 + 4D}\Big) \;=\; tD + O(1), \qquad t = \tfrac{1}{2}\big(\sigma + \sqrt{\sigma^2 - 4}\big), \quad \sigma = \sigma(S) = N/D.$$

The largest admissible tail is therefore the base times the very ratio b/a *that the mass forces. At level* 59 *this is* $9.207 \times 10^{112} = 1.0497\, D$*, a factor* 7.63 *below* $4N$*.*

Proof. Write $x^2 = Aq + D$ and $y^2 = Bq + D$ with $x, y \ge 0$. Since $A - B = 4D$ *exactly,* $x^2 - y^2 = (A-B)q = 4Dq$, that is $(x+y)(x-y) = 4Dq$. The prime q divides exactly one of the two factors: were it to divide both it would divide $2x$ and $2y$, hence x and y as q is odd, hence $q^2 \mid 4Dq$ and $q \mid 4D$, against $q \nmid 2D$. Write the factor divisible by q as qm and the other as c, so that $mc = 4D$ and, in either case, $2x = qm + c$. Multiplying by m,

$$2xm \;=\; qm^2 + mc \;=\; qm^2 + 4D,$$

and squaring against $x^2 = (N+2D)q + D$ gives $4m^2\big((N+2D)q + D\big) = q^2m^4 + 8Dqm^2 + 16D^2$. Substituting $16D^2 = m^2c^2$ and cancelling $m^2 > 0$ leaves $q^2m^2 - 4Nq + c^2 - 4D = 0$; completing the square and using $AB = N^2 - 4D^2$ gives the second display. For the parity, $x + y$ and $x - y$ differ by $2y$ and so have equal parity; were both odd their product would be odd, whereas it is $4Dq$. Hence both are even, and as q is odd, m is even. Finally $m^2 \ge 4$ and $c^2 \ge 0$ give

$4Nq = q^2m^2 + c^2 - 4D \geq 4q^2 - 4D$, that is $Nq \geq q^2 - D$; if $q \geq N+1$ then $q^2 - Nq - D \geq q - D > 0$ because $D < N < q$, a contradiction, so $q \leq N$. (Using only $m \geq 1$ here gives $q \leq 4N$; the parity removes the factor four.) For the sharp form, $m = 2\alpha$ and $c = 2\beta$ with $\alpha\beta = D$ turn $q^2m^2 - 4Nq + c^2 - 4D = 0$ into $\alpha^2q^2 - Nq + \beta^2 - D = 0$, whose larger root is $\big(N + g(u)\big)/(2u)$ with $u = \alpha^2$ and $g(u)^2 = N^2 - 4D^2 + 4uD$. Its derivative in u has the sign of $2uD - g(N+g)$, and $4uD = g^2 - N^2 + 4D^2$ turns that into $-\big((g+N)^2 - 4D^2\big)/2$, negative since $N > 2D$. So the root decreases in α and is largest at $\alpha = 1$. $\square$

Proposition 4.17 (What a barrier improvement must supply)**.** *Let (a,b) be a solution with $a < b$ and put $t = b/a$, so that $t = \sigma(a)$ and, by $\sigma(a)\sigma(b) = 1$,*

$$\sigma(U) \;=\; \sigma(a) + \sigma(b) \;=\; t + \tfrac{1}{t}.$$

For n with $T_n > 2$ let $t_n > 1$ be the root of $t + 1/t = T_n$. Since $\sigma(U) \leq T_{|U|}$,

$$\sigma(a) > t_n \implies |U| \;\geq\; n+1,$$

and this is the only route by which a mass argument can raise the barrier. Numerically

n	59	60	61	62	63	64
T_n	2.0023502	2.0059089	2.0094424	2.0128554	2.0161127	2.0193282
t_n	1.0496677	1.0798804	1.1020081	1.1199914	1.1352477	1.1490254

So the barrier level is decided entirely by how close $\sigma(a)$ may lie to 1, and each further level demands a definite gap: $|U| \geq 61$ needs $\sigma(a) > 1.0799$, a full 8% above 1.

That is why the mass mechanism is exhausted here rather than merely difficult. The only unconditional lower bound on the gap is $\sigma(a) - 1 = (a' - a)/a \geq 1/a$, which is worthless at $a > 10^{56}$; and σ genuinely comes close to 1 from above on very few primes, the Sylvester-type set $\{2, 3, 7, 41\}$ giving $\sigma = 1.00058\ldots$ with four. Nothing in the cycle equations forbids $a' - a = 1$. A barrier past 60 must therefore come from a mechanism that bounds $\sigma(a)$ away from 1 without appealing to the size of U — which is what Theorem 4.3 does in the opposite direction — and we know of none.

The same reduction locates the extreme case. Writing $c = b - a$, the identity $\sigma(U) - 2 = (t-1)^2/t = c^2/D$ is the minus layer $D' - 2D = c^2$, and $\gcd(c, D) = \gcd(b-a, ab) = 1$ because $\gcd(a,b) = 1$. Since an admissible support contains every prime ≤ 167 (Remark 4.7), c is divisible by none of them, so $c = 1$ or $c \geq 173$. The branch $c = 1$ is $D' = 2D + 1$, equivalently $a' = a + 1$ and $(a+1)' = a$; it is the closest a solution could come to $\sigma(a) = 1$, and it is exactly the configuration a mass argument cannot reach. Below 3×10^6 the equation $a' = a + 1$ has only the solutions 30, 858, 1722 and 66,198, and none extends: for the first three $a + 1$ is prime, so $(a+1)' = 1$, and for the last $a + 1$ is not squarefree. The branch is not excluded, only unsearched, since the barrier puts it above 10^{56}. The level–59 computation that gives $|U| \geq 60$ is, in these terms, an exhaustive verification that $\sigma(a) > t_{59}$, and Proposition 4.19 says the same verification is unavailable one level up.

Remark 4.18 (The Lehmer reading of the tail is withdrawn)**.** Remark 4.21 below reads the surviving question as the primality of a term q_n of the *exponential* Pell orbit, and Remark 13.16 names that orbit as the only live branch. Proposition 4.16 withdraws the reading. The orbit is genuinely infinite and both square conditions do hold along all of it; what fails is that no term past N can be prime. The single identity responsible is $A - B = 4D$, which the earlier treatment used only to form the Pell system and never to factor $4Dq$.

Three consequences, in increasing order of what they cost.

The gate is finite, not Lehmer. The tail primes of a base lie in the explicit finite set $\{2(N \pm k)/m^2 : m \mid 4D,\ k^2 = AB + Dm^2\}$. So the second of the three gates is a search over the divisors of $4D$, not a primality question along an exponential sequence, and the classification of the residual difficulty as Lehmer–class rather than Schinzel–class does not survive. Checked sound *and complete* against exhaustive search on 1200 bases with zero mismatches, and stepped out along a real orbit, in `code/tailbound.gp`: on the base $D = 1$, $N = 7$ (bound $N = 7$) the orbit gives $q_1 = 7$ prime, then $q_2 = 741895$, $q_3 = 76921173511$, and every later term composite, with both square conditions holding throughout.

The count reverses sign, and the search has now been run. The expected number of admissible m is $\sum_{m|4D}(AB + Dm^2)^{-1/2} \approx D^{-1/2}\sum_{m|4D} m^{-1}$, which on the first immune base is 2.4×10^{-56} and below 10^{-38} summed over all 34 immune families, the opposite of the "weakly favours YES" of Remark 4.14 on exactly the families that reading was about. That prediction is now checked. The full divisor space is $4 \cdot 2^{58} \approx 1.15 \times 10^{18}$ per family and is not enumerable, but because the hit probability falls like $1/(m\sqrt{D})$ the expected count is carried by small m: restricting to $\omega(m) \le 8$ leaves 99.9998% of the expected count. Over all 34 families that slice is 3.0777×10^{11} candidates, and

$$\text{it contains no } m \text{ with } AB + Dm^2 \text{ a perfect square}$$

(4,933 survive the modular cascade, exact `issquare` kills all of them; the earlier $\omega(m) \le 7$ pass, 4.7081×10^{10} candidates and 771 survivors, agrees).

The same search has been run over *every* admissible base, which matters because Proposition 4.16 requires no primality of A_S or B_S, only the algebra. The 34 immune families are the ones left after the Jacobi kill and the factoring attack, and their immunity is conditional on a Baillie–PSW verdict; the other 3,444 of the 3,478 open families are open precisely because A_S or B_S is a balanced semiprime nobody can split. The tail search does not care. Over all 49,961 admissible bases at $\omega(m) \le 4$, which is 99.2% of the expected count per base, the run is 9.1296×10^{10} candidates, 1,469 survive the modular cascade, and exact `issquare` kills all 1,469. Every survivor was additionally checked to satisfy $m \mid 4D$, the hypothesis the proposition needs. So

$$\text{no admissible base admits a Pythagorean tail prime with } \omega(m) \le 4,$$

unconditionally and with no primality input. Where Proposition 4.8 leaves 3,478 of the one–new–prime families at level 60 open, of which 34 only conditionally immune, this leaves none open on the stated slice. It is a slice result and not a proof: the untested remainder is 0.8% of the expected count per base, and $\omega(m) \ge 5$ is untested. The run is `code/tailsearch.rs` (a cascade of quadratic–residue filters modulo primes coprime to $2D$, 771 survivors) with every survivor then tested exactly by `issquare` in `code/tailsearch_verify.gp` (0 squares). A positive control recovers the known solutions of the toy base $D = 1$, $N = 7$, and an exhaustive exact run at $\omega(m) \le 2$ agrees with the filter, so the cascade is not silently over–rejecting. This is VERIFIED on the stated slice, not on the full divisor set: the residual 0.0018% of the expected count is untested.

The filter primes must be coprime to $2D$, which is itself a structural fact rather than a convenience. Since D contains every prime up to 167, for any $\ell \mid D$

$$AB + Dm^2 - N^2 = D\big(m^2 - 4D\big) \equiv 0 \pmod{\ell},$$

so the value is the square N^2 modulo ℓ for *every* m, and as $\gcd(D, N) = 1$ Hensel lifts that root to every power of ℓ. The tail search therefore has no local obstruction at any prime dividing D, the analogue of Proposition 4.15 for this formulation; the classical 64/63/65/11 pre–filter for squares rejects nothing here, measured at a 75% pass rate modulo 64 and 100% modulo 63. Formalised as `tail_value_mod_dvd` and `hensel_lift_identity`; the induction from the step to all powers is routine and is not formalised.

The paper's own height estimate now says the same thing. Remark 4.21 puts the least point of a nonempty fiber at height $\exp(10^{112})$ on regulator grounds, so its $q \approx x^2/A$ carries some 10^{112} digits, against a bound of 115. Two independent estimates, one from divisor counting and one from the class–number formula, agree that no immune family admits a Pythagorean tail prime. Neither is a proof: the bound $q \leq N$ is proved, the emptiness is HEURISTIC.

Formalised in `lean/Erdos307/TailBound.lean` (`tail_factor`, `tail_quadratic`, `tail_discriminant`, `tail_bound`, `tail_finite`) on the three standard axioms, with a non–vacuity witness at the base above.

Proposition 4.19 (The level–by–level programme terminates at 60)**.** *Let U be a finite set of primes with $\sum_{p\in U} 1/p > 2$ and let B be arbitrary. Then there is a prime $q > B$ with $q \notin U$ such that $U \cup \{q\}$ again has $\sum 1/p > 2$. Consequently, for every $n > 59$ there are infinitely many n–element prime supports of mass exceeding 2.*

Proof. The primes are infinite, so choose a prime q exceeding both B and $\max U$; then $q \notin U$ and $\sum_{p\in U\cup\{q\}} 1/p = \sum_{p\in U} 1/p + 1/q > 2$. Iterating from the first 59 primes, whose reciprocal sum is $N_{59}/P_{59} = 2.00235\ldots > 2$, produces admissible supports of every cardinality above 59, each containing a prime beyond any prescribed bound. □

Remark 4.20 (What the level barrier costs, and what it does not)**.** Proposition 4.5 closes level 59 by enumerating its 49,961 admissible supports, and Proposition 4.8 together with the search of Remark 4.18 closes level 60 by adjoining one prime to each. The obvious continuation is to enumerate the admissible 60–element supports and repeat. Proposition 4.19 says there is no such enumeration.

The finiteness at level 59 is an accident of one inequality being tight: 59 primes reach mass 2 only just, at $2.00235\ldots$, so demanding mass > 2 from exactly 59 primes pins them near the smallest 59. Allow a sixtieth and the first 59 already carry the mass alone, leaving the sixtieth entirely free. The mass condition, having been met, constrains nothing further.

Two consequences, and the second is the one that matters. The searches of Remark 4.18, however far they are pushed in $\omega(m)$, bear on level 60 alone; widening them is not a route upward. And any further progress on #307 must be *non–enumerative*: exhausting supports one level at a time is not merely expensive beyond level 60, it is impossible, so the method behind every unconditional exclusion in this note is now spent. That is a no–go, not a resource limit.

Formalised in `lean/Erdos307/LevelBarrier.lean` (`admissible_extends`, `admissible_of_card_add`, `level_enumeration_terminates`) on the three standard axioms, with a non–vacuity witness exhibiting the first 59 primes as a set the hypotheses apply to.

Remark 4.21 (The immune signal, and where its Lehmer reading broke)**.** Superseded by Proposition 4.16 and Remark 4.18: the orbit described here is real, but the inference that primality along it is the residual question is not, because the tail prime is bounded by N. What follows is the orbit, the three gates, and the structural points that survive.

For a reciprocity–immune base the Pell system of Remark 4.7, $B_S x^2 - A_S y^2 = -4D_S^2$, is locally soluble everywhere ($\kappa_S > 0$), and a Pythagorean pair in the tail family $S \cup \{q\}$ is *exactly* a solution with $q = (x^2 - D_S)/A_S = (y^2 - D_S)/B_S$ a positive prime. Its solutions grow like ε^n in the fundamental unit of the order of discriminant $4A_SB_S$ (224 digits, A_SB_S a nonsquare, on an explicit immune base). Writing $x_{n+1} = tx_n - x_{n-1}$ with $x_n = (\alpha^n + \beta^n)/2$ and $\alpha\beta = 1$, the squares x_n^2 have characteristic roots $\alpha^2, \beta^2, 1$ and the constant shift is absorbed by the root 1, so

$$q_n = \frac{x_n^2 - D_S}{A_S} \qquad \text{satisfies} \qquad q_{n+3} = (t^2-1)q_{n+2} - (t^2-1)q_{n+1} + q_n,$$

a non–degenerate ternary linear recurrence (verified on four Pell parameters). A reciprocity–killed base is the complementary degeneration: there the same system is locally *insoluble* ($\kappa_S = 0$), so no candidate exists at all, and the dichotomy of Proposition 4.8 is Pell–solubility versus Pell–insolubility.

The return is milder than "$k = 0$" suggests: on the Pythagorean locus the deviation $k = (a' - b)/a$ is an *integer* (the identity $v(D(u) - v) = u(u - D(v))$ with $\gcd(u, v) = 1$), forced into $\{0, -1\}$ by the near–balanced immune ratio, so $k = 0$ is a sign choice rather than a measure–zero coincidence, and the naive Wieferich shortcut that would pin q is vacuous, since $x^2 \equiv D_S \pmod q$ holds identically from $x^2 = A_S q + D_S$.

Neither square condition carries the difficulty alone. Imposing only the minus–square, $d^2 = B_S q + D_S$, the admissible d satisfy $d^2 \equiv D_S \pmod{B_S}$, and along such a class $d = d_0 + tB_S$ one has

$$q(t) \;=\; \frac{d_0^2 - D_S}{B_S} \;+\; 2d_0\, t \;+\; B_S\, t^2,$$

a quadratic in t (toy bases: $3 + 54t + 173t^2$, $7 + 238t + 1693t^2$, $2 + 446t + 17357t^2$): a Bunyakovsky question, of the same class as the $p = 5$ family of Proposition 13.15(iii) of the full version, and conjecturally routine. It is the *conjunction* of the two squares that replaces the quadratic parameter by a fundamental unit and turns the orbit exponential.

Where the reading broke. That exponential growth was taken to place the question in the Lehmer class, on two grounds: no algebraic factorisation is available, D_S being squarefree so that $x^2 - D_S$ is irreducible, and no covering congruence can kill the family, by Proposition 4.15. Both observations are correct, and neither is the relevant one. The factorisation that does exist is not of $x^2 - D_S$ but of $x^2 - y^2 = 4D_S q$, available because $A_S - B_S = 4D_S$ *exactly*: the identity was used here only to build the Pell system, never multiplicatively. Proposition 4.16 draws it, and the ternary recurrence above then supplies no prime term past $q \le N$.

Three nested gates separate an immune family from a solution: fiber nonemptiness, a tail prime, and the $k = 0$ return. The first is a class–group condition, since Hasse supplies only rational points, and it is quantitative: immunity makes $A_S B_S$ a product of two distinct primes, hence squarefree, so the conic lives in the near–maximal order of $\mathbb{Q}(\sqrt{A_S B_S})$ of discriminant $\sim A_S B_S \approx 10^{224}$, and the class–number formula puts the expected regulator at $R = \sqrt{\Delta}\, L(1, \chi)/(2h) \approx 10^{112}$; barring an anomalously large class number, the *least* point of a nonempty fiber sits at height $\exp\big(10^{112}\big)$. That estimate is used in Remark 4.18 as independent corroboration that the second gate is empty. A direct sweep to $q \le 10^{12}$ was structurally incapable of reaching even the first gate.

Remark 4.22 (The affine sieve does not reach a rank–one orbit)**.** The modern instrument for primes in orbits is the affine sieve of Bourgain, Gamburd and Sarnak [38], which produces r–almost–primes in orbits of thin groups. It is the one method designed for exactly the shape of question Remark 4.21 isolates, so its failure here is worth recording precisely rather than left implicit.

The affine sieve needs a spectral gap: the congruence quotients of the acting group must form an expander family, and that is what supplies the level of distribution the sieve consumes. Here the q_n run along a Pell orbit, so the acting group is the cyclic group generated by the fundamental unit of the order of discriminant $4A_S B_S$. Cyclic groups are abelian, hence amenable, and have no spectral gap; expansion is precisely the property an abelian group lacks. The sieve therefore returns nothing, and it returns nothing on $2^n - 1$ for the same reason. The obstruction is rank, not thinness as such: non–elementary thin orbits, Apollonian packings among them, do yield almost–primes.

The natural repair fails for a reason worth stating too. One may embed the rank–one orbit in a larger thin group whose orbit does expand, and sieve there. What the affine sieve then delivers is infinitely many almost–primes in the *larger* orbit, inside which our sequence has relative density zero; the sieve does not localise to a prescribed sub–orbit. Expansion is bought at the cost of losing the sequence.

This meets the unit–rank trichotomy of Remark 6.14 from a second direction. There, rank zero gave a classical gate and denied a continuous family, positive rank the reverse. Here, rank one is exactly the cell in which the affine sieve has no traction. Stated as a single missing property: what is wanted is a substitute for expansion at rank one, and that is the Mersenne problem itself, not a lemma about #307.

Remark 4.23 (The gap to the record, measured)**.** It is worth stating how far the required statement is from the best that is known, because the distance is not one of degree.

Write a sequence's *density exponent* θ for $\#\{n \le X\} \asymp X^{\theta+o(1)}$. The sparsest sets currently proved to contain infinitely many primes are the values of binary forms: $a^2 + b^4$ at $\theta = 3/4$, by Friedlander and Iwaniec [33], and $a^3 + 2b^3$ at $\theta = 2/3$, by Heath-Brown [34], the latter being the sparsest polynomial form known to capture its primes. Our q_n grow like ε^{2n}, so $\#\{n : q_n \le X\} \asymp \log X$ and

$$\theta = 0.$$

The distance from 2/3 to 0 is categorical rather than quantitative. Both records are bilinear arguments: they run a sieve of dimension one and close it with a Type II estimate obtained by averaging over a set of size a positive power of X. Every gain in such an argument is a gain of a power of X, and against $\log X$ terms a power of X gains nothing; below $\theta = 1/2$ the standard sieve axioms already fail, and at $\theta = 0$ there is no set left to average over. This is the same wall as Remark 4.22 approached from the analytic rather than the spectral side, and it is why every sequence of exponent zero, Mersenne, Fibonacci, Lucas and ours alike, is open together.

The structure our sequence does carry is the standard one and confers nothing. From $A_S q_n = x_n^2 - D_S$ one gets $x_n^2 \equiv D_S$ for every prime $p \mid q_n$ not dividing D_S, so all prime factors of q_n lie in the density–$\frac{1}{2}$ set where D_S is a quadratic residue. That constrains the factorisations without excluding any, which is exactly the position the primitive divisor theorems already left us in: divisors, never primality.

Remark 4.24 (The residue, atomised, and where its difficulty comes from)**.** Decomposing the question into independently known statements locates the obstruction on a single atom and identifies its source.

Two atoms carry theorems. Every q_n beyond the thirtieth has a primitive prime divisor [41]; and the greatest prime factor satisfies $P^+(q_n) > n\exp\bigl(\log n/(104\log\log n)\bigr)$, by Stewart [35], which settled questions of Schinzel and Erdős and was the first general advance on Bang and Carmichael. Every other atom, that q_n be squarefree infinitely often, that $\omega(q_n)$ be bounded infinitely often, that q_n be a P_k, and primality itself, is open.

The chain that would close the crux is P^+ large, hence ω small, hence prime. It breaks at the first link, and by a measurable amount. Stewart's bound certifies $\log P^+ \ge (1 + o(1))\log n$ against $\log q_n \approx 2n\log\varepsilon$, so the fraction of the term's size that is certified is of order $\log n/n$: 10.6% at $n = 10$, 0.32% at $n = 10^3$, 4×10^{-4}% at $n = 10^6$. Primality is the fraction 1. No sharpening of the constant 104 changes this, since $n^{1+o(1)}$ is the shape linear forms in logarithms produce. The Subspace Theorem is the other available instrument and does not help: it is stronger under weaker hypotheses but ineffective [36], and an ineffective statement cannot exhibit the single prime term a construction needs.

The source is visible in the comparison with the polynomial case. For n^2+1 the greatest prime factor exceeds $n^{1.3}$ infinitely often [37], that is $q^{0.65}$, a *positive proportion* of the term's size, and on such sequences almost-primality is available too. For an exponentially parametrised sequence only $(\log q)^{1+o(1)}$ is available, a vanishing proportion. The gap between the two regimes is exactly the gap between polynomial and exponential parametrisation, and Proposition 8.4 proves #307 admits no polynomial one. So the obstruction recorded in Remark 4.22 as a question of rank, in Remark 4.23 as one of density exponent, and here as one of P^+, is a single obstruction with three faces, and its source is Proposition 8.4: forced exponentiality is what places the residue in the class where nothing of the required strength is known.

Two things sharpen that attribution, and both make the obstruction less incidental rather than more. First, the boundary is exactly where Proposition 8.4 sits. The fraction of a term's size that Stewart's bound certifies is $\log P^+/\log q_n \approx \log n/\log q_n$, so it stays bounded below precisely when $\log q_n \ll \log n$, that is when q_n grows polynomially: degree d gives the constant $1/d$. Every super-polynomial rate sends it to 0, and not only the exponential one: $\exp(\sqrt{n})$ and $n^{\log n}$ do as well. So polynomial growth is not one convenient case among several, it is the whole of the favourable region, and it is exactly what Proposition 8.4 excludes.

Second, the hypotheses of Proposition 8.4 are worth testing, since a gap there would dissolve everything above. It assumes a *fixed* number of polynomials, and its proof uses that finiteness when it splits $\sum_i 1/p_i$ into constants plus a remainder tending to 0. A family whose support grows with the parameter is therefore not covered. That gap is real and useless: a growing support means a growing number of simultaneous primality conditions, which is a harder demand than one, not an easier one, and it does not return the family to the polynomial regime. The proposition forbids the case that would help and leaves uncovered only cases that would not.

Remark 4.25 (The squarefree atom, and what it reduces to)**.** Among the four open atoms one is structurally easier, and it is worth saying why and how far it gets. Squarefreeness of q_n asks only for an *upper* bound on a count of bad events, never for the production of a prime, so it is the one atom not subject to the parity obstruction that blocks bounded ω, P_k, and primality.

The argument it wants is a union bound. For a prime $p \nmid 2A_SD_S$, $p^2 \mid q_n$ forces $x_n^2 \equiv D_S \pmod{p^2}$; the Pell orbit is purely periodic modulo p^2 with period $\pi(p^2)$, there are at most two square roots of D_S, and each is met a bounded number of times per period, so the density of bad n attached to p is at most $C/\pi(p^2)$. Where $\pi(p^2) = p\,\pi(p)$, which is the generic behaviour, and $\pi(p) \asymp p$, this is $\asymp C/p^2$, and the union bound closes as soon as $C\sum_p p^{-2} < 1$. The constant is comfortable: $\sum_p p^{-2} = 0.4522474\ldots$, so any $C < 2.21$ suffices, and a positive proportion of the q_n would be squarefree, hence infinitely many.

What is missing is the hypothesis $\pi(p^2) = p\,\pi(p)$. Its failure is precisely a Wall–Sun–Sun prime for the sequence, and there the density is $\asymp 1/p$ rather than $\asymp 1/p^2$, whose sum diverges and destroys the bound. No such prime is known for any classical sequence and their finiteness is unproved, so the atom does not close: it *reduces* to a Wall–Sun–Sun statement. That is a sharper position than "open", since it names the obstruction and shows it is a single hypothesis rather than a programme, but it is a reduction and not a proof.

The unconditional pieces are formalised. `Erdos307.primeSquareSum_certificate` gives $\sum_p p^{-2} < \frac{1}{2}$ in the finite form used above, the first six primes together with the bound $\frac{1}{2}\sum_{m\geq 16} m^{-2} < \frac{1}{30}$ for the odd primes beyond; `good_density_pos` and `squarefree_positive_density_of_bound` are the union bound and the conditional implication. All carry 0 `sorry` and the three standard axioms (`lean/Erdos307/Squarefree.lean`).

Two further points fix the standing of this atom, and both are negative. First, on the immune bases the argument is blocked at its *lowest* rung as well as its highest. The density attached to a single small p is $\#\{n \bmod \pi(p^2) : p^2 \mid q_n\}/\pi(p^2)$, read off the Pell orbit modulo p^2, and that needs the fundamental solution, the object Proposition 13.4 places at $10^{24.7}$ operations. For a Lucas sequence with a known fundamental unit the small-prime rung is a finite computation; here it is not. The ladder is therefore blocked at both ends, by non-computability below and by Wall–Sun–Sun above. Second, and more to the point, squarefreeness is not what the construction needs: Remark 4.21 requires q to be *prime*, and a squarefree q with several prime factors is useless there, so even the dream lemma of this atom would leave #307 exactly where it stands. Of the four atoms this is the easiest, and it is also the only one whose resolution would not advance the problem.

Theorem 4.26 (Parity dichotomy). *Let $a' = b, b' = a$ be a derivative two–cycle (so a, b squarefree, $\gcd(a, b) = 1$). Then exactly one of the following holds.*

(i) Mixed parity. *Exactly one of a, b is even; say $a = 2k$ with k odd. Then $b = a'$ is odd and $\omega(b)$ is even.*

(ii) All odd. *Both a, b are odd. Then $P \cup Q$ avoids 2, so reaching reciprocal sum > 2 needs ≥ 1412 odd primes, and $\min(a, b) > 10^{2519}$.*

Proof. They cannot both be even, as $\gcd(a, b) = \gcd(a, a') = 1$. In case (i), write $a = 2k$, k odd squarefree; $b = a' = k + 2k'$ is odd. For odd squarefree $b = \prod_i q_i$ one has $b' = \sum_i b/q_i \equiv \omega(b) \pmod 2$ (each b/q_i is odd). Since $b' = a$ is even, $\omega(b) \equiv 0 \pmod 2$. In case (ii) the union of supports omits 2; the reciprocal sum of the first 1411 odd primes is < 2 while that of the first 1412 exceeds 2, so $|P \cup Q| \geq 1412$ and $\prod_{p \in P \cup Q} p \geq 10^{5040}$ (product of the 1412 smallest odd primes, by direct computation). Applying the same rigidity/ratio argument as in Theorem 4.3(b) (now with $2 \notin U$) gives $\min(a, b) \geq \sqrt{R/T(U)} > 10^{2519}$. The all–odd case is therefore vastly more constrained than the mixed case and, heuristically, empty. □

Proposition 4.27 (Longer cycles are harder; two–cycles are extremal). *Every member of a k–cycle of the arithmetic derivative is squarefree, and for $k \leq 3$ the members are pairwise coprime (each pair is consecutive). Their masses $s(a_i) = a_{i+1}/a_i$ multiply to 1, so by AM–GM $\sum_i s(a_i) \geq k$; for $k \leq 3$ this is the union's reciprocal sum, whence the support contains at least m_k primes, where m_k is least with $\sum_{i \leq m_k} 1/p_i \geq k$. Now $m_2 = 59$ and $m_3 = 361{,}139$ (last prime 5,195,977), so a three–cycle has at least 361,139 primes in its support and product exceeding $10^{2.26 \times 10^6}$ (hence its largest member exceeds $10^{752{,}194}$); by Mertens m_k grows doubly exponentially. Two–cycles are thus the extremal, and only practically relevant; case. (For $k \geq 4$ non–consecutive members may share primes; the bound then holds for the multiset reciprocal sum.)*

5 Local anatomy and a heuristic model

Lemma 5.1 (Local anatomy). *Let a be squarefree and q a prime with $q \nmid a$. Then $q \mid a'$ if and only if $\sum_{p|a} p^{-1} \equiv 0 \pmod q$, where p^{-1} is the inverse of p modulo q. If $q \mid a$, then $q \nmid a'$.*

Formalised. *`Erdos307.anatomy_iff`, with the factorisation $a' \equiv a \sum_{p|a} p^{-1}$ proved as `csum_eq_mul_recipSum` rather than assumed; the second half is `anatomy_on_support` and is Rigidity* (`lean/Erdos307/Frame.lean`).

Proof. $a' = a \sum_{p|a} 1/p$. If $q \nmid a$ then $q \mid a'$ iff $q \mid a \sum_{p|a} p^{-1}$ iff $\sum_{p|a} p^{-1} \equiv 0 \pmod q$. If $q \mid a$, then by Lemma 2.1 applied to the prime set of a, $q \nmid a'$. □

Thus the cycle equation $a' = b$, $b' = a$ factors, prime by prime, into $\omega(a)+\omega(b)$ local congruences of exactly the shape that governs amicable–pair heuristics: for each $q \mid b$ one needs $\sum_{p|a} p^{-1} \equiv 0 \pmod{q}$, and symmetrically. These local conditions, however, are individually *empty of obstruction*:

Proposition 5.2 (No single–modulus congruence obstruction)**.** *For every modulus m with $2 \le m \le 210$, the reduced cross–match system $N_P \equiv D_Q$, $N_Q \equiv D_P \pmod{m}$ is realisable by disjoint squarefree prime sets P, Q, in both structural branches:* $\gcd(m, D_P D_Q) = 1$*, and the branch in which a prime divisor of m lies in $P \cup Q$. Hence no individual modulus in this range can rule out a solution of #307.*

Construction. Write $\Sigma_S \equiv \sum_{p\in S} p^{-1} \pmod{m}$, so $N_S \equiv D_S \Sigma_S$. The reduced system is $\Sigma_P \equiv D_Q D_P^{-1}$, $\Sigma_Q \equiv D_P D_Q^{-1} \pmod{m}$; its only coupling is $\Sigma_P \Sigma_Q \equiv 1$, automatic, with D_P, D_Q otherwise free. By Dirichlet each unit class mod m contains primes, fixing D_P, D_Q; appending a prime $r \equiv 1 \pmod{m}$ leaves D_S fixed while sending $N_S \mapsto rN_S + D_S \equiv N_S + D_S$, i.e. incrementing Σ_S by 1, so (when $\gcd(D_S, m) = 1$) every target class is reached with disjoint sets. In the branch $\ell \mid m$, $\ell \in P$, reduction mod ℓ collapses the system to $N_Q \equiv D_P \equiv 0 \pmod{\ell}$, met by inverse–paired residues in Q; composite m needs no separate gluing, the increment acting directly mod m. We verified both branches for all $2 \le m \le 210$ (for prime and prime–power moduli the inside branch requires the directed construction above; seeding a prime divisor of m into P, a blind search missing these as a search–power artefact, not an arithmetic obstruction). □

Remark 5.3 (Precise scope)**.** This excludes only a *single fixed–modulus* congruence disproof. The witnesses are m–dependent (the modulus–6 witness already fails modulo 210), and a single (P, Q) solving the system for *all* m simultaneously is, for bounded integers, equivalent to the integer identities $N_P = D_Q$, $N_Q = D_P$ (that is, to #307 itself) so per–modulus solvability is *no* evidence for existence. Nor does it exclude other finite or local negative arguments (size/Mertens bounds, density, magnitude–congruence hybrids, or a Hasse–type failure across infinitely many moduli); indeed our own $|U| \ge 59$ and 2.09×10^{56} barriers are finite–data results fully consistent with it. Together with Theorem 6.1 (every non–archimedean shadow trivially negative for degree reasons) it brackets the problem: no single–modulus obstruction over $\mathbb{Z}$, fatal non–archimedean analogues over $\mathbb{F}_q[t]$, the difficulty living at the archimedean place.

A first–order count. Model a candidate squarefree a as "P–abundant" ($\sum_{p|a} 1/p > 1$) and ask for $b = a'$ to itself be Q–abundant with $\sum_{q|b} 1/q = 1/s$. Two empirical inputs (Section 9) calibrate this:

- the density of P–abundant n is $\delta = 0.0420$ (and ≈ 0.0176 among squarefree n); a theorem, not merely an observation (Proposition 5.6);
- for a qualifying squarefree a, $b = a'$ is squarefree about 95% of the time, but its mass $\sum_{q|b} 1/q$ has mean ≈ 0.047 against the required ≈ 0.934; the empirical tail probability $\Pr[\sum_{q|b} 1/q \ge 1/s]$ is 0 in 40,920 samples at these scales (Section 9).

The gap is structural: at small scale a P–abundant a hogs the cheap mass–carrying small primes, and $b = a'$, being coprime to a, is both too small and too coprime to assemble reciprocal mass $1/s$. The extremal case is stark: for the primorial $a = \prod_{i \le k} p_i$, though $\sum_{p|a} 1/p$ rises through 2 at $k = 59$, the cofactor sum a' is essentially prime, with $\sum_{q|a'} 1/q < 0.2$ for every $k < 100$ (and only 0.0014 at $k = 59$): the family that *maximises* the forward mass is the most starved on the return. Equivalently, for squarefree n with n' squarefree, $n'' = n\,\sigma(n)\,\sigma(n')$, so the two–step return ratio $n''/n = \sigma(n)\sigma(n')$ is the #307 product itself; over *all* such $n < 10^6$ it stays strictly below 1, never exceeding 0.54 (maximal at $n = 969969 = 3 \cdot 7 \cdot 11 \cdot 13 \cdot 17 \cdot 19$), so D is two–step contracting throughout the tested range and must reverse to unit ratio for a solution to exist. The first scale at which b could plausibly carry the required mass is of the order of the barrier ($\sim 10^{56}$); this dynamical mass–gap is *consistent with* the proved barrier rather than an independent derivation of it (the empirical tail probability 0 at small scale is equally consistent with NO and with YES–beyond–the–barrier). A naive "expected number of solutions" built from the local congruences (one factor $\sim 1/q$ per condition) diverges like a constant times $\log X$; this is a divergent naive count of exactly the type known to be unreliable, so we treat it only as very weak, non–rigorous suggestive evidence that solutions could exist at or beyond the barrier.

Remark 5.4 (Calibration)**.** We present the count of the previous paragraph as a heuristic, not a theorem. Heuristics of amicable–pair type are known to be delicate; we report it as weak evidence, consistent with the divergence of the refined expected count, that the answer to #307 is plausibly YES (and hence that the period–one form of the Ufnarovski–Åhlander conjecture is plausibly false).

Remark 5.5 (A cautionary refinement)**.** One can try to sharpen this into a quantitative rate using a Kubilius/Erdős–Wintner model, stratified over which small primes lie in the support of a versus remain available to b. Two lessons emerge, both methodological. First, a plain Monte–Carlo estimate *undercounts* the rate by some 10^{61}: the expectation is dominated by rare, balanced splits of the small primes that sampling almost never sees, so a naive estimate of this kind is misleadingly small and only an explicit stratified sum corrects it. Second, and this is why we record no numerical prediction, the stratified estimate is itself unstable: refining the stratification from the first 8 to the first 14 primes drops the rate by some 17 orders of magnitude and pushes the implied first occurrence from $\sim 10^{10^{38}}$ out past $\sim 10^{10^{55}}$, with no sign of convergence. The only robust conclusion is negative: every version places a hypothetical first solution *doubly exponentially* beyond any conceivable search, consistent with the constructive deficit of Section 8; the model does not reliably decide existence in either direction.

Proposition 5.6 (The abundance density is a theorem)**.** *The additive function $f(n) = \sum_{p|n} 1/p$ satisfies the Erdős–Wintner three–series criterion [17] (here $\sum 1/p^2$ and $\sum 1/p^3$ converge), so f has a limiting distribution and the density $\delta = \lim_{x\to\infty} \frac{1}{x}\#\{n \le x : f(n) > 1\}$ exists, equal to $\Pr\big[\sum_p X_p/p > 1\big]$ for independent $X_p \sim \mathrm{Bernoulli}(1/p)$. Moreover the constant carries a* certified *enclosure:*

$$0.04202 \;\le\; \delta \;\le\; 0.04223.$$

Writing $S_1 = \sum_{p\le P_1} X_p/p$, $P_1 = 3\times 10^4$, and $T \ge 0$ for the tail, monotone upper and lower cdf envelopes of S_1 are propagated through the Bernoulli convolution on a grid of step 10^{-7} with the shift $1/p$ rounded down for the upper and up for the lower envelope (which preserves them, cdfs being nondecreasing) so $\delta \ge 1 - U(1)$ costs nothing (the tail is nonnegative), while $\delta \le 1 - L(1 - a) + \Pr[T \ge a]$ with the explicit Chernoff bound $\Pr[T \ge a] \le \exp(1.36 - P_1 a)$, $a = 10^{-3}$ (from $\log \mathbb{E}\, e^{\lambda T} \le \lambda s_2 + 0.72\lambda^2 s_3$ at $\lambda = P_1$, using $s_2 < 1/P_1$, $s_3 < 1/(2P_1^2)$ and $e^x - 1 \le x + 0.72x^2$ on $[0,1]$); float slack 10^{-9} exceeds the accumulated rounding by nine orders, and the machinery reproduces an exact four–prime reference to 2×10^{-16} (`code/delta_cert.py`). The point value 0.0420*, at the bottom of the enclosure, agrees to four places with the sieve to 2×10^7; the upper edge is conservative, carried almost entirely by the a–window. As for the density of abundant integers, no closed form is expected.*

Remark 5.7 (Two dynamical probes: no second obstruction)**.** Two measurements sharpen the picture beyond the first–order mass count, and both cut towards rather than against a solution. *(i) Image density.* A two–cycle needs *both* members in the image $D(\mathbb{N})$ (each is the other's derivative), so one might fear a second obstruction; that the high–mass numbers a cycle requires are seldom derivatives. The opposite holds: of the integers below 10^7, 68.3% lie in $D(\mathbb{N})$, and the fraction *rises* with mass, from 61.7% at $\sum_{p|n} 1/p < 0.2$ to 78.2% at $\sum_{p|n} 1/p > 1$ and 80% beyond. The high–mass numbers are thus over–represented among derivatives; the second cycle condition is comfortably met. *(ii) The barrier is where the balanced mass–product reaches* 1. Set $r(n) = \left(\sum_{p|n} 1/p\right)\left(\sum_{q|n'} 1/q\right) = D^2(n)/n$, so $r = 1$ is exactly the cycle condition. Exhaustively over $n \le X$ the maximum of r climbs $0.50 \to 0.62$ as X runs $10^3 \to 10^9$ (with no exact cycle, as the barrier demands), with no ceiling; the climb follows the balanced split $r \approx (S(k)/2)^2$, $S(k) = \sum_{i \le k} 1/p_i$, which reaches 1 precisely when $S(k) = 2$, i.e. $k \approx 59$. The dynamical trajectory thus reproduces the $|U| \ge 59$ barrier from the opposite direction. Neither probe decides existence; both *remove* hypothetical obstructions, leaving the exact archimedean closure as the sole gate.

Remark 5.8 (Why the analytic method is starved: $N(r) \le 1$)**.** The rank–one rigidity (Section 11) has a sharp consequence for the circle–method machinery that resolves the *unrestricted* Egyptian–fraction problem. For a rational r let $N(r) = \#\{\, P : \sum_{p \in P} 1/p = r \,\}$ count the finite prime sets with reciprocal sum r. Then

$$N(r) \le 1 :$$

by the Rigidity Lemma (Theorem 2.1) the sum $\sum_{p \in P} 1/p = N_P/D_P$ is already in lowest terms, so the reduced denominator of r equals $D_P = \prod P$ and P is exactly its set of prime divisors; uniquely determined ($N(r) = 1$ precisely when that denominator is squarefree and the numerator of r is its cofactor sum). The circle method needs the opposite: a target hit by a positive proportion of subsets, so a main term dominates. This is exactly what Bloom's shift to $\{p-1\}$ manufactures, $p-1$ being composite and divisor–rich, a rational is the reciprocal sum of *many* subsets of $\{1/(p-1)\}$ (already among the first fourteen shifted primes the multiplicity reaches 3 and grows), whence [6] represents every positive rational there. Over the primes themselves $N(r) \le 1$ leaves nothing to average: the analytic engine that reaches $\{p - h\}$ is provably starved on $\{p\}$, and this is the method–level reason #307 (the rank–one, exact–prime case) lies beyond it.

6 A function–field lens: the obstruction is archimedean

Define an arithmetic derivative on $\mathbb{F}_q[t]$ by $P' = 1$ for monic irreducibles and the Leibniz rule, so $f' = \sum_i e_i\, f/P_i$ for $f = \prod_i P_i^{e_i}$ (the exponents e_i read modulo the characteristic).

Theorem 6.1. *Over $\mathbb{F}_q[t]$ this derivative has no two–cycles and no nonzero fixed points. Indeed $\deg f' \le \deg f - 1$ for every nonconstant f. The analogues of the integer fixed points p^p also vanish: $(P^p)' = p\,P^{p-1}P' = 0$ in characteristic p. The same holds verbatim for every twisted variant ($P' = u_P \in \mathbb{F}_q^\times$, Leibniz).*

Proof. Each term $e_i\, f/P_i$ has degree $\deg f - \deg P_i \le \deg f - 1$, so $\deg f' \le \deg f - 1$. A two–cycle $f' = g$, $g' = f$ would force $\deg g \le \deg f - 1$ and $\deg f \le \deg g - 1$, hence $\deg f \le \deg f - 2$; a fixed point $f' = f$ is excluded likewise. (Verified exhaustively over $\mathbb{F}_2$ to degree 7, $\mathbb{F}_3$ to degree 5, and $\mathbb{F}_5$ to degree 4; twisted variants over $\mathbb{F}_2$ and $\mathbb{F}_5$ on small supports through 2.3×10^6 twist configurations, none closing.) □

Formalised. `Erdos307.ff_no_cycle` is the theorem, on the single hypothesis $\deg f' + 1 \le \deg f$; `deg_drop_of_terms` is that hypothesis from the term bound, and `no_fixed_point_of_drop` excludes fixed points. The argument is stated on the degrees alone, so it holds verbatim for every twisted variant, the twist multiplying each term by a unit (`lean/Erdos307/FunctionField.lean`).

Corollary 6.2 (The function–field Erdős–Barbeau equation is insoluble)**.** *For every finite field* $\mathbb{F}_q$*, all nonempty finite sets* P, Q *of monic irreducibles, and all unit weights* $u_\pi, v_\rho \in \mathbb{F}_q^\times$*,*

$$\Big(\sum_{\pi\in P} \frac{u_\pi}{\pi}\Big)\Big(\sum_{\rho\in Q} \frac{v_\rho}{\rho}\Big) \neq 1.$$

Proof. Write $f = \prod P$, $g = \prod Q$, $D_u(f) = \sum_\pi u_\pi f/\pi$; the equation reads $D_u(f)\, D_v(g) = fg$. If some $\pi \in P \cap Q$, then $\pi^2 \mid fg$ while $D_u(f) \equiv u_\pi f/\pi \not\equiv 0$ and $D_v(g) \equiv v_\pi g/\pi \not\equiv 0 \pmod{\pi}$, so $\pi \nmid D_u(f)D_v(g)$; impossible; hence $\gcd(f, g) = 1$. The same congruence gives $\gcd(f, D_u(f)) = \gcd(g, D_v(g)) = 1$, so $g \mid D_u(f)$ and $f \mid D_v(g)$, forcing $D_u(f) = c\,g$, $D_v(g) = c^{-1}f$ with $c \in \mathbb{F}_q^\times$: a twisted two–cycle, excluded by the degree drop. The weighted product equation is thus insoluble over every $\mathbb{F}_q(t)$; it *is* soluble over the unit–rich number rings of Theorem 6.11, since any twisted cycle gives $(D_u(a)/a)(D_v(b)/b) = (b/a)(a/b) = 1$, and over $\mathbb{Q}$, where the admissible weights are ± 1, it is #307 itself with the signed barrier of Remark 6.4, open beyond 2.09×10^{56}. The three regimes (non–archimedean monotonicity (closed, negative), unit–broken monotonicity (closed, positive), ordered–archimedean (open, barred)) now have theorems on two of three doors. □

Over $\mathbb{Z}$, by contrast, $n' = n \sum_i e_i/p_i$ can exceed n precisely because the archimedean absolute value lets a sum of strictly smaller terms outgrow the original; the "mass > 1" phenomenon $\sum 1/p > 1$, which has no degree–theoretic analogue. Theorem 6.1 thus localises the entire existence question to the archimedean place: every non–archimedean shadow of #307 is trivially negative. This is consistent with (and partly explains) both the census finding of Section 9 and Proposition 5.2 (no single modulus obstructs the cross–match system in the tested range); the difficulty is not local. Any eventual proof must therefore distinguish $\mathbb{Z}$ from $\mathbb{F}_q[t]$; over the latter the decisive tools are the degree and the Mason–Stothers/Wronskian inequality [18], neither of which has a $\mathbb{Z}$–analogue. The conjectural analogue, *abc*, does not fill the gap, and quantifiably so. Every cycle inherits, from $N' = a^2 + b^2$ with $N = ab$, the relation $(a-b)^2 + 4ab = (a+b)^2$ on coprime squarefree members, but this triple is *abc*–unexceptional, of quality $q = \log(a+b)^2/\log \operatorname{rad}\big((a-b)\cdot ab \cdot (a+b)\big) \approx \frac{1}{2}$ (median 0.543, never above 0.74 across 6×10^4 coprime squarefree pairs; the only $q > 1$ cases are degenerate smooth–gap pairs such as $(5, 3)$, vanishingly far below the 10^{56} scale a solution must occupy). Granting *abc* in its strong explicit form would therefore yield no bound here: the function–field collapse rests on an inequality genuinely *absent* over $\mathbb{Z}$, not merely unproven, and the size mismatch is by a factor of two in the exponent. #307 lives in the *abc*–comfortable zone.

Remark 6.3 (The free grading, and what evaluation destroys)**.** Theorem 6.1 has a structural reading. In the free Leibniz world (the polynomial ring $\mathbb{Z}[x_p]$ with $x_p' = 1$) the derivative of a squarefree monomial of degree n is the elementary symmetric form e_{n-1}, homogeneous of degree $n-1$: the formal derivative strictly lowers the grading, cycles are impossible, and over $\mathbb{F}_q[t]$ the degree realises this grading faithfully. Over $\mathbb{Z}$ the evaluation $x_p \mapsto p$ destroys homogeneity, and the data show it destroys it completely: for random squarefree a of fixed size, $\omega(a')$ is governed by the size of a' alone (Erdős–Kac), flat in $\omega(a)$ within each parity class (measured: mean $\omega(a') =$ 2.7–3.0 as $\omega(a)$ runs over $3, 5, 7$, and 3.6–4.0 over $4, 6, 8$, the offset being the parity anatomy of Theorem 4.26). A two–cycle is therefore a doubly anti–typical event: an additively assembled integer must carry multiplicative rank far in the Erdős–Kac tail *and* on an exactly prescribed support; the two independent prices whose product is the heuristic rarity of Section 5. The deeper additive–multiplicative content collapses, on inspection, into the S–unit form of Proposition 6.7: dividing the cycle equation by D_Q exhibits it as an n–term unit equation; one more coordinate system in which the same exactness reappears unchanged.

Remark 6.4 (The obstruction is *ordered*–archimedean)**.** The barrier is finer than "archimedean": its engine is AM–GM on real, *positive* masses, i.e. the ordering of $\mathbb{R}$. Over $\mathbb{Z}[i]$ (a PID, so the Rigidity Lemma and cross–matching survive verbatim, with cycles defined up to units), the masses $\sum 1/\pi$ are *complex*: phases can align so that $\left|\sum 1/\pi\right| > 1$ with as few as three Gaussian primes (e.g. $\frac{1}{1+i} + \frac{1}{2+i} + \frac{1}{2-i} = 1.3 - 0.5i$), and the 59–prime bound has no analogue. The ordered–archimedean thesis thus predicts that small derivative two–cycles *may* exist over $\mathbb{Z}[i]$ where they are barred over $\mathbb{Z}$, and the prediction is confirmed. Two facts make the contrast precise. *First*, phases do not help over $\mathbb{Z}$: for any *signed* derivative ($p' = \pm 1$, Leibniz) a two–cycle still satisfies $|s||t| = 1$ with $|s| \le \sigma(a)$, $|t| \le \sigma(b)$ and disjoint supports (the rigidity proof is unchanged), so $\sigma(a) + \sigma(b) \ge 2$ and the full 59–prime, 2.09×10^{56} barrier applies to every choice of signs: real phases cannot cancel. *Second*, fourth roots of unity suffice instantly: Leibniz derivatives on $\mathbb{Z}[i]$ with unit prime–values ($\pi' \in \{\pm 1, \pm i\}$) possess genuine two–cycles at norm $\sim 10^4$. The smallest:

$$a = (1+i)(2+7i)(6+11i) = -129 - i, \qquad b = 3\,(2+i)(1+2i)(6+5i) = -75 + 90i,$$

with $a' = b$ under the *standard* values $\pi' = 1$ on the primes of a, and $b' = a$ exactly under $\pi' = i$ on 3, $2+i$, $1+2i$ and $\pi' = 1$ on $6+5i$; seven prime classes ($N(a) = 16{,}642$, $N(b) = 13{,}725$) against the fifty–nine that $\mathbb{Z}$ demands. Exhaustive search finds exactly sixteen such cycles up to conjugation and direction having a member of norm $\le 2 \times 10^7$ with $\omega \le 5$ on the enumerated side (ω–profiles $(3,4)$, $(4,4)$, $(5,4)$, and even $(4,6)$) and the population grows steadily (4 below 3×10^5, 16 below 2×10^7), consistent with an infinite, slowly growing family; in six of the sixteen one side requires no twist at all. All sixteen were verified exactly, by independent recomputation; no twisted *fixed* point ($a' = ua$) exists in the same range. The strict two–sided normalisation $\pi' = 1$ (first–quadrant representatives) has no cycle up to units (nor any fixed point, zero–derivative, or unit–derivative element) with $N(a) \le 2 \times 10^9$ (9.4×10^8 candidates, partners by full factorisation); this is the expected order rather than a separate rigidity, a single fixed phase system being heuristically $4^{\omega-1}$–fold rarer than the union of all systems. The contrast with $\mathbb{Z}$ survives this accounting: over $\mathbb{Z}$ *every* one of the $2^{\omega-1}$ sign systems is barred by the theorem above, while over $\mathbb{Z}[i]$ the union of phase systems is demonstrably populated from norm 1.7×10^4 on. The barrier of #307 is thus localised precisely: it is the impossibility of phase cancellation in the ordered field $\mathbb{R}$, where the only available phases ± 1 provably gain nothing; while the moment fourth roots of unity are admitted, the analogous existence problem closes at seven primes. The cycles found come in conjugate pairs (supports not conjugation–stable); a conjugation–*stable* Gaussian cycle would descend to rational data through the norm form $a^2 + b^2$, i.e. precisely the Pythagorean layer of Proposition 12.2; now a concrete target rather than a speculation.

Remark 6.5 (The descended operator; the symmetric sector)**.** Conjugation–stable supports admit an exact descent to $\mathbb{Z}$. Such a squarefree Gaussian integer is an associate of an odd squarefree rational m (split primes entering as whole conjugate pairs; the $(1+i)$–sector is empty by a short parity argument), and a conjugation–equivariant unit assignment makes the twisted derivative rational:

$$D(m) \;=\; \sum_{\substack{p \mid m \\ p \equiv 1\,(4)}} t_p \,\frac{m}{p} \;+\; \sum_{\substack{q \mid m \\ q \equiv 3\,(4)}} \varepsilon_q \,\frac{m}{q}, \qquad t_p \in \{\pm 2x_p, \pm 2y_p\}, \quad \varepsilon_q = \pm 1,$$

where $p = x_p^2 + y_p^2$; a Leibniz derivative on odd squarefree integers whose prime values have size $2\sqrt{p}$ rather than 1, and whose masses $\sim 2/\sqrt{p}$ reach 2 with four primes, so no 59–prime barrier exists for it. A two–cycle of D is exactly a conjugation–stable Gaussian cycle and would plant the phenomenon on rational soil. Parity constrains the hunt: $D(m)$ is odd iff m carries an odd number of primes $\equiv 3 \pmod 4$, so every member must contain such a prime. One level deeper, since $q^{-1} \equiv q \pmod 8$ for odd q, the descended operator obeys $D(m) \equiv m\,W(m) \pmod 8$ with $W(m) = \sum_{p|m} t_p\, p$, so a two–cycle forces $W(m) \equiv W(m') \equiv mm' \pmod 8$; a mod–8 constraint on the sign assignment that prunes the $4^{\#\text{split}}2^{\#\text{inert}}$ phase enumeration severalfold in future searches (verified on 2×10^5 random assignments; `code/peeling_groupoid.py`). An exhaustive search over members $m \le 10^9$ (Gaussian norm 10^{18}, $\omega \le 8$, 1.4×10^9 closure tests) found no cycle, no fixed point $|D(m)| = m$, and no vanishing $D(m) = 0$. The search has since been extended two decades further in m, on a box narrower in the other parameters: $\omega \in [4, 6]$ and odd primes below 600, with m running over $[10^9, 10^{10}]$ and then $[10^{10}, 10^{11}]$. Both tiers are exhaustive over their 5,778 prime pairs and return nothing: 10,004,024 members and 6.94×10^8 closure tests on the first, 22,218,430 members and 1.82×10^9 closure tests on the second, so 3.22×10^7 members and 2.52×10^9 closure tests in all, with no two–cycle at any point (`hunt/descended_hunt2.rs`, `hunt/run_tier2_par.sh`). The tier boxes are not nested in the one above, being narrower in ω and in the prime range while wider in m, so the two counts complement rather than supersede one another; and the tiers test closure only, not the fixed–point and vanishing conditions recorded above. We claim no obstruction from this: equivariance prunes the available phases fourfold per split prime, and a phase count makes emptiness at this depth the expected order. The descent target remains open over $\mathbb{Z}[i]$, but not in the real–quadratic world, where the corresponding sector contains a fully rational pair (Remark 6.6). Finally, the descended coefficients are classical objects: at a split prime p the four values $\{\pm 2x_p, \pm 2y_p\}$ are exactly the Frobenius traces of the congruent–number curve $y^2 = x^3 - x$ (CM by $\mathbb{Z}[i]$) and of its twists ($a_p \in \{\pm 2x_p, \pm 2y_p\}$ verified by point counting for all $p < 120$) so the conjugation–stable sector is the arithmetic derivative twisted by the Hecke eigensystem of $\mathbb{Q}(i)$, and its mass statistics are governed unconditionally by Hecke's equidistribution theorem: the first point of contact between #307 and automorphic forms. Concretely: over random squarefree m the three–series conditions converge (the variance term is $\sum 2/p^2$), so the descended mass $D(m)/m$ has an Erdős–Wintner *limiting distribution* whose prime components are arcsine laws scaled by $1/\sqrt{p}$; the required input, equidistribution of the Hecke angles, is visible exactly (Kolmogorov–Smirnov distance 0.0041 from uniform over the 8,977 split primes below 2×10^5, well inside the 95% band 0.0144).

Remark 6.6 (The wall is the orderability of $\mathbb{Q}$)**.** The cycles over $\mathbb{Z}[i]$ permit a dissection of the barrier that no amount of theory over $\mathbb{Z}$ could provide. The entire structural chain of this note (rigidity, cross–matching, the Pythagorean identity, the split discriminant) transfers verbatim to twisted derivatives over $\mathbb{Z}[i]$: Leibniz holds for disjoint supports, so a cycle (a, b) has $D(N) = a^2 + b^2$ for $N = ab$, with $D(N) - 2N = (a - b)^2$ and $2N - D(N) = \big(i(a - b)\big)^2$ (all verified exactly on the cycles found). The last equation exposes the wall: over $\mathbb{Z}[i]$, where $-1 = i^2$ is a square, *both* discriminant factors are squares automatically, and the plus/minus distinction of Proposition 13.13 of the full version (which over $\mathbb{Z}$ carries the entire obstruction, via $(a - b)^2 \ge 0 \Rightarrow \sigma(N) \ge 2$) has no content. The unique non–transferring link is the sign–definiteness of squares: the barrier is, in the end, the *formal reality* of $\mathbb{Q}$ in the sense of Artin–Schreier. (The two Artin–Schreier theorems thus bracket the problem: their characteristic–2 map builds the tower of Proposition 13.8 of the full version, their theory of orderable fields names the wall.) This classifies where the mechanism can operate at all: it needs an ordering making squares nonnegative (formal reality, hence a real embedding) *and* bounded twisting phases (finite unit group); by Dirichlet's unit theorem, unit rank $r_1 + r_2 - 1 = 0$ with $r_1 \ge 1$ forces degree one. $\mathbb{Q}$ *is the only number field over which the mass barrier can operate*: over imaginary quadratic fields squares lose sign–definiteness (over $\mathbb{Z}[i]$ even -1 is a square), and over every other field the unit rank is positive and twisted masses are unbounded. This is a classification of the wall's habitat, not an existence claim elsewhere, but over $\mathbb{Z}[i]$ existence is no longer a claim: it is a list of sixteen. The ladder of imaginary quadratic rings confirms the mechanism quantitatively. Over $\mathbb{Z}[\sqrt{-2}]$, whose only units are ± 1, precisely the twisting freedom that provably gains nothing over $\mathbb{Z}$; sign–twisted cycles nevertheless exist from norm 11,462 on ($a = -90 - 41\sqrt{-2}$, $b = 75 - 45\sqrt{-2}$, seven prime classes): the complex embedding alone carries the phenomenon. Over the Eisenstein ring $\mathbb{Z}[\omega]$, with six units, cycles appear at norm 2,964 with *six* prime classes ($a = -40 + 22\omega$, $b = 56 - 21\omega$); the smallest derivative two–cycle exhibited in any ring, and the population at fixed height grows with the unit group: at norm $\le 2 \times 10^7$ the censuses read 14, 56, and 278 twisted–cycle instances for unit groups of order 2, 4, 6 (a factor $\sim$4–5 per two extra units, as a phase count predicts) while the strict canonical systems of all three rings remain empty to norm 2×10^9 (3×10^9 candidates in total, partners by full factorisation). Two further objects appeared over $\mathbb{Z}[\sqrt{-2}]$: a composite with *unit* derivative ($a' \sim 1$; impossible over $\mathbb{Z}$, where $n' = 1$ characterises the primes), and an *anti–holomorphic* cycle: $a = -3163 - 1474\sqrt{-2}$ satisfies $a' = -\bar{a}$ exactly, under a single conjugation–equivariant sign assignment closing both directions ($N(a) = N(b) = 14{,}349{,}921$, five prime classes each side). The equation $a' = u\bar{a}$ is vacuous over $\mathbb{Z}$ only in its *one–dimensional* reading, where it says $n' = \pm n$. In the reading that matches the problem it is not vacuous but central: by Proposition 13.2 of the full version the Pythagorean layer over $\mathbb{Z}$ *is* the anti–holomorphic equation $\mathcal{D}z = \mu\bar{z}$ for the coordinatewise derivative on $z = a + bi$, and #307 is exactly its unit–multiplier case. The ring ladder and the main problem are therefore instances of one equation rather than analogues; what singles out $\mathbb{Z}$ is the extra constraint $\operatorname{Im}\mu = 1$, which confines the multiplier to a line meeting $\mathbb{Z}[i]^\times$ in the single point $\mu = i$. That the anti–holomorphic sector is thin is uniform across the ladder: over $\mathbb{Z}[\sqrt{-2}]$ the cycle above is still the *only* one after widening the census simultaneously in all three directions; prime–norm $\le 2 \times 10^4$, $\omega \le 8$, product–norm $\le 10^{10}$ (a factor 700 past the cycle itself), 8.09×10^8 supports, `hunt/antiholo_hunt.rs`, and over $\mathbb{Z}$ the sector is empty below 10^{112} by Corollary 12.3. We record the plateau as consistent with genuine finiteness, not as evidence for it: a random model for $a' = -\bar{a}$ predicts a count growing like $\log X$, which the tested range cannot separate from a constant. All cycles were verified exactly by independent recomputation. Finally, the second pillar was tested in isolation: over $\mathbb{Z}[\sqrt{2}]$ (*totally real*, so squares are sign–definite in both embeddings, but with infinite unit group $\pm(1 + \sqrt{2})^{\mathbb{Z}}$) twisted cycles appear already at norm 882: with unit twists of height ≤ 2, $a = 21\sqrt{2}$ (classes $\sqrt{2}$, $-1+2\sqrt{2}$, $1+2\sqrt{2}$, 3) cycles exactly with $b = 51 + 13\sqrt{2}$, of norm 2263 and only *two* prime classes (against the fifty–nine that $\mathbb{Z}$ demands) and the composite $369 - 15\sqrt{2}$ has derivative exactly $\varepsilon^6 = 99 + 70\sqrt{2}$, a unit. Both pillars of the classification are thus load–bearing, and each is now confirmed by explicit counterexample on the soil where it alone fails: drop formal reality and cycles appear (imaginary quadratic rings, norms 10^3–10^4); keep formal reality but drop unit finiteness and cycles appear (real quadratic, norm 882, the smallest in any ring); over $\mathbb{Q}$, where both hold, the barrier stands at 10^{112} by theorem. The bracketing of $\mathbb{Q}$ is complete from both axes. The deep census sharpens it: at norm $\le 2 \times 10^7$ the real–quadratic ring carries 30,204 twisted–cycle instances against 14, 56, 278 for the imaginary rings with $2, 4, 6$ units infinite unit rank dominates by two orders of magnitude, and contains both the *minimal* architecture, $a = \sqrt{2}\,(5 - \sqrt{2})$ cycling with $b = 7$ at norms (46, 49), two prime classes per member (the rational prime 7 being composite in the ring), the smallest derivative cycle found in any ring; and a *fully rational* pair:

$$D(3707) = 1547, \qquad D(1547) = 3707 \qquad (3707 = 11 \cdot 337, \quad 1547 = 7 \cdot 13 \cdot 17),$$

exactly, with unit heights ≤ 2 on both sides, verified independently: two ordinary integers in a derivative two–cycle. The descent phenomenon that is empty over $\mathbb{Z}[i]$ to norm 10^{18} is realised over $\mathbb{Z}[\sqrt{2}]$ at four digits. The canonical system of $\mathbb{Z}[\sqrt{2}]$ remains cycle–free to norm 2×10^9 (7.7×10^8 candidates), now with seven unit–derivative composites.

The cycle phenomenon over rings with units is governed by the theory of unit equations, which supplies its rigidity theorem.

Proposition 6.7 (*S*–unit rigidity of twisted cycles)**.** *Let $\mathcal{O}$ be the ring of integers of a number field and S a finite set of primes of $\mathcal{O}$. If (a,b) is a twisted two–cycle with* $\operatorname{supp}(ab) \subseteq S$*, then dividing $D(a) = b$ by b exhibits a solution of*

$$x_1 + \cdots + x_{\omega(a)} = 1, \qquad x_i = u_i\,\frac{a/\pi_i}{b},$$

in the S–unit group of the field, and likewise for b. By the theorem of Evertse–Schlickewei–Schmidt [39] (cf. [32]), such an equation has only finitely many nondegenerate solutions; hence every fixed support carries at most finitely many twisted cycles free of vanishing subsums*, and the species can be infinite only through growing supports; precisely the behaviour of every census above. (Vanishing subsums correspond to partial zero–derivative patterns; none occurred in any search. The relation was verified exactly on the $\mathbb{Z}[\sqrt{2}]$ specimen: the four S–unit terms of $a = 21\sqrt{2}$ sum to 1.)*

Remark 6.8 (Galois folding, and the thinness of symmetric sectors)**.** Over a quadratic field with automorphism σ, a σ–equivariant choice of unit values ($u_{\pi^\sigma} = u_\pi^\sigma$) makes D commute with conjugation, so the *single* equation $D(a) = a^\sigma$ already implies that (a, a^σ) is a twisted two–cycle (checked symbolically and on 1,988 random instances). In the two–prime case the partner is forced: $\rho = -\big(\bar{u}v\,\pi + \bar{v}\,(\pi^\sigma)^2\big)/\big(N(u) - N(\pi)\big)$, with integrality a congruence and the entire residue of the problem the primality of $|N(\rho)|$ along Pell–exponentially growing sequences; Lehmer territory rather than Bunyakovsky. Whether a folded sector is starved or rich is decided by its degrees of freedom. When the fold leaves no slack the sector is thin: the folded *two*–prime family over $\mathbb{Z}[\sqrt{2}]$ is empty in the tested box (36,000 parameter triples, none even integral), the conjugation–stable sector over $\mathbb{Z}[i]$ is empty to Gaussian norm 10^{18}, and the anti–holomorphic cycle over $\mathbb{Z}[\sqrt{-2}]$ is unique in its census. But with *three* primes the folded equation $u_1\pi_2\pi_3 + u_2\pi_1\pi_3 + u_3\pi_1\pi_2 = (\pi_1\pi_2\pi_3)^\sigma$ carries three unit unknowns against two real equations, and this underdetermined system is *populated*: an exhaustive box ($p_i \le 137$, unit heights ≤ 4) contains twenty–four solutions over eleven prime triples, the smallest being

$$a = (3+\sqrt{2})(5-2\sqrt{2})(5-\sqrt{2}) = 57 - 16\sqrt{2}, \qquad D(a) = a^\sigma = 57 + 16\sqrt{2},$$

with unit values ε^2, ε, $-\varepsilon^{-1}$: *the derivative conjugates the element*, at norm 2737 on both sides, the equivariant return being automatic (verified exactly). Existence of twisted cycles over the four rings tested is a theorem by exhibition; for *every* ring with infinite unit group it is now reduced to the abundance of lattice points on these underdetermined unit systems (measured plentiful already at $k = 3$) a geometry–of–numbers question rather than a primality one.

Remark 6.9 (The forced third prime: a Schinzel–type reduction over $\mathbb{Z}[\sqrt{2}]$, unblocked)**.** Within that abundance the arithmetic is sharper than free lattice–counting. Fixing π_1, π_2 and the units u_1, u_2, u_3, the folded equation rearranges to $\pi_3\,(u_1\pi_2 + u_2\pi_1) - (\pi_1\pi_2)^\sigma\pi_3^\sigma = -u_3\pi_1\pi_2$, a 2×2 integer linear system for $\pi_3 = (x, y)$ whose determinant is $N(u_1\pi_2 + u_2\pi_1) - N(\pi_1\pi_2)$ (an identity, verified on all 168 solutions of the box). So π_3 is *forced* by $(\pi_1, \pi_2, u_1, u_2, u_3)$, and infinitude of twisted cycles is exactly the primality of this forced π_3 for infinitely many such data a Schinzel/Bunyakovsky–type condition over $\mathbb{Z}[\sqrt{2}]$. Crucially it is *not* the regime of Proposition 8.4: the exponential unit freedom $u_i = \pm\varepsilon^{k_i}$ keeps the family non–polynomial yet populated, so the polynomial–rigidity that forbids reducing the *rational* #307 to such a hypothesis does not apply here. A Schinzel–type theorem over $\mathbb{Z}[\sqrt{2}]$ would render twisted infinitude unconditional, whereas none touches

#307 itself; the twisted problem sits strictly closer to the reach of current sieve theory than its rational shadow. Concretely the twisted solution set is *abundant* at accessible scale; thousands of cycles at modest norm (Proposition 6.12), and the count of distinct prime–norm signatures grows roughly in proportion to the norm bound in our census (e.g. $6, 12, 20$ at bounds $40, 80, 160$, unit heights saturating); whereas the rational Pythagorean layer is *empty* below 10^{112} (Corollary 12.3). The twisted and rational problems thus differ not in the difficulty of the search but in whether solutions exist to be found at all: for the twist they are plentiful and the only question is their primality, exactly the Schinzel–shaped residue above. That residue is a *pair*, not a triple or a Lehmer sequence: fixing π_1 and bounding the units, the cycle count still grows with $N(\pi_2)$ (verified: $33, 48, 60$ for $\pi_1 = 3+\sqrt{2}$ at $N(\pi_2) \leq 60, 120, 240$), so π_1's primality is free and only the *simultaneous* primality of π_2 and the forced π_3 remains. Setting up the sieve pins the difficulty precisely. Writing $C = u_1\pi_2 + u_2\pi_1$, $E' = (\pi_1\pi_2)^\sigma$, $F = -u_3\pi_1\pi_2$, the forced third prime is $\pi_3 = (\bar{C}F + E'\bar{F})/\det$ with $\det = N(C) - N(E')$ a binary quadratic form in $(c, d) = \pi_2$. The first condition is the clean form $N(\pi_2) = c^2 - 2d^2$; the second, however, is *not* a second form but the rational condition $N(\pi_3) = N(G)/\det^2$ ($G = \bar{C}F + E'\bar{F}$ quadratic, $N(G)$ quartic, $\gcd(N(G), \det) = 1$), evaluated on the integral locus $\det \mid G$, and $N(\pi_3)$ is unbounded, spiking as $\det \to 0$ (values to $\sim 10^{11}$ already at $N(\pi_2) \leq 2 \times 10^4$). On this *frozen* ($\pi_1$, units) slice the question is primality along the folded conic bundle, and the slice is thin: its unimodular fibres $\det = \pm 1$ are *empty* by a genus obstruction (there $\det = \pm 1$ would need $x^2 - 2y^2 \in \{-55, -43\}$, both non–residues). But the thinness is an artefact of freezing exactly the parameters that carry the abundance. The *full* twisted variety $D(a) = a^\sigma$ (π_1, π_2, π_3 and their units all free) is a cubic threefold whose solution count grows with positive exponent; a positive–density attack must run there, not on a fixed fibre, so the fixed–fibre conic sieve is a closed dead end. The honest placement is then: the reachable object is the full threefold at the Heath–Brown $x^3 + 2y^3$ frontier, compounded by a *coupled* second primality ($N(\pi_2)$ and $N(\pi_3)$ simultaneously prime); the two–sparse–conditions/parity barrier where the circle method loses minor–arc cancellation. This is strictly below the exponential/Lehmer tier of the immune families (Remark 4.21, where no sieve reaches at all), but a generation beyond present technology; it is where genuinely new analytic input for the twist would first have to bite.

The natural attempt to *prove* that infinitude (forcing one prime to be the conjugate of another so that a single free prime remains, whence Dirichlet would suffice) is blocked, and the block is a clean structural law.

Proposition 6.10 (No conjugate prime pair in a twisted cycle)**.** *Let D be a Leibniz derivative with unit prime–values over a quadratic ring with automorphism σ, and a a squarefree twisted cycle $D(a) = a^\sigma$. Then no two prime factors of a are σ–conjugate. Equivalently a twisted cycle's support is disjoint from its conjugate; it is never conjugation–stable, which is exactly why the exhibited $\mathbb{Z}[i]$ cycles are asymmetric, coming in conjugate pairs of* distinct *supports rather than on one symmetric support. This* separates *two problems easily conflated: the twisted equality $D(a) = a^\sigma$ treated here, and the symmetric target of Remark 6.5, which is a genuine two–cycle $D(m) = m'$ ($m \neq m'$, both conjugation–fixed) of the* descended *rational operator; not a twisted equality. The present proposition bars the first from the symmetric locus; it does not bear on the second, which remains genuinely open (the exhaustive emptiness there, Remark 6.5, is unexplained by it).*

Proof. Suppose a split prime π and its distinct conjugate π^σ both divide a. Reducing $D(a) = \sum_i u_i\,(a/\pi_i)$ modulo π kills every term but the one at $\pi_i = \pi$, so $D(a) \equiv u_\pi\,(a/\pi) \not\equiv 0 \pmod{\pi}$ (u_π a unit and $\pi \nmid a/\pi$ by squarefreeness). Yet $a^\sigma = \prod_i \pi_i^\sigma$ carries the factor $(\pi^\sigma)^\sigma = \pi$, so $a^\sigma \equiv 0 \pmod{\pi}$; hence $D(a) \neq a^\sigma$. An exhaustive $\mathbb{Z}[\sqrt{2}]$ search confirms no cycle carries a conjugate pair. ∎

We codify what is theorem and what remains open in this circle.

Theorem 6.11 (Existence of twisted derivative two–cycles)**.** *Leibniz derivatives with unit prime–values admit genuine two–cycles over $\mathbb{Z}[i]$, $\mathbb{Z}[\sqrt{-2}]$, $\mathbb{Z}[\omega]$, and $\mathbb{Z}[\sqrt{2}]$. The proof is by exhibited certificate, each verified exactly by independent recomputation: the smallest instances are, in order of norm, $\big(\sqrt{2}(5-\sqrt{2}), 7\big)$ over $\mathbb{Z}[\sqrt{2}]$ at norms $(46, 49)$; the Eisenstein cycle at norm 2,964; the sign–twisted cycle over $\mathbb{Z}[\sqrt{-2}]$ at norm 11,462, and the cycle over $\mathbb{Z}[i]$ at norm 16,642; together with the rational pair $D(3707) = 1547$, $D(1547) = 3707$ over $\mathbb{Z}[\sqrt{2}]$.*

Proposition 6.12 (Geometry of the solution set)**.** *Over $\mathbb{Z}[\sqrt{2}]$, consider the Galois–folded system $u_1\pi_2\pi_3+u_2\pi_1\pi_3+u_3\pi_1\pi_2 = (\pi_1\pi_2\pi_3)^\sigma$ of Remark 6.8.* (i) *For* every *triple of split primes the system is solvable with the u_i on the real unit hyperbola $\{|st| = 1\}$: solving the second embedding for u_2 and letting the third coordinate tend to zero, the first–embedding residual sweeps $\pm\infty$ while the remaining terms stay bounded, so a root exists by continuity; the real solution set is a nonempty curve (verified numerically on 200 random triples, all solvable). The volume of the solution variety never vanishes.* (ii) *By Proposition 6.7 each fixed support carries finitely many integral solutions, so infinitude can only come from growing supports.* (iii) *The measured integral hit rates are 11 of 455 prime triples in the smallest box, and 30,204 cycles of norm $\le 2\times 10^7$ at twist height ≤ 2.* (iv) *What remains open is precisely whether the discrete unit lattice meets the real curve in integer points for infinitely many supports, inhomogeneous Diophantine approximation on exponentially parametrised curves. Two shortcuts are closed: bounded–unit polynomial families are obstructed (the rational component of the return forces unit coefficients to track growing prime coordinates, which bounded units cannot), and even the cleanest forward family ($D\big(\sqrt{2}(x-\sqrt{2})\big) = x+2$* identically in *$x$, supplying a candidate cycle whenever x^2-2 and $x+2$ are prime) has sporadic returns: the closing two–term unit equation, effectively finite for each x, is solvable at $x = 5$ and provably insolvable at $x = 15$ and $x = 21$. Those failures are, however, exactly the rigid case:* a prime *return $x\pm 2$ leaves the closing system no slack. The identity has four unit branches $D\big(\sqrt{2}(x\pm\sqrt{2})\big) = x\pm 2$ (independent signs, twists 1 and $\pm\varepsilon^{\pm 1}$), and when $x\pm 2$ is composite squarefree the closing system gains one unit unknown per prime class against the same two equations, and does close: $x = 107, 121, 387, 1693$ yield cycles with rational partners $b = 105, 119, 385, 1695$, each verified exactly. Infinitude along this family is thereby reduced to Bunyakovsky for x^2-2 together with recurring composite–return closure: the softest form the gap takes anywhere in this paper. Even the primality half of that reduction can be removed. By the half–dimensional sieve [14, 15] (applied to $(2y+1)^2-2 = 4y^2+4y-1$ to keep x odd), x^2-2 is a P_2 infinitely often; every prime factor of x^2-2 has 2 as a quadratic residue and therefore splits in $\mathbb{Z}[\sqrt{2}]$, so $\sqrt{2}(x-\sqrt{2})$ is infinitely often squarefree and all–split with $\omega \in \{2, 3\}$. The candidate supports thus exist* unconditionally*; all that remains open on this route is whether the unit gates open along that supply infinitely often. Exactness migrates; it does not disappear.*

Proposition 6.13 (No vanishing derivatives, no transport, and the price of exactness)**.** (i) *No squarefree z with $\omega(z) \ge 1$ has $D(z) = 0$, for any twisted derivative over any of the rings considered (indeed over any UFD): modulo a prime $\pi \mid z$ every cofactor term but one vanishes, leaving $D(z) \equiv u_\pi\, z/\pi \not\equiv 0 \pmod{\pi}$. The zero–derivative counters of every census above are therefore theorems, not observations.* (ii) *Consequently cycles cannot be transported multiplicatively: for squarefree c coprime to ab, Leibniz gives $D(ca) = D(c)\,a + c\,D(a)$, so (ca, cb) is a cycle exactly when $D(c) = 0$; impossible. Cycles do not breed; each one requires fresh exactness.* (iii) *The two cycle equations can be bought by construction, at a conserved price. Buying* neither *leaves the generic stratum: two exact unit conditions (30,204 specimens). Buying* one *is free in two dual ways; forward, $\pi := q-\varepsilon^m\sqrt{2}$ makes $D(\sqrt{2}\,\pi) = q$ an identity (the four–branch family above is the case $m = \pm 1$); return, $\pi := (\rho+\varepsilon^{-1}\rho^\sigma)/\sqrt{2} = (c+2d)-d\sqrt{2}$ for $\rho = c+d\sqrt{2}$ closes $D(b) = \sqrt{2}\,\pi$ identically* *for every associate b of $\rho\rho^\sigma$, and the residue is one primality, $|N(\pi)|$, plus one exact unit gate, measured at* 2 *of* 50 *pairs with both primalities in the smallest box. The two hits re–derive, with no search, precisely the two census minimal–shape cycles with prime rational partner: norms* $(46, 49)$ *and* $(206, 49)$*. Buying* both *collapses the parameters to unit–indexed quadratics in q with a square–discriminant condition; the resulting $q = 7$ line carries the known cycles at $m = 1$ and $m = -2$ (discriminants* 25 *and* 81*) together with one Galois image, and every other prime–norm rung out to* $|m| \le 14$ *fails its return, verified exactly. Each purchased identity is repaid as a sparser arithmetic condition elsewhere: the hardness of exactness is conserved, never destroyed.*

Remark 6.14 (The two faces of the gap: unit rank dichotomises the obstruction)**.** The invariant that governs abundance in Remark 6.6 (the unit rank $r = r_1 + r_2 - 1$ of Dirichlet) also governs the *type* of the residual open problem, and the two types are mutually exclusive. When $r = 0$ (the rings $\mathbb{Z}$, $\mathbb{Z}[i]$, $\mathbb{Z}[\sqrt{-2}]$, $\mathbb{Z}[\omega]$) the unit group is finite, so a given pair of supports admits only $|\mathcal{O}^\times|$ twist assignments and hence finitely many candidate cycles: the solution locus is a set of *isolated points*, zero–dimensional, and infinitude can arise only from infinitely many *distinct* supports independently passing a closing test; a Bateman–Horn/Schinzel–type recurrence over prime tuples, with no continuous family to deform along. The constituent primalities are individually classical (each support prime is a value of a binary norm–form, so by Fermat–Dirichlet (and, for $\mathbb{Z}[i]$, by the Green–Sawhney theorem with *both* coordinates prime) there is no shortage of supports); it is their *simultaneous* closure that is not classical. This was tested directly: the minimal shape $a = (1 + i)\pi$ over $\mathbb{Z}[i]$ yields *no* cycle for any Gaussian prime π of norm ≤ 4000 under any of the sixteen twists, and the sixteen genuine cycles are irregular mixtures of inert and split classes ($\omega = 3, 4, 5$) with no one–parameter family among them; the signature of isolated points, not a curve. When $r \ge 1$ (every real quadratic ring, e.g. $\mathbb{Z}[\sqrt{2}]$), Proposition 6.12 makes the real solution locus a positive–dimensional curve in each support, but the unit lattice is an infinite rank–r discrete set, and infinitude requires it to meet that curve in *integer* points infinitely often; inhomogeneous Diophantine approximation on an exponentially parametrised curve. No unit rank delivers both a classical gate and a continuous family: rank zero gives the former and denies the latter, positive rank the reverse. This is why no single ring lets the infinitude be *finished* by present methods; a structural account of the obstruction's universality, not a barrier theorem. It also locates $\mathbb{Z}$ exactly: the unique ring carrying *both* the rank–zero sporadicity *and*, by formal reality, the mass barrier; the worst cell of the trichotomy of Corollary 6.2, which is precisely why #307 is the hard representative of its family.

7 A Thābit–type rule and its honest limits

The bridge recasts the search for a solution as the search for a derivative two–cycle, and the derivative's Leibniz structure makes the cycle equations *linear* in the "closing" primes. This yields a constructive calculus analogous to Thābit ibn Qurra's rule for amicable pairs and to Euler's divisor method.

Theorem 7.1 (Frame Rule)**.** *Let M, N be coprime squarefree integers with derivatives M', N', and set $\Delta := MN - M'N'$. Suppose*

$$p = \frac{MN' + N^2}{\Delta}, \qquad q = \frac{M'N + M^2}{\Delta}$$

are integers, both prime, each coprime to MN, with $p \ne q$. Then $a := Mp$, $b := Nq$ form a derivative two–cycle $a' = b$, $b' = a$; equivalently (P, Q) with P the prime set of Mp and Q that of Nq solves Erdős #307.

Formalised. *`Erdos307.frame_cycle` verifies both cycle equations from the two divisibility hypotheses, and `frame_solve` runs the derivation backwards, so the two quotients are* forced *rather than guessed and the rule is an equivalence. The Leibniz step $a' = M'p + M$ is `csum_insert_prime`. What no algebra reaches is the primality of the two quotients* (`lean/Erdos307/Frame.lean`).

Proof. With p prime and $p \nmid M$, $a = Mp$ is squarefree and $a' = M'p + M$; similarly $b' = N'q + N$. The cycle conditions $a' = b$, $b' = a$ read

$$M'p + M = Nq, \qquad N'q + N = Mp,$$

a linear system in (p, q) with determinant $-(MN - M'N') = -\Delta$. Solving gives exactly $p = (MN' + N^2)/\Delta$ and $q = (M'N + M^2)/\Delta$; substituting back verifies both equations identically. (Both identities were checked symbolically.) $\square$

A two–primes–per–side variant ("Rule 2") is obtained identically: with $a = Mp_1p_2$, $b = Nq_1q_2$ and elementary symmetric functions e_1, e_2, f_1, f_2, the cycle equations are linear in the f_i,

$$f_2 = \frac{Me_1 + M'e_2}{N}, \qquad f_1 = \frac{MNe_2 - MN'e_1 - M'N'e_2}{N^2},$$

and a solution is realised whenever $f_1^2 - 4f_2$ is a perfect square with prime roots.

An Euler-style divisor trick. Fix N and let $M = M_0 r$ with r a new prime, so $M' = M_0' r + M_0$. Put $\Delta_0 := M_0 N - M_0' N'$.

Proposition 7.2 (Divisor trick)**.** *With the above notation and $p_{\mathrm{num}} := MN' + N^2$:*

(a) $\Delta(r) := MN - M'N' = r\,\Delta_0 - M_0 N'$ (linear in r)*;*

(b) $\Delta_0\, p_{\mathrm{num}} - M_0 N'\,\Delta(r) = K$ *where* $K := \Delta_0 N^2 + (M_0 N')^2$ *is* constant in r*;*

(c) *consequently* $\Delta(r) \mid p_{\mathrm{num}} \Rightarrow \Delta(r) \mid K$*, and the converse holds provided* $\gcd(\Delta(r), \Delta_0) = 1$ *(equivalently* $\gcd(M_0 N', \Delta_0) = 1$*, since* $\Delta(r) \equiv -M_0 N' \bmod \Delta_0$*);*

(d) *each positive divisor $d \mid K$ with $d \equiv -M_0 N' \pmod{\Delta_0}$ yields a candidate $r = (d + M_0 N')/\Delta_0$ and recovers $\Delta(r) = d$; this enumeration is complete (it captures every genuine solution). The resulting $p = p_{\mathrm{num}}/d$ is integral when* $\gcd(M_0 N', \Delta_0) = 1$*, and otherwise must be checked.*

Proof. (a) and (b) are direct expansions (verified symbolically; $\partial K/\partial r = 0$). (c): from (b), $\Delta(r) \mid p_{\mathrm{num}} \Rightarrow \Delta(r) \mid \Delta_0 p_{\mathrm{num}} = M_0 N' \Delta(r) + K \Rightarrow \Delta(r) \mid K$. For the converse, $\Delta(r) \mid K$ gives $\Delta(r) \mid \Delta_0 p_{\mathrm{num}}$, whence $\Delta(r) \mid p_{\mathrm{num}}$ iff $\gcd(\Delta(r), \Delta_0) = 1$. (d) inverts (a). Excluded degeneracies: $\Delta_0 = 0$ (r undefined) and $K \le 0$. $\square$

Thus, exactly as in Euler's amicable–pair machinery, one factors the single number K once and reads off candidates from its divisors lying in a fixed residue class modulo Δ_0; only the primality of r, p, q then remains. Two caveats deserve emphasis: the converse in (c) and the automatic integrality of p require $\gcd(M_0 N', \Delta_0) = 1$ (it fails, e.g., for $M_0 = 2$, $N = 19 \cdot 47$, $r = 5$, where $\Delta(r) \mid K$ but $\Delta(r) \nmid p_{\mathrm{num}}$), and the count of admissible classes is governed by Δ_0, not merely by $\tau(K)$.

When the closing side carries a *single* new prime, the rule collapses to a normal form that separates the two layers of the problem.

Proposition 7.3 (One–prime normal form; the exactness stratum)**.** (a) *Erdős #307 has a solution if and only if there exist squarefree M, N with*

$$MN - M'N' = M^2,$$

such that $p := N'/M$ is a prime not dividing M. Indeed the equation forces $M \mid M'N'$, hence $M \mid N'$ by $\gcd(M, M') = 1$*, and $N = M + M'p$; then $a = Mp$, $b = N$ satisfy $a' = M'p + M = N$ and $b' = N' = Mp = a$ (coprimality and $a \neq b$ are automatic). Conversely every solution (a, b) and every prime $p \mid a$ yield such a pair with $M = a/p$, so each solution carries $\omega(a)$ witnesses. The problem thus splits into an* exactness *layer (the Diophantine equation) and a single* primality *layer (one prime value of the determined quantity N'/M).* (b) The exactness layer alone carries the full barrier. *Call (M, N) a* relaxed witness *if $MN - M'N' = M^2$ with $p = N'/M$ composite. The identity $\big(\sigma(M) + \frac{1}{p}\big)\sigma(N) = \frac{N}{Mp} \cdot \frac{Mp}{N} = 1$ holds for* every *witness, so by strict AM–GM ($\sigma(N) = 1$ being impossible) $\sigma(M) + \sigma(N) > 2 - \frac{1}{p}$. Rigidity forces three structural facts. The supports of M and N are* automatically *disjoint: a common prime q would divide $N' = Mp$ through $q \mid M$, against* $\gcd(N, N') = 1$*. Likewise* $\gcd(p, MN) = 1$*: a prime shared by p and M propagates to $N = M + M'p$ and lands in the previous case, while one shared by p and N divides N'. And if p is even, both members are odd: $2 \mid N' = Mp$ forces N odd by the same lemma, while M even would make $N = M + M'p$ even. The union of supports therefore avoids every prime divisor of p, and for even p avoids 2; exact greedy computation then gives, for* every *composite p,*

$$\omega(MN) \;\geq\; 59, \qquad MN \;>\; 8.77 \times 10^{112}$$

the cycle barrier itself; with strictly stronger floors at every finite p: $\omega \geq 194$, $MN > 10^{497}$ at $p = 9$; $\omega \geq 230$, $MN > 10^{609}$ at $p = 4$; the even–p floors rising towards the all–odd bound 10^{5040}, and the infimum 8.77×10^{112} approached only as $p \to \infty$ through composites free of prime factors ≤ 277. Deleting primality thus lowers the barrier by nothing*: the 59–prime, 10^{112} wall is carried entirely by the exactness equation, and the primality of N'/M is a separate demand on top of it. (No witness, prime or composite, exists with $M \leq 120$, $p \leq 2000$, exhaustively.)*

The decisive obstruction: a circular objective. The heuristic yield of the divisor trick from one frame is

$$\mathbb{E}[\text{cycles}] \;\approx\; A \cdot \frac{1}{\ln r \, \ln p \, \ln q}, \qquad A = \#\{d \mid K :\ d \equiv -M_0 N' \pmod{\Delta_0}\} \;\approx\; \frac{\tau(K)}{\Delta_0},$$

the last step being an equidistribution heuristic. To make $\mathbb{E} \geq 1$ at the barrier scale ($1/\ln^3 \approx 5 \times 10^{-7}$) one needs $A \gtrsim 2 \times 10^6$ admissible divisors per frame, hence $\Delta_0 = O(1)$ with $\tau(K) \sim 10^6$. The natural objective is therefore "engineer frames with Δ_0 small." But here is the obstruction, an exact identity:

$$\boxed{\Delta_0 \;=\; M_0 N\big(1 - s_{M_0}\, s_N\big), \qquad s_{M_0} = \sum_{p \mid M_0} \frac{1}{p}, \quad s_N = \sum_{p \mid N} \frac{1}{p}.}$$

So Δ_0 is small *relative to* $M_0 N$ exactly when $s_{M_0} s_N \approx 1$ (which is the Erdős #307 condition for the frames themselves) and $\Delta_0 = O(1)$ in absolute terms at scale requires $s_{M_0} s_N = 1 - O(1/M_0 N)$, i.e. a reciprocal–product within $O(1/M_0 N)$ of 1, *strictly harder* than #307. The frame–engineering objective is thus circular: the Frame Rule is a faithful *parametrisation* of solutions, not a reduction in difficulty.

Remark 7.4 (Honest status of the constructive program)**.** The Frame Rule (Theorem 7.1) and divisor trick (Proposition 7.2) are exact theorems; they make the existence of a solution *falsifiable in principle by finite computation per frame*, and they place #307 in the same constructive lineage as Thābit's and Euler's rules. They do not, as they stand, lower the difficulty below #307 itself: by the boxed identity the bottleneck (small Δ_0) is the original problem in disguise, and a complete small–frame sweep (2.2×10^5 frames) produces zero cycles with total expected count < 1, as the barrier demands. We regard the calculus as an exact parametrisation of solutions whose central obstruction we have now identified precisely; not a pathway that lowers the difficulty, and not evidence that a solution is within reach.

8 A breeder form and the feasibility of certificate search

Allowing *two* free primes on one side turns the divisor trick into a cleaner bilinear form, Euler's amicable–pair method, in the shape used by te Riele's breeding algorithms [12], and lets us state precisely the strongest honest conclusion: a *conditional* reduction, which nonetheless does not become a feasible finite search.

Proposition 8.1 (Bilinear breeder form)**.** *Fix coprime squarefree M_0, N and seek a two–cycle $a = M_0rp$, $b = Nq$ with r, p, q distinct primes coprime to M_0N. Put $G := NM_0 - N'M_0'$ $(= \Delta_0)$, $c := M_0N'$, and $K := GN^2 + c^2$. Then the cycle equations $a' = b$, $b' = a$ are equivalent to*

$$(Gr - c)(Gp - c) = K, \qquad q = \frac{M_0rp - N}{N'}.$$

Proof. Expanding, $(Gr - c)(Gp - c) - K = -N'G\,(a' - b)$ (verified symbolically), so the bilinear equation is equivalent to $a' = b$; the second cycle equation $b' = a$ then determines q as stated. □

Proposition 8.2 (The breeder is integral)**.** *Keep the notation of Proposition* 8.1 *and let $G \neq 0$. Then*

$$(Gr - c)(Gp - c) = K \iff G\,rp - c(r + p) = N^2,$$

and every solution of that equation has

$$N \mid a', \qquad q = \frac{M_0rp - N}{N'} = \frac{a'}{N} \in \mathbb{Z}.$$

Proof. Expanding, $(Gr - c)(Gp - c) - c^2 = G\bigl(G\,rp - c(r + p)\bigr)$ while $K - c^2 = GN^2$, so the two equations differ by the factor $G \neq 0$. Substituting $G = NM_0 - N'M_0'$, $c = M_0N'$, $a = M_0rp$ and $a' = M_0'rp + M_0(r + p)$ into $G\,rp - c(r + p) = N^2$ gives $NM_0rp - N'M_0'rp - M_0N'(r + p) = N^2$, that is

$$N\,(a - N) \;=\; N'\bigl(M_0'rp + M_0(r + p)\bigr) \;=\; N'a'.$$

Hence $N \mid N'a'$, and N squarefree gives $\gcd(N, N') = 1$, so $N \mid a'$ and $q = a'/N$. □

This is worth isolating because the displayed $q = (M_0rp - N)/N'$ is a quotient by a quantity of the size of the barrier, and one might expect a candidate to be usable only when N' happens to divide $M_0rp - N$ — a further demand of probability about $1/N'$, which would leave the construction hopeless for reasons having nothing to do with primality. There is no such demand. The expectation $\mathbb{E} \approx \bigl(\tau(K)/G\bigr)(\ln r \ln p \ln q)^{-1}$ is therefore the correct count: the breeder loses candidates only to primality, never to integrality. A scan of 51,132 frames (`code/breeder_integral.gp`) produced 9 admissible candidates, every one with integral q. Formalised in `lean/Erdos307/Breeder.lean`.

Each factorisation $K = d \cdot (K/d)$ with $d \equiv -c \pmod{G}$ proposes a candidate $(r,p) = \big((d+c)/G, (K/d+c)/G\big)$; a solution is any such candidate whose r, p, q are all prime. A dual, partition–side form is occasionally useful:

Proposition 8.3 (Subset form)**.** *For a support U with $D = \prod_{p\in U} p$ and $\Sigma = \sum_{p\in U} D/p$, a partition $U = P \sqcup Q$ solves #307 iff $S = P$ satisfies $A(\Sigma - A) = D^2$, where $A := \sum_{p\in S} D/p$. Moreover $A(\Sigma - A) - D^2 = -(1 - s_P s_Q)\, D^2$.*

(Both identities verified symbolically and on 2×10^4 random supports.) The last identity says the "analytic" residual $1 - s_P s_Q$ and the "arithmetic" residual $A(\Sigma - A) - D^2$ are one equation up to the factor D^2: the two obvious steering handles carry no independent information.

A conditional reduction. Several ingredients of a construction are unconditional. Greedy Egyptian–fraction choice gives disjoint P, Q with $0 < 1 - s_P s_Q = O(1/T)$ (solvability to any precision); a prime r nearest $M_0 N'/\Delta_0$ gives $G \le \Delta_0^{0.475}(M_0 N')^{0.525}$ unconditionally (Baker–Harman–Pintz [13]), and each small prime ℓ can be forced to divide K by one congruence on a free prime slot of N (Dirichlet then supplies the prime). Granting these, #307 reduces to a single prime–density statement,

(H) among the admissible candidates so produced, some triple (r, p, q) is prime,

and (H) $\Rightarrow$ #307 is YES. This (H) is a Hardy–Littlewood/Bateman–Horn–type assertion for an explicit, computable, *non–polynomial* family; we know of no named conjecture implying it. That is not an accident of our construction:

Proposition 8.4 (No polynomial families)**.** *Let $p_1(t), \dots, p_k(t)$, $q_1(t), \dots, q_l(t)$ be integer polynomials, each taking* positive *prime values on a common infinite branch $t \to +\infty$ (a fixed–shape Schinzel–type family), such that $\big(\sum_i 1/p_i(t)\big)\big(\sum_j 1/q_j(t)\big) = 1$ for infinitely many t. Then every p_i and q_j is constant: the family is a single fixed solution. Consequently no polynomial parametrisation can reduce #307 to Bunyakovsky, Schinzel's Hypothesis H, or Bateman–Horn.*

Formalised. *`Erdos307.const_prod_eq_one_of_tendsto` is the limit step, `eps_eq_zero_of_identity` the vanishing step, and `nonconstant_empty` the conclusion that both index sets of nonconstant members are empty; `nopoly` chains them. The polynomial input enters only as the two hypotheses it supplies, that each nonconstant member contributes a strictly positive reciprocal and that the remainders tend to* 0*, so what is imported is visible.* 0 *`sorry`, three standard axioms* (`lean/Erdos307/NoPoly.lean`).

Proof. Holding at infinitely many t, the relation holds identically as rational functions. A nonconstant p_i taking infinitely many prime (hence positive) values has positive leading coefficient on an infinite branch, so $p_i(t) \to +\infty$ there; split each side into constants and nonconstant members, $\sum_i 1/p_i = A_P + \varepsilon_P(t)$ with $\varepsilon_P(t) \to 0^+$. Letting $t \to \infty$ in the identity gives $A_P A_Q = 1$ (in particular both sides contain constants), and subtracting, $A_P \varepsilon_Q(t) + A_Q \varepsilon_P(t) + \varepsilon_P(t)\varepsilon_Q(t) = 0$ for all large t. Every term on the left is nonnegative, and strictly positive as soon as one nonconstant member exists; hence there are none. ☐

Proposition 8.5 (No fixed–shape family, of any growth rate)**.** *Let $P(t) = \{p_1(t), \dots, p_k(t)\}$ and $Q(t) = \{q_1(t), \dots, q_l(t)\}$ be families of integers of* any *shape — no polynomial, algebraic or analytic hypothesis is imposed — indexed by $t \in \mathbb{N}$, with k, l fixed. Suppose each member takes positive prime values, each nonconstant member tends to $+\infty$, and $\big(\sum_i 1/p_i(t)\big)\big(\sum_j 1/q_j(t)\big) = 1$ for* infinitely many *t. Then every member is constant.*

Proof. Let φ enumerate, in increasing order, the t at which the relation holds; φ is strictly monotone, so $\varphi(n) \to \infty$. Split each side into its constant members, contributing A_P and A_Q, and its nonconstant ones, contributing $\varepsilon_P(t), \varepsilon_Q(t) > 0$ which tend to 0 because each nonconstant member tends to $+\infty$ and there are finitely many. Along φ the relation holds at every index and the remainders still tend to 0, so Proposition 8.4's two steps apply verbatim: the limit gives $A_P A_Q = 1$, and subtracting leaves $A_P \varepsilon_Q + A_Q \varepsilon_P + \varepsilon_P \varepsilon_Q = 0$ with every term nonnegative and some term strictly positive if a nonconstant member exists. $\square$

The proof of Proposition 8.4 uses polynomiality in exactly one place, to pass from "holding at infinitely many t" to "holding identically", and restricting to the subsequence removes that need. What the argument actually consumes is three hypotheses — finitely many members, positive values, nonconstant members growing — and none of them is about shape.

The strengthening is not cosmetic. Excluding polynomial families rules out a reduction of #307 to Bunyakovsky, Schinzel or Bateman–Horn, but it says nothing about *exponential* families, and those are what the classical technique for problems of this kind actually uses: Thābit's rule for amicable pairs runs on primes $3 \cdot 2^n - 1$, $3 \cdot 2^{n-1} - 1$, $9 \cdot 2^{2n-1} - 1$, and Euler's generalisation is of the same shape. A member $3 \cdot 2^n - 1$ is positive and tends to infinity, so it satisfies the hypotheses above and the family is excluded. Proposition 8.5 therefore closes the whole Thābit/Euler programme for #307, not merely its polynomial shadow: no fixed–shape rule of any growth rate can generate solutions, and the constructive calculus of Section 7 is not failing for want of a cleverer ansatz.

Two limits on the statement should be read with it. The hypothesis that the families have *fixed* cardinality is used, in the splitting into constants plus a remainder tending to 0, and a family whose support grows with t is not covered; that gap is real, and harmless, since a growing support is a growing number of simultaneous primality demands rather than a weaker one. And positivity is a genuine hypothesis, not a consequence, for the reason recorded in Remark 8.6. Formalised as `Erdos307.no_family_of_any_shape` on three standard axioms, in the same shape as Proposition 8.4: the abstract core is proved, and the two facts the family supplies — that each nonconstant member contributes a strictly positive reciprocal and that the remainders tend to 0 — enter as hypotheses rather than being derived, so what is imported stays visible. The theorem's statement quantifies over arbitrary index types and arbitrary $f_P : \mathbb{N} \to \iota \to \mathbb{R}$; no polynomial occurs anywhere in it.

Remark 8.6 (Erratum: positivity is a hypothesis, not a consequence)**.** Earlier versions asked only that each polynomial take prime values for infinitely many integers t, without requiring the values to be positive on a common branch. As printed that statement is *false*: take $P = (2, 2, t, -t)$ and $Q = (2, 2)$. Each member takes prime values for infinitely many integers, and

$$\left(\tfrac{1}{2} + \tfrac{1}{2} + \tfrac{1}{t} - \tfrac{1}{t}\right)\left(\tfrac{1}{2} + \tfrac{1}{2}\right) = 1$$

identically, with two nonconstant members present. The proof's step "every term on the left is nonnegative" silently assumes positivity, which is exactly the missing hypothesis; with it, as now stated, the proof is correct and the counterexample is excluded.

The formalisation carries the repaired statement, not the printed one: `Erdos307.nonconstant_empty` takes $0 < f_P(i)$ for every nonconstant member as an explicit hypothesis, so the Lean development was already stating the corrected proposition. This is the one place in the paper where formalisation caught a missing hypothesis before review did.

This upgrades the earlier computational observation (that the breeder family is not polynomially parametrised) to a theorem for *all* fixed–shape families: the prime–density hypothesis behind #307 is irreducibly non–polynomial, which is why no standard conjecture applies.

Proposition 8.7 (Near–miss frames are extremal for G)**.** *Let a be squarefree with a' squarefree, and take the frame $(M_0, N) = (a, a')$, so that $M_0' = a'$ and $N' = a''$. Then*

$$G \;=\; NM_0 - N'M_0' \;=\; a'\,(a - a''),$$

and since $a'' \neq a$ unless #307 is solved, $|a - a''| \geq 1$ and therefore $|G| \geq a'$.

Proof. Direct substitution; the inequality is $|a - a''| \geq 1$ for the nonzero integer $a - a''$. □

The point is not that the bound is large but that it runs the wrong way. Writing $r = a''/a$ for the near–miss ratio, $G = aa'(1 - r)$, so a frame built from a good near–miss has $1 - r$ small and aa' correspondingly large, and the two cancel *exactly*, because $a(1 - r) = a - a''$ is an integer and a nonzero one. Improving the near–miss therefore does not shrink G; it drives a' up and G with it. The witness of Proposition 13.9, with $|1 - r| = 1.65 \times 10^{-13}$, gives $|G| \geq a' > 10^{112}$, against the $\log_{10} G \approx 13$ that the Baker–Harman–Pintz route reaches at the barrier. So the natural reading of "steer Δ_0 small by steering $s_{M_0} s_N$ to 1" is not merely circular in the sense of Section 7: on near–miss frames it is inverted.

Small $|G|$ does occur, but only on small frames: over 1,143,888 coprime squarefree frames with $M_0, N \leq 2000$ the minimum is $|G| = 5$, at $(M_0, N) = (2, 3)$. That is the supply–demand conflict in raw form, and it can be measured. Binning by $|G|$ and taking the best frame in each bin (`code/breeder_supply.gp`) gives

$\log_{10}\lvert G\rvert$	0.5	1.5	2.5	3.5	4.5	5.0
$\max \tau(K)/\lvert G\rvert$	0.600	0.313	0.214	0.053	0.0065	0.0030

The ratio is largest on the smallest frames, where it is already below 1, and decays monotonically from there. Since a productive frame needs $\tau(K)/|G|$ well above 1, no frame in the range supplies one, and the trend is against scale rather than with it.

The descent, and why it cannot be iterated. Adjoining a prime w to M_0 acts on G *linearly*: with $M_0 \mapsto M_0 w$ one has $M_0' \mapsto M_0' w + M_0$ and hence

$$G \;\longmapsto\; wG - M_0 N', \qquad \text{so} \qquad |G_{\text{new}}| = |G| \cdot |w - w^*|, \qquad w^* := \frac{M_0 N'}{G} = \frac{\sigma(N)}{\varepsilon},$$

writing $\varepsilon = 1 - \sigma(M_0)\sigma(N) = G/(M_0 N)$. Since $G = 0$ is #307, this is a descent towards a solution, and on the ε scale it is quadratic: $\varepsilon_{\text{new}} = \varepsilon^2 \theta / \sigma(N)$ with $\theta = w - w^*$. It is a Newton iteration for #307.

Three facts close it. First, it cannot start below the barrier: $w^* = \sigma(N)/\varepsilon \geq 2$ needs $\varepsilon \leq \sigma(N)/2$, i.e. $\sigma(M_0)\sigma(N)$ near 1, which by AM–GM needs $\sigma(M_0 N) \geq 2$ and so $\omega(M_0 N) \geq 59$. Over 4,284,896 coprime squarefree frames with $M_0 \leq 5000$, $N \leq 3000$ the smallest $|\varepsilon|$ is 0.470, and $w^* < 2$ throughout.

Second, only $w = \lfloor w^* \rfloor$ or $\lceil w^* \rceil$ can shrink $|G|$, since $|G_{\text{new}}| < |G|$ demands $|w - w^*| < 1$. So a step descends only when one of those two integers is prime, with probability $\approx 2/\log w^*$.

Third, and decisively, a successful step squares the difficulty. From $w^* = M_0 N'/G$,

$$w^*_{\text{new}} = \frac{M_0 w N'}{G\theta} = w^* \cdot \frac{w}{\theta} \;\approx\; \frac{(w^*)^2}{\theta},$$

so $\log w^*$ more than doubles at every step while $\log |G|$ falls by only $|\log \theta| \approx 0.69$. The gain is linear in the number of steps k and the cost is quadratic in the exponent: $|G|$ falls like 2^{-k}, whereas the probability of surviving k steps is $\prod_{j<k} 2/\log w_j^* \asymp 2^{-k^2/2}$. Starting from the greedy frame $N = 30$, $M_0 = \prod_{7\le p\le 271} p$, where $\log|G| = 249$ and $w^* = 428.2$ (`code/breeder_descent.gp`):

paths available	10^2	10^4	10^6	10^8	10^{10}	10^{12}
reachable depth k	3	5	6	7	8	8
$\lvert G\rvert$ shrinks by	8	32	64	128	256	256

against the factor 10^{108} required to reach $G = 0$. Even 10^{12} independent descent paths buy eight steps and a factor 256. This is the quantitative form of the circularity of Section 7: the Newton iteration for #307 exists, is exact, and converges quadratically, and each step costs a prime whose size is the square of the last one's.

Why (H) **is not a finite search.** It is tempting to read (H) as "run a certificate hunt that is expected to succeed." It is not, for a quantitative reason. The expected number of certificates from one frame is $\mathbb{E} \approx \big(\tau(K)/G\big) \cdot (\ln r\, \ln p\, \ln q)^{-1}$, and *both* factors are pinned by the geometry.

- *Demand.* $G = \Delta_0 = M_0N(1 - s_{M_0}s_N)$ exactly, so G is small only in the circular regime $s_{M_0}s_N \approx 1$; over 2×10^4 random coprime squarefree frames the ratio $G/(M_0N)$ has median 1 and best value $\approx 10^{-0.23}$, and the honest Baker–Harman–Pintz reduction still leaves $\log_{10} G \approx 13$ at the barrier.

- *Supply.* $\tau(K)$ is bounded by the *size* of $K = GN^2 + c^2$ ($\ge c^2$), which is an *output* of the frame, not a free dial. The maximal order of the divisor function [16, Thm. 317] caps $\log_2 \tau(K)$ at ≈ 36 for $K \le 10^{54}$ (the barrier geometry) and at ≈ 112 even for $K \le 10^{240}$; reaching $\tau(K) = 2^{120}$ needs $K \gtrsim 10^{263}$, i.e. frame scale far beyond the barrier.

Enlarging a frame grows $K \propto N^2$ but gains divisors only at rate $\log 2/\log\log K < 1$ per log–unit of K, while it grows G at rate 1 per log–unit of N; so $\tau(K)/G$ cannot be driven up. Optimising $\mathbb{E}$ over all (M_0, N) splits with geometry–consistent K and the maximal–order τ, and granting the most favourable demand and singular–series constants, yields $\log_{10} \mathbb{E} \le -4$ at the barrier and *decreasing* with scale; empirically 385 factorable frames realise a median of 0 and a maximum of 1 admissible divisor, against the $\sim 10^5$ a single productive frame would need. A self–consistent accounting; capping the number w of forced prime factors of K by the frame's own residue freedom (counted generously), sharpens this to $\log_{10} \mathbb{E} \le -8$ at $B = 56$, falling to ≈ -260 at $B = 3000$: the deficit is *scale–invariant*, so no choice of scale, slot count, or budget split reaches expectation 1. (The construction is even self–similar: engineering K's algebraic; e.g. Gaussian–integer; factorisation to supply divisors reduces to prescribing N and N' simultaneously, an arithmetic–antiderivative inverse problem of the same difficulty class as #307.) The breeder thus makes #307 conditional on (H) but does *not* render it a feasible finite computation: the supply–demand conflict through $K = GN^2 + c^2$ is exactly the circularity of Section 7 in quantitative form. There is also a structural reason the analogy with amicable pairs was bound to fall short: te Riele's breeding amplifies a *known seed* pair into new ones, whereas the barrier (Theorem 4.3) forbids any seed below 2.09×10^{56} the method imports a precondition the problem cannot supply.

9 Computations

All verdicts use exact integer/rational arithmetic; floating point is used only for pre–screening.

- *Exhaustive non–existence below* 10^8*.* Enumerating all prime sets P with $\sum 1/p > 1$ and $\prod P \le 10^8$ (forcing Q as the prime support of N_P and testing $N_Q = D_P$): 1,075,419 candidate sets, no solution. Subsumed by Theorem 4.3 but an independent sanity check.

- *Derivative sieve.* No solution of $n'' = n$ with $n' \ne n$ and both members $\le 2 \times 10^7$ (smallest–prime–factor sieve of the derivative).

- *Function field.* The arithmetic derivative on $\mathbb{F}_q[t]$ has no two–cycles and no nonzero fixed points, verified exhaustively over $\mathbb{F}_2$ (degree ≤ 7), $\mathbb{F}_3$ (≤ 5), $\mathbb{F}_5$ (≤ 4), confirming Theorem 6.1.

- *Density.* The sieve gives density 0.042029 to 2×10^7 (0.017623 among squarefree); the independent Erdős–Wintner convolution over primes $\le 10^5$ gives $\delta = 0.04202$, agreeing to four places (Proposition 5.6).

- *Conditional mass.* Of 43,087 qualifying squarefree a sampled, $b = a'$ is squarefree 95.0% of the time; on the resulting 40,920 squarefree b the mean reciprocal mass is 0.047 vs. the required $1/s \approx 0.934$, with empirical tail probability $\Pr[\sum_{q|b} 1/q \ge 1/s] = 0/40{,}920$.

- *Frame/ledger sweep.* Over 2.2×10^5 factorable small frames: the divisor–trick identities hold on every instance; the smallest Δ_0 observed is 22; the equidistribution $A \approx \tau(K)/\Delta_0$ holds only in aggregate ($\sum A = 127$ vs. $\sum \tau(K)/\Delta_0 = 39.8$) and fails per frame ($A = 0$ for 99.95% of frames); total expected cycles < 1; zero cycles, as the barrier requires.

- *μ–Sondow line census.* Enumerating the lines $n' = n \pm d$ (squarefree members coprime to d) finds no adjacent opposite–sign pair at additive distance d: exhaustively to 2×10^7 for $|d| \le 200$, and along the small–$|\delta|$ genealogy tree out to members of more than a thousand digits. Every $\delta = \pm 1$ member encountered is a known Giuga or primary pseudoperfect number; the closest opposite–sign approach recorded is $|y - (x + d)| = 11$ (at $d = 1$, 30 vs. 42, and $d = 37$, 210 vs. 258), attained only among small members; the two trees do not come close at height.

Code and logs accompany this note.

10 Formalisation in Lean 4

The rigidity and barrier results are formalised against mathlib (Lean 4, toolchain `v4.30.0`). The following are machine–checked and `sorry`–free, depending only on the standard axioms `propext`, `Classical.choice`, `Quot.sound`:

- Lemma 2.1 (`rigidity_coprime`) and Theorem 2.3 (`solution_structure`);

- Lemma 4.1 (`reciprocal_sum_gt_two`); the exact thresholds $\sum_{\text{first } 58} 1/p < 2 < \sum_{\text{first } 59} 1/p$ grounding $|U| \ge 59$ (`recipSum58_lt_two`, `recipSum59_gt_two`); the numeric core $\Pi_{59}^2 \ge (4 \times 10^{112}) N_{59}$ (`barrier_numeric`), and the algebraic core of the barrier (`barrier`), giving $D_P^2 \ge 4 \times 10^{112}$;

- Theorem 13.20 (`log_lt_half_sub`, `delta_decreasing`, `two_le_add_of_mul_eq_one`) and Theorem 13.24 (`criterion_fails_of_pos`, `criterion_fails_of_nonpos`, `no_lambda_works`, `lyapunov_criterion_fa`
neither of which uses `native_decide`;

- Proposition 13.27 (`half_works`, `no_f_above_two`, `mass_sum_threshold`) and Proposition 13.28 (`exists_f_of_injective`, `increment_iff`, `no_two_cycle_of_decreasing`), which together machine-check the closure of the sign branch in Theorem 13.22;

- Proposition 13.14 (`exists_block_sum_near`, `infimum_zero`);
- Theorem 13.22(1) in its mechanism (`Erdos307.Congruential`) and Proposition 15.28 of the full version (`live_slot_threshold`, `threshold_bracket`).

The single non–elementary ingredient, that the k smallest primes minimise $\prod$ and maximise $\sum 1/p$ among k–element prime sets, is isolated as an explicit hypothesis of `barrier` (the ratio bound $R/T(U) \geq \Pi_{59}/T_{59}$), so the dependency is transparent rather than hidden. It is no longer an open dependency: `Erdos307.Extremal` proves it (`prod_first_primes_le`, `recipSum_le_first_primes`, `hRatio_of_extremal`), which is why Theorem 4.3 appears above in the hypothesis-free form `erdos307_barrier_clo`

Two remarks on the trust boundary, since it moved at v1.4.6. The numeral bridge `np0`...`np59` is checked by the Lean kernel through `decide`, not by `native_decide`, so `card_ge_59` and `erdos307_barrier_closed` carry only the three standard axioms. The whole development contains exactly one `native_decide` site, `dfs_run`, the pruned-search execution behind Proposition 4.5; it is the only non-logical input anywhere, and `erdos307_sixty` is the only theorem that carries it. Proposition 4.5's product bounds are proved in this paper and not in Lean: `erdos307_sixty` establishes $|P \cup Q| \geq 60$ alone.

11 Status and questions

Erdős #307 remains **open**. We have shown it is equivalent to the existence of a non–trivial two–cycle of the arithmetic derivative; established rigidity and a barrier (Rosen's $|P \cup Q| \geq 59$, together with our $\min(\prod P, \prod Q) \geq 2.09 \times 10^{56}$) that voids all direct search and improves the $n'' = n$ record by ~ 47 orders of magnitude; given a parity dichotomy, a proof that longer cycles are strictly harder, an Erdős–Wintner density theorem, and a function–field theorem localising the obstruction to the archimedean place; developed an exact Thābit/Euler–type constructive calculus and a bilinear breeder form whose central obstruction we identify precisely, and extracted the maximal honest consequence of the construction; a conditional reduction (H) $\Rightarrow$ YES to an explicit prime–density hypothesis, together with a quantitative ledger showing that (H) is *not* a feasible finite search (the supply–demand conflict through $K = GN^2 + c^2$ keeps the expected certificate count per reachable frame below 1). The rigidity result and the barrier are fully Lean–verified (extremality proved, not assumed), and the finite verification of Proposition 4.5 sharpens the barrier to $|P \cup Q| \geq 60$, $\min(\prod P, \prod Q) > 3.50 \times 10^{57}$. The heuristics weakly favour YES (existence), hence weakly against the period–one Ufnarovski–Åhlander conjecture, but we claim no proof in either direction.

Open problems: (i) close the residual 3,478 single–tail families at level 60, and close the pair sector of Proposition 4.9; for which Proposition 4.11 rules out every congruence kill, so only global, per–family input (factorisation or direct search) can serve (the canonical tail family is closed, Proposition 4.8; the finite verification of Proposition 4.5 is Lean–checked, `erdos307_sixty`; the conditional 3/4 law is explained; it is the reciprocity switch of Remark 4.12, made exact by the class–vector functional there); (ii) decide the parity dichotomy's all–odd case (we expect it empty); (iii) sharpen the certified enclosure $0.04202 \leq \delta \leq 0.04223$ of Proposition 5.6, whose width is carried almost entirely by the tail window a; three digits are certified, and a longer envelope convolution with a smaller window certifies more; (iv) settle the density hypothesis (H) of Section 8, or exhibit a non–circular frame family escaping the $\tau(K)/G$ ceiling; (v) find a $\mathbb{Z}$–side structure detecting the archimedean obstruction of Section 6; one that must be non–local, since by Proposition 5.2 no single modulus obstructs; (vi) settle #307.

Remark 11.1 (The place of the problem: a local–global dictionary)**.** The pieces above assemble into a dictionary. (i) For a fixed support U, the subset form of Proposition 8.3, $A(\Sigma - A) = D^2$, satisfies the *Hasse principle as an equation*: an integer solution A exists iff one exists over $\mathbb{R}$ and over every $\mathbb{Z}_\ell$ (the discriminant $\Sigma^2 - 4D^2$ must be a nonnegative square, and squares obey local–global). Its real–place condition is $\Sigma \geq 2D$, i.e. $\sigma(U) \geq 2$: *the barrier is precisely the real–place solvability condition of the subset form.* (ii) The barrier never needed primality: pairwise coprime integers have distinct smallest prime factors, so $\sum 1/a_i \leq \sum 1/p_i$, and *any* pairwise–coprime support of mass ≥ 2 has at least 59 elements and product at least Π_{59}. Coprimality and the ordering of $\mathbb{R}$ carry the wall; primality enters only through rigidity. (iii) What remains unsolved is the *realization* of the root $A^* = \big(\Sigma + \sqrt{\Sigma^2 - 4D^2}\big)/2$ as a sub–multiset sum of the cofactors D/p. Its local conditions are exactly the local anatomy of Lemma 5.1: modulo ℓ^2 for $\ell \in U$ the subset sums occupy precisely the two cosets $\ell\mathbb{Z} \cup (D/\ell + \ell\mathbb{Z})$ (2ℓ residues, observed exactly (6 of 9, 10 of 25, 14 of 49, 22 of 121)) and these are the same constraints the equation itself forces: no completion separates the equation from its realization. In the language of integral points, then: #307 is everywhere locally soluble, with the barrier as its archimedean height condition, and its resolution is a *strong approximation* question for the thin, prime–structured set of subset sums a failure mode invisible to Brauer–Manin–type obstructions, consistent with every no–go above and with the existence of cycles over the neighbouring rings. We record this as a dictionary, not a method.

Attribution. We are careful to separate prior, concurrent, and present contributions (details in Section 12). *Prior/concurrent:* the two–cycle reformulation, and its equivalence with the arithmetic derivative, are due to Ufnarovski–Åhlander [7] (2003, Conjecture 4); the naming of it as #307 to Seva [19] (2019); an independent rederivation to Bado [20]; the bound $|P \cup Q| \geq 59$ to J. Rosen and the disjointness valuation $v_p(\sum 1/p_i) = -1$ to R. Israel [9]; the source identification to "Woett" [9]. Bado [20] independently obtained the rigidity cross–matching, the local anatomy, the parity restriction, a one–sided exact test, approximate solvability, and the union criterion of Proposition 12.1. *Present note:* the identification of #307 with Ufnarovski–Åhlander's Conjecture 4 and the resulting two–way bridge to the $n'' = n$ literature; the product barrier 2.09×10^{56}; the function–field theorem and archimedean localisation; the constructive deficit analysis and seed obstruction; the Erdős–Wintner density theorem, and the k–cycle barriers. Both external sources [19, 20] have been examined directly and the overlaps above reflect their actual contents.

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12 Prior and concurrent work

We record the overlaps with four items, each examined directly.

Ufnarovski–Åhlander (2003) [7]. Their Theorem 5 gives $p^k \| n$, $0 < k < p \Rightarrow p^{k-1} \| n'$, and their Theorem 4 gives $n^{(j)} \to \infty$ when $p^p \mid n$ properly; together with Corollary 2 (n squarefree $\Leftrightarrow (n, n') = 1$) these force a period–two point to have squarefree members with disjoint supports. The argument is sound: for $2 \leq k < p$ two applications of Theorem 5 give $v_p(n'') = k - 2 \neq k$, while $k \geq p$ falls to Theorem 4 or to the fixed point $n = p^p$, and the edge cases $p = 2$, $k \in \{2, 3\}$ are covered by the latter because $4 = p^p$. They then state the period–two case as **Conjecture 4**: $\big(\sum_{i \leq k} 1/p_i\big)\big(\sum_{j \leq l} 1/q_j\big) = 1$ has no solution in distinct primes. This is #307, and priority for the two–cycle reformulation belongs here rather than to 2019: what [19] adds is the identification of that equation as #307, which [7] does not make — they cite [1] for the derivative, not [2] for the problem. The consequence is that #307 has been an open conjecture in the arithmetic–derivative literature since 2003 under another name, and that the bounds of Section 4 are bounds on Conjecture 4.

Seva (MathOverflow 323194, 2019) [19]. Defines $s(Q) = Q\sum_{p|Q} 1/p$, derives the valuation property $v_p(s(Q)) = v_p(Q) - 1$, asks for $s(P) = Q$, $s(Q) = P$, and states this is Problem #307 "in disguise." Priority for *naming* the two–cycle question as #307 belongs here. He further asks whether every s–trajectory stabilises at 0; the s–analogue of the Ufnarovski–Åhlander conjecture, posed independently of it. (Note $s \neq n'$ globally, e.g. $s(p^\alpha) = p^{\alpha-1}$, but $s = n'$ on squarefree integers, where both two–cycle questions live.)

Kovič (2012), Propositions 17 and 18 [8]. Proposition 18 claims $r + s \geq 34$ unconditionally (≥ 57 if $\min\{p_1, q_1\} \geq 3$, ≥ 110 if ≥ 5), together with $p_1 < r$, $q_1 < s$ and $\max\{m, n\} \geq \prod_{i\leq 17} p_i \approx 1.92 \times 10^{21}$. Its proof does not establish them. Substituting $n' = m$, $m' = n$ into its two correct pairs of bounds gives

$$\text{(A)}\quad \frac{rn}{p_r} < m < \frac{rn}{p_1}, \qquad \text{(B)}\quad \frac{sm}{q_s} < n < \frac{sm}{q_1},$$

after which it asserts "Hence: $p_1q_s < rs$ and $q_1p_r < rs$", and deduces everything from these via $2\max\{p_r, q_s\} < N^2/4$ against $\max\{p_r, q_s\} \geq P_N$. But products of inequalities are valid only in matching directions, and (A) with (B) admits exactly two: upper × upper gives $p_1q_1 < rs$, lower × lower gives $rs < p_rq_s$. The asserted $p_1q_s < rs$ pairs an upper bound on m, carrying p_1, with a lower bound on n, carrying q_s. Nor is it recoverable: the cycle says precisely $\sigma(n) \in (r/p_r,\, r/p_1)$ and $\sigma(m) = 1/\sigma(n) \in (s/q_s,\, s/q_1)$, so $\sigma(n)$ lies in $(r/p_r,\, r/p_1) \cap (q_1/s,\, q_s/s)$; nonemptiness of that intersection is equivalent to those two products and to nothing further, whereas $p_1q_s < rs$ reads $q_s/s < r/p_1$, a comparison of the two *upper* endpoints. In $n = 2\cdot 5\cdot 13\cdot 37\cdot 53\cdot 73\cdot 79\cdot 89\cdot 97\cdot 101$, $m = 3\cdot 7\cdot 19\cdot 31\cdot 43\cdot 59\cdot 67\cdot 71\cdot 83\cdot 103$ — squarefree, disjoint supports, $r = s = 10$ — all four inequalities hold while $p_1q_s = 206 > 100 = rs$ and $q_1p_r = 303 > 100$ (`code/kovic_prop18_audit.gp`; the valid products and the invalidity are machine–checked in `lean/Erdos307/KovicProp18.lean`). Two arithmetic slips sit alongside: Table 2 lists $P_{11} = 27$, and the "product of the first 17 primes" is printed with 27 for 29.

What fails is the *inference*, not the *statement*: no two–cycle is known, so no counterexample to $r+s \geq 34$ exists to be given, and none is claimed. The conclusion is unproven rather than false, and is in fact true, since Theorem 4.3 gives $|U| \geq 60$ by a route touching none of this. Propositions 16 and 19 are sound, and so is 17, but 17 assumes the *smaller* member is odd and [8] notes that no estimate follows when it is even. The strongest valid unconditional bound in the prior literature is therefore that paper's computer search $x > 10^4$; Theorem 4.3 improves it by 52 orders of magnitude, and $|U| \geq 60$ is the first valid cardinality bound for arithmetic–derivative two–cycles rather than an improvement on an earlier one. None of this diminishes [8], which remains a standard reference and whose Propositions 16, 17 and 19 we use.

Bado (preprint, Aix–Marseille, 2026) [20]. Working with $A(S) = \sum_{p\in S} M(S)/p$ and $M(S) = \prod_{p\in S} p$ (our N_S, D_S), Bado independently obtains: the lowest–terms lemma (his Lemma 3.1, our Lemma 2.1); the forcing identities $A(P) = M(Q)$, $A(Q) = M(P)$ with $P \cap Q = \varnothing$ (his Thm. A, our Theorem 2.3); the two–cycle reformulation $T(T(P)) = P$ (his Thm. B); the valuation signature (his Thm. 6.1, = Israel's); the local zero–sum congruences (his Prop. 8.1, our Lemma 5.1); the parity restrictions (his Prop. 8.2, the parity part of Theorem 4.26); an exact one–sided test (his Thm. 9.1);

the deviation parameter, i.e. the integer t with $A(P) = n + tm$, $A(Q) = m - tn$ and solubility exactly at $t = 0$ (his Thm. 7.6, our rung k of Proposition 13.1 of the full version); the intrinsic union criterion (his Thm. 7.3, our Proposition 12.1 in *iff* form); uniqueness of the splitting (his Cor. 7.5); the approximation theorem (his Thm. 10.2, sharper than our greedy version in that all primes exceed a given bound), and the union discriminant / Pythagorean structure (his Thms. 7.1, 7.4). He cites only Barbeau, Bloom, and Erdős–Graham: crucially, he does *not* identify the map with the arithmetic derivative, and his note contains none of the present note's $n'' = n$ bridge, explicit barrier constant, function–field theorem, density theorem, k–cycle bounds, or constructive deficit analysis. We regard the overlap on the shared structural layer as strong independent validation, and import his union criterion with credit. Bado has confirmed this account in correspondence (August 2026): he agrees that the results listed above were present in his preprint and that the overlap is concurrent and independent, and he has confirmed the identification $A(S) = M(S)'$, which he had not made, and is revising his historical discussion accordingly. The concurrency is therefore recorded on both sides rather than asserted on one:

Proposition 12.1 (Single–integer test; derivative form of Bado's Thms. 7.1, 7.4)**.** *If (a, b) is a two–cycle of the arithmetic derivative, then $N = ab$ is squarefree and satisfies $N' = a^2 + b^2$, whence $N'^2 - 4N^2 = (a^2 - b^2)^2$ is a perfect square. Conversely, a squarefree N with $N'^2 - 4N^2 = \square$ determines a unique unordered factorisation $N = ab$ with $N' = a^2 + b^2$, on which the cycle condition reduces to a single equation (Bado's deviation parameter $t = 0$, his Thm. 7.6). This gives a one–integer necessary test; computationally, no squarefree $N < 2 \times 10^5$ with $\omega(N) \geq 2$ passes it (*0 *of* 103,596*, consistent with Theorem 4.3, which forces $N > 4 \times 10^{112}$).*

Proof. $N = ab$ with a, b coprime squarefree gives N squarefree and, by Leibniz with $a' = b$, $b' = a$, $N' = a'b + ab' = b^2 + a^2$; then $N'^2 - 4N^2 = (a^2 + b^2)^2 - 4(ab)^2 = (a^2 - b^2)^2$. The converse is Bado's [20]; the emptiness of the test below 10^{112} is upgraded from a sieve to a theorem in Corollary 12.3. $\square$

The Pythagorean equation $N' = a^2 + b^2$ deserves to be isolated from the (harder) cycle condition.

Proposition 12.2 (Pair form; the one–equation relaxation)**.** *For coprime squarefree a, b, the single equation $a^2 + b^2 = (ab)'$ holds if and only if there is an integer k with $a' = b + ak$ and $b' = a - bk$; a two–cycle is exactly the case $k = 0$. We call a solution of $a^2 + b^2 = (ab)'$ (any k) a* Pythagorean pair*.*

Proof. The Leibniz identity $a^2 + b^2 - (ab)' = a(a - b') - b(a' - b)$ (a polynomial identity, verified symbolically and on 2000 random pairs) makes the equation equivalent to $a(a - b') = b(a' - b)$; since $\gcd(a, b) = 1$ this forces $a \mid a' - b$ and $b \mid a - b'$ with equal quotients k, i.e. $a' = b + ak$, $b' = a - bk$. Setting $k = 0$ recovers $a' = b$, $b' = a$. (Equivalently, in $\mathbb{Z}[i]$: #307 asks for $z = a + bi$ with coprime coordinates and ab squarefree satisfying $|z|^2 = (\operatorname{Im} z^2/2)'$ and $k = 0$.) $\square$

Corollary 12.3 (The single–integer test is provably empty below 10^{112})**.** *Every Pythagorean pair satisfies $\sum_{p|a} 1/p + \sum_{q|b} 1/q = a/b + b/a \geq 2$ (divide $a^2 + b^2 = (ab)'$ by ab and apply AM–GM), so its support carries at least* 59 *primes and $N = ab \geq \prod(\text{first } 59 \text{ primes}) > 7.9 \times 10^{112}$. Hence no squarefree $N < 10^{112}$ with $\omega(N) \geq 2$ satisfies $N'^2 - 4N^2 = \square$: the empirical "*0 *of* 103,596*" of Proposition 12.1 is a theorem, and the barrier of Theorem 4.3 extends from two–cycles to the strictly weaker one–equation (Pythagorean) problem.*

Formalised. *`Erdos307.pyth_sum_of_squares` is $N' = a^2 + b^2$ from Leibniz and the cycle, `pyth_discriminant` the resulting square, `split_identity` and `split_product` the two–layer form of Proposition 13.13 of the full version, and `mass_ge_two_of_pythagorean` with `card_ge_59_of_pythagorean`* *this corollary, glued by* `mass_ge_two_of_pyth_support` *into* `emptytest`*, which takes the Pythagorean equation itself rather than an assumed mass bound. Earlier editions proved the two ends and not the link, so this tag overstated what was present. The AM–GM step of Proposition 4.27 is* `sum_ge_card_of_prod_eq_one`*, valid for every k at once and proved from $\log x \le x - 1$ rather than a k–term AM–GM. Bado's converse is cited, not reproved* (`lean/Erdos307/Pythagorean.lean`)*.*

Remark 12.4. (a) As $N' = a^2 + b^2$ is a sum of two coprime squares, every odd prime dividing N' is $\equiv 1 \pmod 4$; a cheap union–side pre–filter complementing Bado's congruence filters. (When exactly one of a, b is even this is the hypotenuse of the primitive triple $(|a^2 - b^2|, 2ab, a^2 + b^2)$.) (b) Since Seva's s agrees with n' on squarefree integers, and his valuation property forces two–cycle members squarefree, his question is *exactly* #307, not merely an analogue. (c) Primitive Pythagorean triples form the Barning–Hall ternary tree (the orbit structure of a reflection group in $O(2,1;\mathbb{Z})$), and a #307 solution's triple $(|a^2-b^2|, 2ab, a^2+b^2)$ occupies one node with generator $(a,b) = (\prod P, \prod Q)$. The tree nonetheless yields no descent on the problem: its moves $(a,b) \mapsto (2a \mp b,\ a)$, $(a+2b,\ b)$ are linear, preserving coprimality but not squarefreeness (over random squarefree coprime generators only 37% of children keep both members squarefree) so the locus of #307–eligible nodes (squarefree generators of disjoint prime support meeting $a^2+b^2 = (ab)'$) is not tree–invariant. The additive tree is transverse to the multiplicative constraint; the arithmetically natural recursion is instead the prime–append δ–tree of Proposition 15.33 of the full version, $\delta(Mp) = p\,\delta(M) + M$, which preserves squarefreeness and tracks the line parameter, which is why the classical members organise on *that* tree, not Barning–Hall.

The deviation parameter of Proposition 12.2 organises the one–equation problem into a short graded family.

13 Two closures on the existence route

Atom A10 asks for a prime term of the ternary recurrence of Remark 4.21. Before attacking it, two families of approach can be removed: the relaxations that would give the construction more freedom, and the algebraic methods that would avoid the archimedean place. Both are closed outright.

Proposition 13.1 (No free slot)**.** *Let P, Q be finite sets of primes with $\sigma(P)\sigma(Q) = 1$, and put $a = \prod P$, $b = \prod Q$. Then $b = a'$ and $a = b'$ exactly. In particular Q is* determined *by P: it is the set of prime factors of a', and the existence route carries exactly one parameter, the set P.*

Formalised. `Erdos307.Q_determined`*: given $a' = b$, the set Q is exactly the set of prime factors of a'. The one–prime normal form of Proposition 7.3 is* `oneprime_forced`*,* `oneprime_cycle`*,* `oneprime_converse` (*which closes the equivalence; earlier editions formalised one direction only*)*,* `oneprime_mass_identity` *and* `oneprime_mass_bound`*, the last two giving part* (b)*: the mass identity holds for* every *witness, prime or composite, so the exactness layer alone carries the barrier. The slot calculus of Proposition 15.17 of the full version is* `slot_layers` *and* `slot_recovery`*, the filter of Corollary 15.16 of the full version is* `qr_filter`*, resting on* `cofactor_survival`*, the congruence $N' + 2N \equiv N/\ell \pmod{\ell}$, which is proved rather than assumed (an earlier edition stated* `qr_filter` *with that arithmetic as a hypothesis on abstract residues, which left it contentless); Lemma 15.1 of the full version is* `Frame.csum_eq_mul_recipSum` *for its first half and* `symbolfact` *for the Jacobi factorisation; and the reduction mechanism behind Proposition 6.10, Rigidity and Lemma 5.1 alike is isolated once as* `dvd_sum_erase_iff` (`lean/Erdos307/NormalForm.lean`)*.*

Proof. $\sigma(a) = a'/a$ and $\sigma(b) = b'/b$, both in lowest terms since $\gcd(m, m') = 1$ for squarefree m. Then $\sigma(a)\sigma(b) = 1$ reads $a'b' = ab$. From $\gcd(a', a) = 1$ and $a' \mid ab$ we get $a' \mid b$, say $b = a'k$; symmetrically $b' \mid a$, say $a = b'j$. Substituting, $a'b' = ab = b'j \cdot a'k$, so $jk = 1$ and $j = k = 1$. □

Corollary 13.2 (The multi-slot relaxations are vacuous)**.** *There is no version of the construction in which two or more primes of Q may be chosen freely. In particular one cannot fix a base $B \subset Q$ and solve $1/q_1 + 1/q_2 = r/s$ for the remaining two primes by ranging over the $\tau(s^2)$ factorisations $(rq_1 - s)(rq_2 - s) = s^2$: by Proposition 13.1 the pair (q_1, q_2) is already forced to be the corresponding part of the factorisation of a'. The exponential parametrisation of Remark* 4.21 *is therefore intrinsic, not an artefact of the one-free-prime ansatz, and no amount of extra slots converts it into a Bunyakovsky one. What does dispose of the exponential growth is not extra slots but Proposition 4.16, which bounds the tail prime outright and so replaces the orbit by a finite divisor search.*

The second closure is not new here, but it belongs in this list because it is what forces the whole existence route to be analytic. Theorem 6.1 of Section 6 shows that over $\mathbb{F}_q[t]$ the arithmetic derivative satisfies $\deg f' \leq \deg f - 1$, hence has no two–cycles and no nonzero fixed points, and Corollary 6.2 concludes that the function–field Erdős–Barbeau equation is insoluble outright. (An independent brute-force recheck, 8192 squarefree monics over $\mathbb{F}_2$ to degree 13 and 6561 over $\mathbb{F}_3$ to degree 8, 0 violations and 0 two–cycles: `code/ff_nocycle.rs`.) What is added here is not that statement but its consequence for *method*, recorded next.

Remark 13.3 (Why the existence route cannot be algebraic)**.** The mechanism of Theorem 6.1 is the ultrametric inequality. The cycle equation is $\sigma(a)\sigma(b) = 1$, so one of the two masses must reach 1; over $K[t]$ the mass satisfies $|\sigma(f)| = \max_i |1/\pi_i| < 1$ by ultrametricity and can never do so, whereas over $\mathbb{Z}$ the archimedean absolute value permits $\sum_{p|n} 1/p > 1$. So #307 is possible at all only because $|\cdot|_\infty$ is not ultrametric.

This has a consequence for method. Any argument for *existence* that is purely algebraic, in the sense that it applies verbatim with $\mathbb{Z}$ replaced by $K[t]$, would prove a statement that Theorem 6.1 refutes; hence no such argument exists. Together with Proposition 8.4 (no polynomial parametrisation) and Proposition 13.11 of the full version (no place of $\mathbb{Q}$ carries the descent) this closes the algebraic and geometric toolkit on *both* routes: the negative route has no place to descend at, and the positive route has no place to construct at. What remains for A10 is archimedean and analytic, which is precisely the regime in which lower bounds for primes in sparse sequences are unavailable. That is a sharper statement of the obstruction than Remark 4.21 gives, and it is the reason the present work does not resolve the existence direction.

There is a third closure, and it removes the one concrete attack that survives the other two. The 34 immune families of Remark 4.10 are precisely the tail families the congruence campaign cannot kill, so each is a live candidate and the obvious move is to test them one at a time. That move is not available, and the reason is quantitative rather than a want of effort.

Proposition 13.4 (The immune families cannot be tested)**.** *Let S be one of the* 34 *immune bases. Deciding whether the tail family $S \cup \{q\}$ contains a Pythagorean pair is, by Remark* 4.21*, deciding whether $B_S x^2 - A_S y^2 = -4D_S^2$ has an integral solution with $q = (x^2 - D_S)/A_S$ prime. Neither of the two routes to that decision is within reach.*

(i) Global route. *Producing a solution requires the fundamental unit, equivalently the class group, of the real quadratic order of discriminant $4A_SB_S$. On the first immune base that discriminant has* 225 *digits and is a nonsquare, so the subexponential index-calculus cost $L_\Delta(1/2) = \exp\big(\sqrt{\log \Delta \, \log\log \Delta}\big)$ evaluates to $10^{24.7}$ operations.*

(ii) Parametric route. *Since A_S is prime and $(D_S \mid A_S) = +1$, a square root $x_0^2 \equiv D_S \pmod{A_S}$ exists and the family is genuinely parametrised by $x = x_0 + tA_S$, with $q(t) = \big((x_0 + tA_S)^2 - D_S\big)/A_S$. But a solution additionally needs $B_S q(t) + D_S$ to be a square, and that quantity has 223 digits already at $t = 0$, so the heuristic hit rate is 10^{-112} per trial. Checked at $t = 0, 1, 2, 3$: not a square* (`code/a10_pell_feasibility.gp`).

There is a third route, which is to decline the computation rather than perform it, and it is worth costing because it is the natural thing to try next and because nothing structural forbids it. The size of the first candidate is governed by the regulator: $q_1 \approx \varepsilon^2/A_S$ with $R = \log\varepsilon$, so a base whose regulator happens to be near its minimum $R_{\min} \approx \frac{1}{2}\log\Delta$ would give q_1 of about 10^{113}, which an Atkin–Morain test settles in minutes. The obstruction is not that such bases are useless but that they are rare. Near-minimal regulator means the fundamental solution has $v = 1$, that is $\Delta = u^2 - 4$, and the $\Delta \le X$ of that shape number about $\sqrt{X}$; so the fraction of discriminants with a testable regulator is about $\Delta^{-1/2} = 10^{-112.5}$, against a typical $R \approx \sqrt{\Delta} \approx 10^{113}$ which puts q_1 at $\exp(10^{112})$ *and out of reach of notation, let alone computation. Since $\Delta = 4A_S B_S$ is determined by the base and not chosen, one may only sample: the 34 immune bases give an expected 10^{-111} hits and all* 49,961 *give 10^{-108}, against the $10^{112.5}$ bases one expected hit would need. The shortfall against the immune family is 10^{111}. So base selection does not circumvent the two routes above; the construction is one fortunate discriminant away from being decidable, and the fortune required is $10^{112.5}$. The estimate is* HEURISTIC, *being a density count for the regulator.*

One hope survives all of this on its face: a lower bound for *some term of some sequence in a family* is formally weaker than one for a single sequence, and Proposition 4.5 supplies 49,961 of them. It fails, and it fails for a reason no enlargement of the family can repair.

Proposition 13.5 (Multiplicity cannot help)**.** *Every known lower-bound method for primes in a set requires that set to have counting function at least X^θ for some fixed $\theta > 0$. The union over all* 49,961 *bases of the terms of their sequences below X has counting function $C\log X$ with $C = 49{,}961/\log\varepsilon$, since each sequence grows like ε^n and so contributes only $O(\log X)$ terms. Hence*

$$\frac{\log(\textit{union below } X)}{\log X} = \frac{\log\big(C\log X\big)}{\log X} \longrightarrow 0,$$

and the family size C is powerless because it sits inside a logarithm.

Measurement. On an immune base ε has 224 digits, so at $X = 10^{500}$ each sequence contributes 2.2 terms and the whole family contributes $10^{5.05}$, a density exponent of 0.0101. Reaching $\theta = \frac{1}{2}$ at that X would need $10^{249.7}$ bases against the $10^{4.7}$ available, a shortfall of 10^{245}. Even the absolute ceiling on the family at *every* level combined, all $2^{139} = 10^{41.8}$ subsets of the primes below 800, falls short by $10^{207.8}$ (`code/a10_multiplicity.gp`). □

Each of the four blocks was then attacked on its own terms. None fell; each is now located more precisely, and two further routes are closed.

Proposition 13.6 (The number–field barrier transfer is FALSE)**.** *Theorem 6.1 kills the algebraic route by ultrametricity, which suggests replacing $K[t]$ by a number ring $\mathcal{O}_K$, where the absolute value is archimedean and the obstruction disappears. It does disappear. Earlier versions of this paper asserted that nothing is gained, namely that for every number field K the least $\prod N(\pi)$ over prime ideals with $\sum 1/N(\pi) \ge 2$ is at least that of $\mathbb{Q}$.* That assertion is false. *It is withdrawn, and what replaces it is the following.*

(i) The true infimum is 16. *In the biquadratic field $K = \mathbb{Q}(\sqrt{17}, \sqrt{33})$, of discriminant* 314,721, *the prime* 2 *splits completely: there are four prime ideals of norm* 2*. They carry mass* $4 \cdot \frac{1}{2} = 2$ *exactly, and their product of norms is* $2^4 = 16$*. Conversely* 2 *is the least possible ideal norm, so mass* 2 *needs at least four ideals of norm* 2 *and any admissible family has product at least* 16*; a field can carry at most* $[K : \mathbb{Q}]$ *ideals of norm* 2*, so degree* 4 *is also minimal. Hence* 16 *is the exact infimum over all number fields, against the claimed* $\prod_{i \le 59} p_i > 8.77 \times 10^{112}$*: the withdrawn statement was wrong by a factor* 10^{112}*.*

(ii) Where the argument broke. *The Chebotarev computation is correct as a statement about* density*: in a degree-d Galois field the completely split primes have density* $1/d$*, so they carry mass* $d \cdot (1/d) \log\log Y = \log\log Y$*, exactly as over* $\mathbb{Q}$*, while each costs* $d \log p$*, and inert primes contribute* $1/p^d$*, that is, nothing. The error was to read a statement about the tail as a statement about the minimum. The barrier consumes only* finitely many *primes, and for any finite set of small primes one may choose K in which all of them split; Chebotarev constrains how often splitting happens, not which particular primes split.*

(iii) What survives, and what closes the route instead. *For a* fixed *field the measurement stands:* $\mathbb{Q}$ *needs* 59 *primes and* $10^{112.9}$*, whereas* $\mathbb{Q}(i)$ *needs* 226 *prime ideals and* $10^{735.1}$ (`code/a10_fieldbarrier.gp`). *What does not survive is the use of this proposition as a* block*. The route is nevertheless closed, and by a sounder argument, which is Proposition 13.7 below: the ideal–theoretic #307 is* trivially soluble*, so it carries no information about the rational problem.*

Proposition 13.7 (The difficulty of #307 is distinctness, not mass)**.** *Allow P and Q to be finite* multisets *of primes rather than sets. Then the #307 equation is soluble with four primes in total:* $\left(\frac{1}{2} + \frac{1}{2}\right)\left(\frac{1}{2} + \frac{1}{2}\right) = 1$*. With P, Q genuine sets the same equation forces* $|P \cup Q| \ge 59$ *and* $\prod > 10^{112.9}$ *by Theorem 4.3. The barrier is therefore a statement about the injectivity of* $p \mapsto 1/p$ *on the primes, not about the arithmetic of the mass condition.*

Two consequences, and the second is the operative one.

(i) The ideal–theoretic problem is trivial. *In* $K = \mathbb{Q}(\sqrt{17}, \sqrt{33})$ *the prime* 2 *splits completely into four pairwise coprime prime ideals of norm* 2*; taking* $P = \{\pi_1, \pi_2\}$ *and* $Q = \{\pi_3, \pi_4\}$ *gives* $\sigma(P) = \sigma(Q) = 1$ *and* $|P \cup Q| = 4$ (`code/ideal307.gp`)*. Nor does the derivative transfer:* $\pi_1 + \pi_2$ *is the unit ideal, so* a' *collapses. This is the correct reason the number–field route is closed, and it explains why refuting the previous proposition reopened nothing: the transferred problem was never the same problem.*

(ii) A no–go for transfers generally. *Any setting admitting two distinct objects of the same norm* $n \ge 2$ *on each side solves the mass equation at size four, whatever the setting is. So a dictionary permitting repeated denominators* dissolves *#307 rather than relocating its difficulty, and can carry no information about it. This prunes a class of attacks in one line, and it is the test any future transfer should face first. Corollary 4.6 passes it only because pairwise–coprime integers still have distinct least prime factors, which is distinctness re-entering through the back door.*

Formalised. *`Erdos307.multiset_solution` is the four–element witness, `distinctness_carries_the_barrier` the contrast against `card_ge_59_of_recipSum_ge_two`, stated as a conjunction so the two halves cannot drift apart, and `transfer_dissolves` the general no–go* (`lean/Erdos307/Distinctness.lean`)*.*

Remark 13.8 (The three other blocks, attacked)**.** (i) Against Proposition 13.1. The rigidity $\prod Q = a'$ would dissolve if disjointness were an assumption one could relax. It is not: $P \cap Q = \varnothing$ is *derived* in Theorem 2.3 by the Israel mechanism, $r \in P \cap Q$ forcing $v_r(st) = -2 \neq 0$. There is nothing to relax.

(ii) Against Proposition 13.4. Homogenised, $B_S x^2 - A_S y^2 = -4D_S^2$ is the ternary form $B_S x^2 - A_S y^2 + 4D_S^2 z^2$, a *conic*, and conics obey Hasse–Minkowski. Since Remark 4.7 already gives local solubility everywhere, *rational* solubility follows unconditionally, with no class group anywhere in the argument. This sharpens the block rather than removing it: what is missing is not solubility but *integrality*, and extracting integral points from the rational parametrisation returns the fundamental unit. Worse, the constructive step is itself blocked, since Simon's algorithm must factor the determinant $-4A_SB_SD_S^2$, whose cofactor after removing the known primes of D_S is the 224-digit balanced semiprime A_SB_S: the same cryptographic wall as Remark 4.10, reached from a new direction.

(iii) Against Proposition 13.5. The template that exploits a family without density is Heath-Brown's: Artin's conjecture holds for all but at most two primes, with no means of saying which. It does not transfer. That argument needs each failure to impose a *detectable* structure, which several failures then cannot share. Here a failing family would be one all of whose q_n are composite, and the only known mechanism for that is a covering system, which Proposition 4.15 proves does not exist. A failure would therefore be structureless, and structureless failures cannot be played against one another.

All of the above bounds the negative direction. The positive direction admits a quantitative shadow that had not been measured: a solution is exactly $r(a) := \sigma(a)\sigma(a') = 1$, so one may ask how close r actually comes to 1. The answer is instructive, and it disposes of the usual way such heuristics are calibrated.

Proposition 13.9 (The near-miss spectrum, and why the heuristic cannot be calibrated)**.** *Over all* 4,385,727 *integers* $a \le 10^7$ *with* a *and* a' *both squarefree, the maximum of* $r(a)$ *is* 0.5535*, attained at* $a =$ 3,972,254*; the frontier advances by decade as* 0.333, 0.345, 0.502, 0.502, 0.536, 0.554*. In particular* $r \ge 1$ *never occurs, the defect* $|a'' - a|$ *is of size* $a^{0.9}$ *or larger in all but three cases, and the smallest relative defect is* 0.4465*. Consequently the constant in the heuristic count of two-cycles,* $f(1)\log X$ *with* f *the density of* r *at* 1*, is* not measurable*, and the obstruction is two–sided.*

By Theorem 4.3 the region $r \ge 1$ *lies above* $10^{112.9}$*. The window immediately to the* left *of* 1 *is barred as well, by AM–GM rather than by the barrier hypothesis:* $\sigma(a)+\sigma(a') \ge 2\sqrt{\sigma(a)\sigma(a')} = 2\sqrt{r}$*, and* $2\sqrt{r} > T_{58} = 1.99874\ldots$ *as soon as* $r > (T_{58}/2)^2 = 0.998740\ldots$*, whence* $|P \cup Q| \ge 59$ *and the product again exceeds* $\Pi_{59} > 8.77 \times 10^{112}$*. So the whole neighbourhood* $r \in (0.998740, \infty)$ *of the value* 1 *lies above the barrier, not merely the half-line* $r \ge 1$*, and no sweep reaches any of it* (`code/a10_nearmiss.rs`, `code/a10_density1.rs`)*. Earlier editions barred only* $r \ge 1$*, which left the left window formally open and the unmeasurability argument one–sided.*

Proposition 13.10 (An effective approximation theorem, with a certified witness)**.** *Bado's Theorem 10.2 [20] asserts that for every* Y *and every* $\varepsilon > 0$ *there are disjoint finite sets of primes* P, Q*, all greater than* Y*, with* $0 \le \sigma(P)\sigma(Q) - 1 < \varepsilon$*. The proof is a greedy accumulation and is ineffective. Made effective it is expensive: to land in* $[T, T+\eta)$ *the accumulation must start above* $\approx 1/\eta$*, and Mertens gives* $\sum_{Y_1 < p \le W} 1/p \approx \log\log W - \log\log Y_1 = T$*, so* $W = Y_1^{e^T}$*; two stages with* $T \approx 1$ *put the largest prime and the cardinality both at* $\approx \varepsilon^{-e^2} = \varepsilon^{-7.389\ldots}$*.*

A Newton ladder does the same work with a handful of primes. Take $Q = \{2, 3, 5\}$*, so* $\sigma(Q) = 31/30$ *and* P *must approach* $30/31$*; from a pool, repeat* $p \leftarrow$ *the least prime* $\ge 1/d$ *and* $d \leftarrow d - 1/p$*, where* d *is the running deficit. Each rung squares it,* $d_{\text{new}} = d - 1/p \approx t\, d^2$ *with* t *the local prime gap, so the largest prime is* $\approx \varepsilon^{-1/2}$ *and the number of rungs is* $O(\log\log(1/\varepsilon))$*. Taking the pool to be the primes* $7 \le p \le 271$ *and eleven rungs gives* $|P| = 66$*,* $|Q| = 3$*, largest prime of* 2,010 *digits, and, in exact rational arithmetic,*

$$\big|\sigma(P)\sigma(Q) - 1\big| \;=\; 4.98523\ldots \times 10^{-4016},$$

a rational whose numerator has 129 *digits and whose denominator has* 4,145*. Every rung prime is* proved *prime by* APR–CL*, not merely a pseudoprime* (`code/nearmiss_effective.gp`)*. The certificate is tiered, and a reader may stop where they like:*

rungs	9	10	11
largest prime (digits)	505	1,007	2,010
$\lvert\sigma(P)\sigma(Q)-1\rvert$	9.57×10^{-1007}	2×10^{-2010}	4.99×10^{-4016}
cumulative certification	< 10 *s*	≈ 110 *s*	≈ 27 *min*

The gap is not marginal. At $\varepsilon = 10^{-13}$ the greedy needs a largest prime near 10^{94} and some 10^{92} primes; the ladder needs a largest prime near 10^7 and three rungs. The ladder is exponentially better in the size of the largest prime and doubly exponentially better in the count, and the reason is visible: greedy convergence is linear in the number of primes, Newton convergence is quadratic. Truncating gives certificates of any size — five rungs and a 34–digit largest prime already give 6.07×10^{-66}, and seven rungs give 3.22×10^{-254}. The witness above approaches 1 from below; taking the last rung's prime just under $1/d$ instead overshoots, so either side is available and the statement of Theorem 10.2 as printed is met.

What this is not: progress towards a solution. A free pair of disjoint prime sets carries none of the rigidity an exact hit requires — $\sigma(P)\sigma(Q)=1$ forces $\prod P = (\prod Q)'$ and $\prod Q = (\prod P)'$, and nothing in the ladder produces that — and by Proposition 13.9 the size of $|\sigma(P)\sigma(Q)-1|$ does not measure progress towards the integer identity at all. The value of the construction is that it makes an existence theorem effective and exhibits a witness anyone can check.

Proposition 13.11 (The near–miss frontier obeys a double–logarithmic law)**.** *For $X \geq 2$ put $R(X) = \max\{r(a) : a \leq X$ squarefree$\}$, where $r(a) = \sigma(a)\sigma(a')$. Let $\omega_{\max}(X)$ be the largest k with $\Pi_k \leq X$, let $\sigma_{\max}(X) = T_{\omega_{\max}(X)}$, and let $n(X)$ be the largest n with $\Pi_n \leq X^2\sigma_{\max}(X)$. Then*

$$R(X) \;\leq\; \left(\tfrac{1}{2}\,T_{n(X)}\right)^2.$$

Proof. Rigidity gives $\gcd(a,a')=1$, so $U = \operatorname{supp}(a)\sqcup\operatorname{supp}(a')$ has $|U| = \omega(a)+\omega(a')$ distinct primes and $\prod_{p\in U} p \mid aa'$. By AM–GM, $\sum_{p\in U} 1/p = \sigma(a)+\sigma(a') \geq 2\sqrt{\sigma(a)\sigma(a')} = 2\sqrt{r}$, while the same sum is at most $T_{|U|}$ by the extremality of the smallest primes; hence $r \leq (T_{|U|}/2)^2$. For the size, $\Pi_{|U|} \leq \prod_{p\in U} p \leq aa' = a^2\sigma(a) \leq X^2\sigma_{\max}(X)$, so $|U| \leq n(X)$, and T is increasing. $\square$

The bound is unconditional, needs no squarefreeness of a', and is close: against the measured frontier of Proposition 13.9 it reads

X	10^2	10^3	10^4	10^5	10^6	10^7
bound	0.4014	0.4920	0.5296	0.5879	0.6129	0.6342
$\max r$ observed	0.333	0.345	0.502	0.502	0.536	0.554

(`code/frontier_bound.gp`). Two consequences. First, since $T_{n(X)} = \log\log\bigl(X^2\sigma_{\max}\bigr) + M + o(1)$ by Mertens, the frontier can climb only like

$$R(X) \;\leq\; \tfrac{1}{4}\bigl(\log\log X^2 + M + o(1)\bigr)^2,$$

a *doubly logarithmic* law: this is why the measured frontier crawls, and why an extrapolation of its per–decade advance is worthless. Second, the bound reaches 1 exactly when $T_n \geq 2$, that is $n \geq 59$, that is $X^2\sigma_{\max} \geq \Pi_{59}$, giving $X \geq 2.0942\times 10^{56}$ — the constant of Theorem 4.3, recovered here from the frontier side rather than from the cycle equations. The two derivations are independent and agree to all printed digits.

Corollary 13.12 (The cost of a near–miss; the barrier as a function of r)**.** *For* $0 < r \le 1$ *put* $n(r) = \min\{n : T_n \ge 2\sqrt{r}\}$. *Every squarefree* a *with* $r(a) \ge r$ *has* $|U| \ge n(r)$, *hence* $aa' \ge \Pi_{n(r)}$ *and*

$$\min(a, a') \;\ge\; \sqrt{\Pi_{n(r)}/T_{n(r)}}.$$

Proof. The first display of the proof of Proposition 13.11 gives $2\sqrt{r} \le \sum_{p\in U} 1/p \le T_{|U|}$, so $|U| \ge n(r)$; the rest is Theorem 4.3 verbatim with 59 replaced by $n(r)$. □

At $r = 1$ this returns $n = 59$ and 2.0929×10^{56}, so Theorem 4.3 is the endpoint of a one–parameter family rather than an isolated statement. The family is what makes the crawl quantitative (`code/nearmiss_cost.gp`):

r	0.5	0.554	0.6	0.7	0.8	0.9	0.99	1
$n(r)$	8	9	11	16	24	37	56	59
$\min(a, a') \ge$	$2.6{\cdot}10^3$	$1.2{\cdot}10^4$	$3.6{\cdot}10^5$	$4.4{\cdot}10^9$	$1.2{\cdot}10^{17}$	$4.3{\cdot}10^{30}$	$4.7{\cdot}10^{52}$	$2.1{\cdot}10^{56}$

Each increment in r is paid for exponentially in the product, which is the same double–logarithmic law seen from the other side: r enters the bound only through $2\sqrt{r}$, and T_n grows like $\log\log p_n$. It also explains the shape of the swept data. The record $r = 0.5535$ needs $\min(a, a') \ge 1.2 \times 10^4$ and was found at $a = 3{,}972{,}254$; the next decade of r costs four decades of a, and by $r = 0.7$ the requirement has passed 10^9, beyond any sweep that has been run.

The region is not unreachable, only unsweepable. The witness n_0 of Theorem 13.24 has n_0 and n_0' both squarefree and $r(n_0) = \sigma(n_0)\sigma(n_0') = 1.03944\ldots$, so $n_0'' > n_0$: the value 1 that a solution must hit is interior to the range of r, not extremal, and no inequality on r alone can settle #307. The same construction approaches 1 from below. Tuning $\sigma(a)$ against the forced $\sigma(a') = \frac{31}{30} + \varepsilon$ gives

$$a = 3359 \prod_{\substack{7\le p\le 269\\ p\notin\{229,271\}}} p, \qquad a' = 2\cdot 3\cdot 5\cdot 383\cdot 113717\cdot P_{99},$$

with P_{99} prime, both a and a' squarefree, 108 digits, and

$$r(a) = 0.99207796\ldots, \qquad |r(a) - 1| = 7.9220 \times 10^{-3},$$

against the smallest relative defect 0.4465 over the swept range (`code/nearmiss_tune.gp`).

The same tuning with its acceptance window mirrored, $\sigma(a)$ just *above* 30/31 rather than just below, reaches 1 from the other side and closer. With

$$a = 3301 \prod_{\substack{7\le p\le 293\\ p\notin\{137,263\}}} p, \qquad a' = 2\cdot 3\cdot 5\cdot P_{117},$$

where P_{117} is a 117–digit prime, one has a of 118 digits and, exactly,

$$r(a) = 1.001014244\ldots, \qquad |r(a) - 1| = 1.0142 \times 10^{-3}, \qquad a'' > a.$$

This is exact rather than a lower bound, since a' factors completely: the cofactor after trial division is prime, so $\sigma(a')$ is known and not merely bounded below. It improves the record on both sides at once, against 7.9220×10^{-3} from below and 3.944×10^{-2} from above, the latter being $r(n_0)$ of Theorem 13.24 (`code/nearmiss_above.gp`, re–checked independently in `code/nearmiss_above_verify.gp`).

That record is not the natural limit, and seeing why identifies what the near–miss problem actually is. In every witness of this shape $a' = 2 \cdot 3 \cdot 5 \cdot P$ with P prime, so $\sigma(a') = \frac{31}{30} + P^{-1}$ with P^{-1} of size 10^{-118}, and therefore

$$|r(a) - 1| = \tfrac{31}{30}\left|\sigma(a) - \tfrac{30}{31}\right|$$

to any precision that matters. The quantity a' has dropped out. What remains is the Diophantine question of how closely a sum of *distinct prime reciprocals* approaches 30/31 from above, and the 10^{-3} above is only the granularity of adjusting by a single prime below 4000.

Closing the deficit in stages removes that ceiling. With $d = \frac{30}{31} - \sigma(S)$ for S the primes in $[7, 271]$, take p_1 the least prime exceeding $1/d$, then p_2 the least exceeding $1/(d - p_1^{-1})$, and finally p_3 the greatest prime below the reciprocal of what remains; the overshoot is then of order $\mathrm{gap}(p_3)/p_3^2$. Each rung squares the precision. Sweeping the choice at the first two rungs and retaining those that satisfy the forcing congruence and have $a' = 2 \cdot 3 \cdot 5 \cdot P$ gives, with $(p_1, p_2, p_3) = (569, 1949, 15461)$,

$$a = 569 \cdot 1949 \cdot 15461 \prod_{7 \le p \le 271} p \quad (58 \text{ primes}, \ 120 \text{ digits}), \qquad a' = 2 \cdot 3 \cdot 5 \cdot P_{118},$$

and, exactly,

$$r(a) = 1 + 6.278713 \times 10^{-9}, \qquad a'' > a.$$

This is 1.6×10^5 times closer to 1 than the preceding record.

That record is on the wrong side of 1, and identifying the family says why. Every witness here has $a' = 2 \cdot 3 \cdot 5 \cdot P$ with P prime, which is exactly the stratum $M = 30$ of Proposition 7.3: with $M' = 31$ the equation $MN - M'N' = M^2$ reads $30N - 31N' = 900$, and the proposition's $N = M + M'p$, $N' = Mp$ become $a = 31P + 30$, $a' = 30P$. Two identities follow, both exact: $a'' = (30P)' = 31P + 30$, so $r = a''/a = (31P + 30)/a$, and a solution of the stratum is precisely $30a - 31a' = 900$. Hence

$$\sigma(a) = \frac{30P}{31P + 30} = \frac{30}{31} - \frac{900}{31a} \ < \ \frac{30}{31}$$

for every solution of this stratum, strictly. Approaches to 1 from *above*, which is what the 6.28×10^{-9} above and the 3.76×10^{-9} of a wider sweep are, therefore lie on a side the stratum cannot populate. They stand as records for the spectrum of r and cannot converge on a solution.

Aimed below instead, where the stratum's solutions must lie, the same ladder gives

$$a = 431 \cdot 68111 \cdot 2792213 \prod_{7 \le p \le 271} p \quad (58 \text{ primes}, \ 123 \text{ digits}), \qquad a' = 2 \cdot 3 \cdot 5 \cdot P_{122},$$

and, exactly,

$$r(a) = 1 - 1.654663 \times 10^{-13}, \qquad \sigma(a) < \tfrac{30}{31}, \qquad a'' < a,$$

which is 4.8×10^{10} times closer to 1 than the previous record from below. For this witness $30a - 31a'$ is a 112–digit number where a solution needs 900, which is the honest measure of the distance remaining: the ratio is close, the integer is not.

Neither construction is exhausted. Over the same sweeps the smallest *available* deficit was 2.043×10^{-18} from below and 1.209×10^{-14} from above; what limits the reported values is the rate at which a' takes the form $2 \cdot 3 \cdot 5 \cdot P$, not the tuning (`code/nearmiss_ladder.gp`, `code/nearmiss_below.gp`, re–checked in `code/nearmiss_ladder_verify.gp` and `code/nearmiss_below_verify.g`

We stop here, and the reason is a measurement rather than a budget. Between the two from–below witnesses the ratio improved by three orders of magnitude while $30a - 31a'$ fell from 114 digits to 112, against the 3 a solution requires. Exhausting the available deficit would gain perhaps five more digits of the 112. The near–miss frontier measures $|r-1|$, a ratio; membership of the stratum is the integer identity $30a - 31a' = 900$, and the two are not commensurable at this scale. Driving the ratio further is therefore not evidence about the integer, and the line is closed as a line of attack. What survives it is structural and is stated above: the family is the $M = 30$ stratum of Proposition 7.3, its solutions satisfy $\sigma(a) < 30/31$ strictly, and approaches from above cannot reach one. Those three facts are machine–checked in `lean/Erdos307/StratumM30.lean`.

The constant 30/31 is not special. In the normal form of Proposition 7.3 a solution of the M–stratum is $a = Mp$, $b = N = M + M'p$, so $\sigma(a) = \sigma(M) + 1/p$ and, by the identity $(\sigma(M) + 1/p)\sigma(N) = 1$ of part (b), $\sigma(b) = 1/\sigma(a) < 1/\sigma(M) = M/M'$, with defect exactly $M^2/(M'(M + M'p))$. Hence *every* stratum is one–sided: for no squarefree M does the half–space $\sigma(b) \geq M/M'$ contain a solution, and $M = 30$ recovers 30/31. This is a corollary of Proposition 7.3(b) rather than a new result, but it is the constraint that governs search direction, so it is machine–checked in `lean/Erdos307/StratumGeneral.lean`. The consequence for the near–miss programme is that the mirrored tuning of the previous paragraph is not merely unlucky at $M = 30$; the corresponding tuning is empty in every stratum. Consistently with the two–sided statement above, a lies past the barrier: $118 > 113$ digits, and r sits inside the barred neighbourhood $(0.998740, \infty)$.

What remains unmeasurable is the density of r at 1, since constructed points do not estimate a density, and that is what the heuristic constant needs. The frontier reported above is therefore superseded as a record and unchanged as an argument.

The situation is not the usual one. Heuristics of Bateman–Horn or Hardy–Littlewood type are normally checked against data, and the agreement is what justifies confidence in them. Here the prediction that solutions exist rests on a constant that cannot be estimated by sampling, because the barrier and the sampleable range are disjoint. The expected-count heuristic for #307 is therefore unverifiable in the usual way.

It is not, however, beyond computation, and the reason is that the constant is the barrier.

Proposition 13.13 (The heuristic constant is the barrier's own probability)**.** *Model a random squarefree a by $\mathbb{P}(p \mid a) = 1/(p+1)$, and a' by the coprimality model that Proposition 13.17 validates, so that $\mathbb{P}(q \mid a') = 1/q$ for $q \nmid a$. Then each prime independently lies in a, in a', or in neither, with probabilities $\frac{1}{p+1}, \frac{1}{p+1}, 1 - \frac{2}{p+1}$, and*

$$\sigma(a) + \sigma(a') = \sum_{p \mid aa'} \frac{1}{p}, \qquad \mathbb{P}(p \mid aa') = \frac{2}{p+1}.$$

Since a two-cycle forces $\sigma(a)\sigma(a') = 1$ and hence $\sigma(a) + \sigma(a') \geq 2$ by AM–GM*, the density f of $r = \sigma(a)\sigma(a')$ is supported in the event $B : \sigma(a) + \sigma(a') \geq 2$, so $f(1) \leq C\,\mathbb{P}(B)$ with C the conditional density of r at 1 given B. Numerically*

$$\mathbb{P}(B) = 2.74 \times 10^{-90}.$$

The point is the order and not the digits. $\mathbb{P}(B)$ is a *tail* probability of a one-dimensional sum, not a density, which is what makes it stable: it moves from 4.09 to 3.01 to 2.74×10^{-90} as the grid is refined from 2×10^4 to 8×10^4 bins and the prime range from 10^4 to 10^6, while a direct attempt at $f(1)$ on a two-dimensional grid moves by four orders of magnitude per refinement and converges to nothing. Two checks support it. The same computation with $\mathbb{P}(p \mid n) = 1/p$ returns $\mathbb{P}(\sigma(n) \geq 1) = 0.0420$, reproducing the measured density of prime-abundant integers to four places; and the single configuration in which aa' absorbs exactly the first 59 primes has probability $\prod_{p\le 277} 2/(p+1) = 10^{-96.0}$, below $\mathbb{P}(B)$ by the factor one expects from the many other prime sets of mass 2 (`code/barrier_probability.rs`).

One more thing follows, and it disarms the near-miss spectrum entirely.

Proposition 13.14 (The near-miss infimum is zero, so closeness is not evidence)**.**

$$\inf\Big\{ \big|\sigma(P)\sigma(Q) - 1\big| \ : \ P, Q \text{ finite disjoint prime sets} \Big\} = 0,$$

and #307 asks exactly whether the infimum is attained.

Proof. Split the primes into two infinite classes, say by parity of index. Each class has terms tending to 0 and divergent reciprocal sum, so by the classical subseries theorem each carries a subseries summing to exactly 1. This produces disjoint *infinite* P_∞, Q_∞ with $\sigma(P_\infty) = \sigma(Q_\infty) = 1$. Truncating both at height y gives finite disjoint sets with $\sigma(P_y)\sigma(Q_y) \to 1$. □

The analytic content is formalised, and in a slightly stronger form than the proof above uses. For the infimum one does not need the subseries theorem, only that each mass can be placed in $(1-\delta, 1)$: `Erdos307.exists_block_sum_near` takes the first consecutive block whose sum exceeds $T - \varepsilon$ and observes that it overshoots by less than one term, and `infimum_zero` assembles two such blocks into a product within ε of 1. Both carry 0 `sorry` and the three standard axioms (`lean/Erdos307/InfZero.lean`). The arithmetic input, that the primes split into two infinite classes each with reciprocal terms tending to 0 and unbounded partial sums, is cited rather than reproved, as in Theorem 13.20.

The greedy form of that argument reaches $|\sigma(P)\sigma(Q)-1| = 6.50\times 10^{-9}$ with 144 primes, against 0.4465 for the swept frontier of Proposition 13.9 (`code/infimum_zero.gp`). So the analytic half of #307 is empty: the mass equation can be approximated as closely as one likes, and no near-miss, however good, is evidence that a solution exists.

What the near-misses do measure is a trade-off, and it is the honest picture of the obstruction. A solution must kill two defects at once, the mass defect $|\sigma(P)\sigma(Q) - 1|$ and the size defect $\log_{10}\big(\prod Q/\sigma(P)\prod P\big)$, the latter vanishing exactly when $\prod Q = N_P$, which is Theorem 2.3. Each is reachable alone and neither construction moves the other:

construction	mass defect	size defect
greedy, Q free	6.50×10^{-9}	$10^{341.3}$
$Q = \text{primes}(a')$, `nearmiss_tune.gp`	7.92×10^{-3}	0

Fixing Q as the prime set of a' makes the size defect vanish identically and leaves the mass defect at 10^{-3}; choosing Q freely drives the mass defect to 10^{-9} and costs 341 orders of magnitude in size. #307 is the demand that both columns read zero simultaneously, and nothing in this paper moves them together.

Remark 13.15 (What the calibrated heuristic says, and it is not encouraging)**.** With $\mathbb{E}(X) \sim f(1)\,\delta\log X$ and $\delta = 0.3721$, Proposition 13.13 gives an expected number of two-cycles below the proved barrier of about 10^{-88}, and $\mathbb{E}(X)$ does not reach 1 until X is of order $10^{4\times 10^{89}}$. So the heuristic favours YES only in the limiting sense that $\log X$ diverges; at every scale that any search or any barrier argument will reach, it predicts nothing. Two consequences, both negative and both worth stating. A proof of the negative direction would have to contradict a divergence, and a search for a solution is hopeless not merely at present but at any scale expressible in the notation of this paper. The constant is small for a reason that is not incidental: it *is* the barrier, seen as a probability rather than as a counting bound, which is the third independent appearance of the same Mertens threshold in this work.

The label is HEURISTIC and the model is doing the work. What is claimed is only that the constant, said above to be unavailable to sampling, is available to computation once the coprimality model of Proposition 13.17 is granted.

Remark 13.16 (What is left, on both sides)**.** The two directions can now be specified rather than merely called open.

Negative. Theorem 13.22 closes the certificate route on both branches, so a proof must count rather than certify; and by Remark 13.23 the count it must defeat diverges. A proof of the negative direction would therefore have to contradict a divergent expected count, which is to say that the direction is not merely unproved but probably false.

Positive. Three routes are available in principle, and two are closed quantitatively.

(a) *Exhibit a witness.* Excluded by Remark 13.15: the first is expected near $10^{4\times10^{89}}$.

(b) *Bound the count below by analytic means.* Excluded by size. At the proved barrier the main term $f(1)\delta \log X$ lies $10^{200.6}$ below the trivial count X, and the saving a method would need in order to keep the main term above its own error grows like X itself. Sieve and circle-method errors save a power of X or a power of $\log X$. So an analytic proof here would have to be *exact*, with error identically zero, rather than sharp.

(c) *Construct exactly.* This is the only branch alive, and it exists: the Pell orbit of Remark 4.21. It is alive but narrower than it was, and the narrowing should not be mistaken for proximity. Two cautions, both quantitative. The searches of Remark 4.18 came back empty over 3.99×10^{11} candidates, but empty was the *predicted* outcome: the expected count was 2.4×10^{-56}, so the computation confirms the heuristic model rather than deciding anything, and a confirmed model is not a decided problem. And by Proposition 4.19 those searches cannot be continued upward at all, since the admissible supports are infinite at every level past 59. The unconditional exclusions of this note stop where enumeration stops, which is now. Proposition 4.16 bounds the tail prime by $4N$, so the orbit contributes nothing past a bounded initial segment and the construction is a finite search over the divisors of $4D$ rather than a Lehmer primality question; Remark 4.18 then makes that finite search heuristically empty on all 34 immune families, by two independent estimates. What survives is the branch itself, not its expectation: the construction must now either come from a base outside the immune list or exploit the parametrisation $q = 2(N \pm k)/m^2$ directly.

The two exclusions interlock, and the interlock is the real statement. Exactness is what algebra supplies, and Theorem 6.1 excludes every argument that survives the passage to $K[t]$, where the conclusion is false. The Pell construction escapes that exclusion for one reason only: its final input is the primality of a determined term, and primality does not survive to $K[t]$. So a successful method must be *archimedean and exact at the same time*, a combination which is unusual but not unprecedented, continued fractions solving Pell being the standard instance, and Remark 4.21 is exactly the analogue here.

The labels differ across the three branches and the difference matters. The exclusions of (a) and (b) are HEURISTIC: both rest on the value of $f(1)$ from Proposition 13.13, hence on the coprimality model, and a reader who rejects that model recovers both branches. The interlock itself is not heuristic: Theorem 6.1 and Proposition 8.4 are PROVED, and so is the reduction of branch (c) to a primality question. So what follows is stated at the strength of its weakest link, which is the model.

What remains of #307 is therefore one question and not a programme: whether some base admits a Pythagorean tail prime, together with the sign selection $k = 0$, which Remark 4.21 shows is a choice between $\{0, -1\}$ rather than a coincidence of measure zero. By Proposition 8.4 the orbit parametrising the Pythagorean locus is forced to be exponential rather than polynomial. That was previously read as placing the question in the Lehmer class rather than the Schinzel class, hence open for every non-degenerate sequence; but by Proposition 4.16 a prime tail satisfies $q \leq 4N$, so the exponential orbit supplies no candidate past a bounded initial segment and the question is neither Lehmer nor Schinzel. It is a finite search over the divisors $m \mid 4D$ with $AB + Dm^2$ a square, which Remark 4.18 makes heuristically empty on all 34 immune families. The residue is thus smaller and better located than it was, and still undecided: the bound is proved, the emptiness is not, and no base outside the immune list has been examined at all. That is why nothing in this paper decides the problem.

The measurement does expose a sharp structural fact, which turns out to be exactly the barrier seen statistically.

Proposition 13.17 (The barrier is statistically complete)**.** $\mathbb{E}[\sigma(a')]$ *falls by an order of magnitude as* $\sigma(a)$ *grows, from* 0.2035 *unconditionally to* 0.0207 *on* $\sigma(a) \geq 1.3$*. The whole of this suppression is coprimality. Writing the prediction for an integer chosen at random subject to* $\gcd(n, a) = 1$*, namely* $\sum_{p \nmid a} p^{-2}$*, the ratio of measured to predicted is*

$$0.65, \quad 0.82, \quad 0.91, \quad 0.92, \quad 0.95, \quad 1.01$$

at $\sigma(a) \geq 0$, 0.5, 0.9, 1.0, 1.2, 1.3*: it converges to* 1 *precisely in the regime a solution would inhabit* (`code/a10_anticorr.rs`)*.*

So on the sampled range a' is mass-poor for one reason only, that it is coprime to a and a has absorbed the small primes, and there is no further effect to exploit. This is an independent confirmation of Proposition 15.25 of the full version: the barrier is not merely near-optimal as a bound, it already contains the entire statistical structure of the obstruction, and no refinement by correlation is available.

The range matters, and the statement is about averages. Every $a \leq 4 \times 10^8$ with $\sigma(a) > 1$ is even, so the measurement never sees an a that omits the small primes. Such a exist, since the prime reciprocals diverge, and for them a' is free to absorb 2, 3 and 5 and is not mass-poor at all: Theorem 13.24 exhibits one with $\sigma(a') = \frac{31}{30}$. Proposition 13.17 bounds an expectation on a range, and nothing here bounds a supremum off it.

A Lyapunov candidate for the negative direction

Theorem 6.1 rules out cycles over $\mathbb{F}_q[t]$ by exhibiting a function that provably decreases, namely the degree. Proposition 13.11 of the full version shows no *absolute value* on $\mathbb{Q}$ plays that role. It does not exclude functions that are not absolute values, and the following is the outcome of a search among such functions. It is a conjecture, not a theorem, and it is recorded because it is falsifiable and because the mechanism protecting it is one of the paper's own results.

Definition 13.18. For $\lambda > 0$ and squarefree n set $\delta_\lambda(n) = \log n + \lambda\, \sigma(n)$.

Proposition 13.19 (A decreasing δ_λ would settle #307 negatively)**.** *If for some* $\lambda > 0$ *one has* $\delta_\lambda(n') < \delta_\lambda(n)$ *for every squarefree* n *with* n' *squarefree, then the arithmetic derivative has no cycle of any length among such* n*, and in particular #307 has answer* NO*.*

Proof. A cycle would give $\delta_\lambda(n) < \delta_\lambda(n)$. □

Since $n' = n\sigma(n)$, the requirement is the pointwise inequality

$$\log \sigma(n) \ < \ \lambda\big(\sigma(n) - \sigma(n')\big). \tag{$*$}$$

Around a two-cycle the two instances of $(*)$ have increments summing to $\log(\sigma(a)\sigma(b)) = 0$, so they cannot both hold: the admissible range of λ closes exactly at a cycle. The content is therefore whether that range is non-empty.

Below the barrier the criterion is not conjectural: it holds with the explicit value $\lambda = \frac{1}{2}$, by an elementary inequality.

Theorem 13.20 (The half-mass Lyapunov function)**.** *Put $\delta(n) = \log n + \frac{1}{2}\sigma(n)$. Let n be squarefree with n' squarefree. If $\sigma(n\,n') < 2$ then $\delta(n') < \delta(n)$.*

Proof. By Theorem 2.3 n and n' are coprime, and both are squarefree, so $\sigma(n\,n') = \sigma(n) + \sigma(n')$. The hypothesis therefore gives $\sigma(n') < 2 - \sigma(n)$, hence

$$\tfrac{1}{2}\big(\sigma(n) - \sigma(n')\big) \ > \ \sigma(n) - 1 \ \geq \ \log \sigma(n),$$

the last step by $\log x \leq x - 1$. If $\sigma(n) = 1$ the two ends read 0 and the first inequality is still strict, since $\sigma(n) + \sigma(n') < 2$ forces $\sigma(n') < 1$. Now $n' = n\sigma(n)$ gives $\delta(n') - \delta(n) = \log \sigma(n) - \frac{1}{2}(\sigma(n) - \sigma(n')) < 0$. □

Verified over all 7,480,605 squarefree $n \leq 2 \times 10^7$ with n' squarefree lying in the region: 0 violations, least slack 0.242 (`code/invent_lyapunov.rs`). The analytic content is formalised: `Erdos307.log_lt_half_sub` proves $\log s < (s-t)/2$ for $s > 0$ and $s + t < 2$, `delta_decreasing` gives the increment form, and `two_le_add_of_mul_eq_one` is the AM–GM step of Corollary 13.21; all three carry 0 `sorry` and only the three standard axioms (`lean/Erdos307/HalfLyap.lean`). The arithmetic inputs, that n and n' are coprime and that $n' = n\sigma(n)$, are Theorem 2.3 and the definition, so the statement is VERIFIED modulo those.

Corollary 13.21 (A Lyapunov proof of the barrier)**.** *The arithmetic derivative has no cycle, of any length, all of whose consecutive pairs satisfy $\sigma(n\,n') < 2$. In particular a two-cycle $a' = b$, $b' = a$ with $a \neq b$ has $\sigma(ab) = \sigma(a) + \sigma(b) \geq 2\sqrt{\sigma(a)\sigma(b)} = 2$, with equality only if $\sigma(a) = \sigma(b) = 1$ and hence $a = b$; so $\sigma(ab) > 2$ and therefore $ab \geq \Pi_{59}$.*

Proof. A cycle would give $\delta(n) < \delta(n)$ by Theorem 13.20. The second statement is AM–GM together with the mass estimate of Theorem 4.3. □

There are exactly two ways to exploit the fact that increments cancel around a cycle, and the second is foreclosed. Around a two-cycle the two increments of any function sum to zero, so an obstruction must pin down either their *sign*, which is the Lyapunov route above, or their *value in a finite group*: if some φ satisfied $\varphi(n') - \varphi(n) \equiv c \pmod{m}$ for all n with $2c \not\equiv 0$, a two-cycle would give $0 \equiv 2c$. No such φ exists, and not for want of searching: Proposition 4.15 proves the cycle system soluble modulo every m, so any obstruction expressible as a congruence is empty a priori. Measured, the increments of $\omega(n)$, of $\#\{p \mid n : p \equiv 3\ (4)\}$, of n, and of $n + \omega(n)$ each hit every residue class modulo 2, 3 and 4 (`code/invent_lyapunov.rs`). The sign route is therefore the only one of the two available, which is another form of the thesis that the obstruction is ordered–archimedean rather than congruential (Remark 6.4).

The sign route is available but, as it turns out, not weaker than the problem. Both branches of the dichotomy can now be closed at once.

Theorem 13.22 (Both branches are closed)**.** *Let S be the set of squarefree $n \geq 2$ with n' squarefree.*

(1) (Finite-group branch.) Conjecture, *downgraded. The claim that there is no φ on S and modulus m with $\varphi(n') - \varphi(n) \equiv c \pmod{m}$ for all n and $2c \not\equiv 0 \pmod{m}$ is* not *established. Proposition 4.15 settles only the moduli it covers, $m \leq 210$, and the general statement is equivalent to the target rather than independent of it: on a functional graph with no periodic orbit one may build a constant-increment φ for any (m, c) by assigning φ along depth, so nonexistence for all m is equivalent to the existence of a periodic orbit. Branch (1) therefore has exactly the status of branch (2), which Proposition 13.28 already proves for the sign side. What is proved is the mechanism recorded after the proof.*

(2) (Sign branch, not weaker than the target.) *A δ on S with $\delta(n') < \delta(n)$ throughout exists if and only if $n \mapsto n'$ has no periodic orbit in S; that condition implies a negative answer to #307 and is not implied by it.*

Proof. (1) is not proved; see the statement. Proposition 4.15 makes the cycle system soluble modulo every $m \leq 210$, which does not reach the quantifier the branch needs. (2) is Proposition 13.28; a two-cycle is a periodic orbit, and periodic orbits of length other than two are not excluded by a negative answer to #307. □

The mechanism of branch (1) is formalised. `Erdos307.two_nsmul_eq_zero_of_period_two` proves that a constant increment c in any abelian group forces $2c = 0$ at a point of period two, `const_increment_iterate` gives the general $\varphi(D^{[k]}n) = \varphi(n) + kc$, `no_period_two_of_two_nsmul_ne_zero` is the certificate such a φ would supply, and `zmod_two_mul_eq_zero_of_period_two` states it over $\mathbb{Z}/m$; all with 0 `sorry` and on `propext` alone (`lean/Erdos307/Congruential.lean`). What is cited rather than formalised is the *emptiness* of the class, that no such φ with $2c \neq 0$ exists for the arithmetic derivative, which is Proposition 4.15 and is arithmetic. Branch (1) is therefore VERIFIED in its mechanism and PROVED in its emptiness, and the two should not be conflated.

The taxonomy that makes this a closure rather than two isolated facts is the displayed thesis above, that an invariant whose increments cancel around a cycle can be exploited only through their sign or their value in a finite group. That thesis is not a theorem, and an argument outside it is not excluded by anything here.

This is a statement about methods, not about #307, and it is the natural endpoint of the structural programme: Proposition 13.11 of the full version removes the absolute values, Theorem 6.1 removes the arguments that survive to $K[t]$, Proposition 4.15 removes the congruences, and Theorem 13.22 removes what is left of the pointwise route. A proof of the negative direction, if there is one, must be global: it must count, and not certify.

Remark 13.23 (What a counting proof would have to defeat)**.** The count it would have to defeat diverges. A two-cycle is exactly $r(a) = 1$ for $r(a) = \sigma(a)\sigma(a') = a''/a$, so if r has a limiting density f on the class of squarefree a with a' squarefree, then $\mathbb{P}(a'' = a) = \mathbb{P}\big(r \in [1, 1 + \frac{1}{a})\big) \sim f(1)/a$ and

$$\mathbb{E}\big(\#\{a \leq X : a'' = a\}\big) \ \sim\ f(1)\,\delta \log X, \qquad \delta = 0.3721\ldots,$$

δ being the measured density of the admissible class (`code/region_shape.rs`, 3,721,148 admissible $a \leq 10^7$). This diverges, so the heuristic predicts infinitely many two-cycles, and any proof of the negative direction must account for a divergent expected count producing no solutions at all.

The same formula locates them, and shows why the location is not knowable. Setting the expectation to 1 gives a first solution near $\exp\big(1/(\delta f(1))\big)$, which is exponential in $1/f(1)$: at $f(1) = 10^{-2}$ it is $10^{116.7}$, at $f(1) = 5 \times 10^{-3}$ it is $10^{233.4}$, at $f(1) = 10^{-3}$ it is 10^{1167}. Halving the constant squares the answer. Consistency with Theorem 4.3 needs $f(1) < 0.01034$, which is evidence of the softest kind, since an expectation near 1 with no solution observed is an ordinary event and not a contradiction. What Proposition 13.9 shows is that $f(1)$ is the one quantity here that cannot be sampled (`code/heuristic_location.gp`). That $f(1) > 0$ is not established either; the constructed witnesses of Theorem 13.24 and of the near-miss frontier show only that r takes values on both sides of 1 and within 8×10^{-3} of it.

Corollary 13.21 recovers the barrier by a route independent of the minimisation used in Theorem 4.3: no extremality argument, no enumeration, only $\log x \leq x - 1$ and the coprimality of Theorem 2.3. It is also stronger in one direction, since it excludes cycles of every length rather than two-cycles alone. What it does not do is decide #307: the region it covers is exactly the region where the barrier already says nothing can happen, and above Π_{59} the hypothesis $\sigma(n\,n') < 2$ fails, which is where any solution must live. The general conjecture is what remains.

Over all 153,397,754 such $n \leq 4 \times 10^8$, $(*)$ holds for every $\lambda \in (0.275438, 1.250828)$ and fails for no step (`code/lyap_battery.rs`). The interval narrows with the range, of width 1.654, 1.267, 1.237, 1.116, 1.092, 1 at cutoffs 10^4 through 4×10^8; the binding lower constraint is at $n = \Pi_9 = 223{,}092{,}870$. Six candidate functions were tested and five died, among them the control $\delta = \log n$, which fails first at $n = 30$ where $\sigma = 1.0333$; a battery that did not kill it would not be testing anything (`code/invent_lyapunov.rs`).

That interval is an artefact of the range swept. No λ works.

Theorem 13.24 (The criterion fails, for every λ)**.** *There is no real λ for which $(*)$ holds at every squarefree n with n' squarefree. Two witnesses suffice. Let*

$$n_0 = \prod_{\substack{7 \leq p \leq 373 \\ p \notin \{307, 317, 359\}}} p,$$

a squarefree integer with 68 prime factors and 142 decimal digits. Then $n_0' = 2 \cdot 3 \cdot 5 \cdot C$ with C a prime of 141 digits, so n_0' is squarefree and $\sigma(n_0') = \frac{31}{30}$ exactly, while $\sigma(n_0) = 1.0059067\ldots$. Since $\sigma(n_0) > 1$ and $\sigma(n_0') > \sigma(n_0)$,

$$\log \sigma(n_0) \;>\; 0 \;>\; \lambda\big(\sigma(n_0) - \sigma(n_0')\big) \qquad (\lambda > 0),$$

so $(*)$ fails at n_0 for every positive λ. At $n = 30$, with $30' = 31$ prime, $\sigma(30) = \frac{31}{30} > 1$ and $\sigma(30) - \sigma(31) = \frac{931}{930} > 0$, so $\log \sigma(30) > 0 \geq \lambda(\sigma(30) - \sigma(31))$ for every $\lambda \leq 0$. Together the two exclude every real λ.

Proof. Both statements are finite verifications in exact rational arithmetic. For n_0: $\sigma(n_0)$ and $\sigma(n_0')$ are computed as rationals, $\gcd(n_0, n_0') = 1$ and $\sigma(n_0) = n_0'/n_0$ confirm Theorem 2.3, the factorisation $n_0' = 2 \cdot 3 \cdot 5 \cdot C$ multiplies back to n_0' and has all exponents 1, and C carries an Atkin–Morain certificate, generated and validated (`certs/lyap_refute_cofactor_ecpp.txt`). The comparisons $\sigma(n_0) > 1$ and $\sigma(n_0') > \sigma(n_0)$ are exact, and $\sigma(n_0') = \sigma(n_0)$ is impossible in any case by Lemma 15.2 of the full version. The case $n = 30$ is arithmetic (`code/lyap_refute_verify.gp`). ☐

The argument is formalised. `Erdos307.criterion_fails_of_pos` and `criterion_fails_of_nonpos` give the two sign clashes, `no_lambda_works` combines them, and `lyapunov_criterion_false` instantiates the combination at $\sigma(n_0)$, carried as the exact quotient n_0'/n_0, and at the pair $(\frac{31}{30}, \frac{1}{31})$ of $n = 30$; the two numeric comparisons $1 < \sigma(n_0)$ and $\sigma(n_0) < \frac{31}{30}$ are checked on the 142-digit numerals themselves. All six declarations carry 0 `sorry` and only `propext`, `Classical.choice`, `Quot.sound`, with no `native_decide` (`lean/Erdos307/LyapFalse.lean`). What Lean does not see is the factorisation $n_0' = 2\cdot 3\cdot 5\cdot C$ and the primality of C, which is the Atkin–Morain certificate; the status is therefore VERIFIED modulo that certificate, in the same sense that Theorem 13.20 is VERIFIED modulo Theorem 2.3.

The witness is built, not found. For squarefree n and a prime $w \nmid n$ one has $n' \equiv n \sum_{p|n} p^{-1} \pmod{w}$, hence

$$w \mid n' \iff \sum_{p|n} p^{-1} \equiv 0 \pmod{w}. \tag{F}$$

Imposing (F) for $w \in \{2,3,5\}$ forces $\sigma(n') \geq \frac{31}{30}$, and building n from primes ≥ 7 with $\sigma(n)$ just above 1 gives $\sigma(n') > \sigma(n)$. Reaching mass 1 without $2,3,5$ needs primes only to about 360, which is what keeps n_0' small enough to factor completely; the squarefreeness of n_0' is therefore verified rather than assumed. Three of the larger pool primes are dropped to satisfy (F), at negligible cost in mass (`code/lyap_refute.gp`).

The sweep could not have seen this. Every $n \leq 4\times 10^8$ with $\sigma(n) > 1$ is even, so the battery measured only n containing 2, whereas the failure needs n to omit 2, 3 and 5 so that n' may absorb them. By Theorem 13.25 below, any failure must have $n\,n' \geq \Pi_{59}$, and $n_0 n_0'$ has 284 digits: the region was out of reach of any sweep, and only a construction enters it.

One route to the criterion had looked provable rather than measured, and it fails for the same reason. Write $\rho = \sup\{\sigma(n')/\sigma(n) : \sigma(n) > 1\}$, the supremum being over the same class of n. If $\rho \leq c < 1$ then $\sigma(n) - \sigma(n') \geq (1-c)\sigma(n)$, and since $\log x/x \leq 1/e$,

$$L := \sup_{\sigma(n)>1} \frac{\log \sigma(n)}{\sigma(n) - \sigma(n')} \leq \frac{1}{e(1-c)}.$$

Measured over $n \leq 4\times 10^8$, $\rho = 0.519746$, which would give $L \leq 0.766$ against a measured upper demand of 1.274; a proof of $\rho < 1 - 1/(eU)$, for U a lower bound on the upper demand, would then have given $(*)$. The measurement is not the truth. Already

$$a_0 = \prod_{5\leq p\leq 163} p, \qquad a_0' = 2\cdot 3\cdot 173\cdot 227\cdot 1973\cdot 82890547\cdot 68126695363\cdot P_{36},$$

with P_{36} prime, is squarefree with squarefree derivative, $\sigma(a_0) = 1.0723027\ldots > 1$ and $\sigma(a_0') = 0.8440258\ldots$, so $\rho \geq 0.7871152\ldots$; the bound the route would yield is $1/(e(1-\rho)) \geq 1.7280$, already above the measured upper demand, so it constrains nothing. The witness n_0 of Theorem 13.24 gives $\rho \geq 1.02726$, so $\rho < 1$ is false outright (`code/rho_verify.gp`).

The mechanism advanced for $\rho < 1$ was that $\sigma(n) > 1$ forces n to absorb the small primes while n' stays coprime to n. It does not: the reciprocals of the primes diverge, so $\sigma(n) > 1$ is reachable from primes ≥ 7 alone, and then n' is free to take 2, 3 and 5. Both a_0 and n_0 are of that shape. The earlier probe of this regime evaluated $\sigma(n')/\sigma(n)$ at the least odd n with $\sigma(n) > 1$, obtaining 0.0312, and read the regime off that single point; the quantity to be bounded is a supremum, and the least member of a family does not report it. This is the error recorded in Remark 15.26 of the full version, in a second instance.

What survives is the location of the failure, and it is exactly where the barrier puts it.

Theorem 13.25 ($\rho < 1$ below the barrier)**.** *Let n be squarefree with n' squarefree and $\sigma(n) > 1$. If $\sigma(n') \geq \sigma(n)$ then $n\,n' \geq \Pi_{59} > 8.77\times 10^{112}$. Consequently $\rho < 1$ for every n with $n\,n' < \Pi_{59}$, and since $\sigma(n) > 1$ gives $n' > n$, for every $n < \sqrt{\Pi_{59}}$, that is for every $n < 10^{56.5}$.*

Proof. $\sigma(n') \geq \sigma(n) > 1$ gives $\sigma(n) + \sigma(n') > 2$. By Theorem 2.3 n and n' are coprime, and both are squarefree, so $\sigma(n\,n') = \sigma(n) + \sigma(n') > 2$. A squarefree integer of mass at least 2 has at least

59 prime factors and is therefore at least Π_{59}, since $\sigma(\Pi_{58}) = 1.998740 < 2 \le 2.002350 = \sigma(\Pi_{59})$; this is the mass estimate underlying Theorem 4.3. Finally $\sigma(n) > 1$ gives $n' = n\sigma(n) > n$, so $n^2 < n\,n'$. □

The counting step is formalised. `Erdos307.card_ge_59_of_recipSum_ge_two` isolates the mass estimate, that a set of primes of reciprocal sum at least 2 has at least 59 elements, and `Erdos307.card_union_ge_59_` applies it to two disjoint sets each of mass exceeding 1; both carry 0 `sorry` (`lean/Erdos307/RhoBarrier.lean`). Their axiom lists are those of Theorem 4.3 itself, the three standard axioms together with the `native_decide` evaluation of the first 58 primes. The same core proves `card_ge_59` of the barrier, which assumes $\sigma(P)\sigma(Q) = 1$ instead; the two hypotheses differ and the shared step has been extracted rather than duplicated.

Theorem 13.25 therefore locates the failure before exhibiting it: $\rho \ge 1$ forces $n\,n' \ge \Pi_{59}$, so no sweep can meet a counterexample, and any counterexample must have n of at least 57 digits. The witness n_0 has 142, and $n_0 n_0'$ has 284. The regime in which $\rho \ge 1$ is possible is the regime in which a two-cycle is possible, since both need two coprime squarefree integers of total mass at least 2; what Theorem 13.24 shows is that the first of those is genuinely non-empty, which says nothing about the second.

Corollary 13.26 (The half-mass hypothesis cannot be relaxed)**.** *For every $\lambda > 0$ and every $c > 2.03925$, the implication "$\sigma(n\,n') < c$ implies $\delta_\lambda(n') < \delta_\lambda(n)$" is false. Indeed $\sigma(n_0\,n_0') = 2.039240\ldots$ while $\delta_\lambda(n_0') - \delta_\lambda(n_0) = \log\sigma(n_0) + \lambda(\sigma(n_0') - \sigma(n_0)) > 0$ for every $\lambda > 0$.*

Two is also the exact threshold for the argument of Theorem 13.20, in the following sense, which concerns the inequality alone and not the arithmetic of σ.

Proposition 13.27 (The mass-sum relaxation stops at 2)**.** *Let $c > 0$. There is $f : (0,\infty) \to \mathbb{R}$ with $f(x) - f(y) > \log x$ for all $x, y > 0$ satisfying $x + y < c$ if and only if $c \le 2$. For $c = 2$ the linear solutions are exactly $f(x) = \frac{1}{2}x + const.$*

Proof. The region $x + y < c$ is symmetric, so a solution gives both $f(x) - f(y) > \log x$ and $f(y) - f(x) > \log y$; adding, $0 > \log(xy)$. Taking $x = y \uparrow c/2$ forces $c^2/4 \le 1$, that is $c \le 2$. For $c \le 2$ take $f(x) = \frac{1}{2}x$: if $x + y < 2$ then $\frac{1}{2}(x - y) > x - 1 \ge \log x$. If $f = \lambda x$ then letting $y \uparrow 2 - x$ gives $2\lambda(x - 1) \ge \log x$ for all $x \in (0, 2)$, and dividing by $x - 1$ and letting $x \to 1$ from each side gives $2\lambda \ge 1$ and $2\lambda \le 1$. □

Both halves are formalised: `Erdos307.half_works` for $c \le 2$ and `no_f_above_two` for $c > 2$, the latter by the diagonal point $x = y$ with $1 < x < c/2$, where the right side vanishes and the left is positive; `mass_sum_threshold` states the two together. The uniqueness of the constant is formalised as well: `lambda_eq_half` shows that a linear solution at $c = 2$ forces $\lambda = \frac{1}{2}$, by the local-minimum route rather than a two-sided limit, since $g(x) = 2\lambda(x - 1) - \log x$ is nonnegative on $(0, 2)$ and vanishes at 1, so its derivative $2\lambda - 1$ vanishes there (`lean/Erdos307/NoInvariant.lean`).

Any strengthening must therefore use an input about n beyond the pair $(\sigma(n), \sigma(n'))$, and by the next proposition there is no room in the choice of function either.

Proposition 13.28 (The ansatz is universal, so the search was the problem)**.** *Let S be the set of squarefree $n \ge 2$ with n' squarefree. The following are equivalent:*

(1) *$n \mapsto n'$ has no periodic orbit inside S;*

(2) *there is $\delta : S \to \mathbb{R}$ with $\delta(n') < \delta(n)$ whenever $n, n' \in S$;*

(3) *there is f on $\sigma(S)$ with $f(\sigma(n)) - f(\sigma(n')) > \log\sigma(n)$ for all such n.*

Proof. (2) implies (1), since a periodic orbit would give $\delta(n) < \delta(n)$. For (1) implies (2), the transitive closure of $n \mapsto n'$ is irreflexive by (1) and transitive by construction, hence a strict partial order on the countable set S; it has a linear extension, and a countable linear order embeds order-preservingly in $\mathbb{Q}$, which supplies δ. For (2) and (3), σ is injective on squarefree integers by Lemma 15.2 of the full version, so every δ on S is $\log n + f(\sigma(n))$ for exactly one f, namely $f(\sigma(n)) = \delta(n) - \log n$; and $n' = n\sigma(n)$ turns $\delta(n') < \delta(n)$ into the inequality of (3). $\square$

The collapse of the ansatz is formalised. `Erdos307.exists_f_of_injective` proves that injectivity of σ forces *every* δ into the shape $\log n + f(\sigma(n))$, `increment_iff` converts $\delta(n') < \delta(n)$ into $(*)$ verbatim, and `no_two_cycle_of_decreasing` is the implication of Proposition 13.19; all with 0 `sorry` and the three standard axioms (`lean/Erdos307/NoInvariant.lean`).

So Definition 13.18 restricts nothing. The tournament that produced δ_λ was searching the space of all functions on S, and by Proposition 13.28 the existence of any member of that space is equivalent to the absence of cycles, which is strictly stronger than a negative answer to #307. The criterion was never a weakening of the problem, and Theorem 13.24 settles the one form of it that had been left open. What remains true, and is the residue of the whole construction, is Theorem 13.20 together with Corollary 13.21: an elementary Lyapunov proof of the barrier, valid exactly up to mass 2 and, by Corollary 13.26, not one step further.

Remark 13.29 (Four independent blocks on A10)**.** The existence route is therefore obstructed at four separate levels, and no two of them are the same obstruction. Proposition 13.1 removes the freedom: the construction carries one parameter and cannot be relaxed into a denser family. Theorem 6.1 removes the method: no argument that survives the passage to $K[t]$ can prove existence, because there the statement is false. Proposition 13.4 removes the computation: even the 34 most reduced objects in the whole problem, the ones every sieve has failed to kill, cannot be examined. Proposition 13.5 removes the last refuge, the hope that a large family of sequences is easier than one: the union stays logarithmically thin however many sequences it has. What is left is a lower bound for primes in a sparse exponential sequence, which exists for no non-degenerate linear recurrence at all. We state plainly that the present work does not decide the existence direction and does not contain a route to deciding it.

References

[1] E. J. Barbeau, *Remarks on an arithmetic derivative*, Canad. Math. Bull. **4** (1961), 117–122.

[2] E. J. Barbeau, *Computer challenge corner: Problem 477*, J. Recreational Math. **9** (1976/77), 30.

[3] P. Erdős and R. L. Graham, *Old and New Problems and Results in Combinatorial Number Theory*, Monogr. Enseign. Math. **28**, Geneva, 1980, p. 38.

[4] R. L. Graham, *Paul Erdős and Egyptian fractions*, in *Erdős Centennial*, Bolyai Soc. Math. Stud. **25**, Springer, 2013, pp. 289–309.

[5] T. Watanabe, *New examples of the representation of* 1 *by the sum of reciprocals of semiprime numbers*, arXiv:2009.03275 (2020).

[6] T. F. Bloom, *Unit fractions with shifted prime denominators*, Proc. Roy. Soc. Edinburgh Sect. A (2024); arXiv:2305.02689.

[7] V. Ufnarovski and B. Åhlander, *How to differentiate a number*, J. Integer Seq. **6** (2003), Article 03.3.4.

[8] J. Kovič, *The arithmetic derivative and antiderivative*, J. Integer Seq. **15** (2012), Article 12.3.8.

[9] MathOverflow, *Can the product of two sums of prime reciprocals be 1?*, question 320838 (comments by R. Israel and J. Rosen, answer by "Woett"), 2019. `https://mathoverflow.net/questions/320838`.

[10] T. F. Bloom, *Erdős problem #307*, `https://www.erdosproblems.com/307`.

[11] OEIS Foundation, *Sequence A003415 (arithmetic derivative)*, `https://oeis.org/A003415`.

[12] H. J. J. te Riele, *Computation of all the amicable pairs below* 10^{10}, Math. Comp. **47** (1986), 361–368.

[13] R. C. Baker, G. Harman, and J. Pintz, *The difference between consecutive primes, II*, Proc. London Math. Soc. (3) **83** (2001), 532–562.

[14] H. Iwaniec, *Almost-primes represented by quadratic polynomials*, Invent. Math. **47** (1978), 171–188.

[15] R. J. Lemke Oliver, *Almost-primes represented by quadratic polynomials*, Acta Arith. **151** (2012), 241–261.

[16] G. H. Hardy and E. M. Wright, *An Introduction to the Theory of Numbers*, 6th ed., Oxford Univ. Press, 2008.

[17] G. Tenenbaum, *Introduction to Analytic and Probabilistic Number Theory*, 3rd ed., Grad. Stud. Math. **163**, Amer. Math. Soc., 2015 (Erdős–Wintner theorem).

[18] W. W. Stothers, *Polynomial identities and Hauptmoduln*, Quart. J. Math. Oxford (2) **32** (1981), 349–370; R. C. Mason, *Diophantine Equations over Function Fields*, Cambridge Univ. Press, 1984.

[19] Seva (`mathoverflow.net` user), *A recursion with a number–theoretic function*, MathOverflow question 323194 (2019). `https://mathoverflow.net/questions/323194`.

[20] I. O. Bado, *Structural constraints for an Erdős unit–fraction problem over primes*, preprint, Aix–Marseille University (May 2026). `https://doi.org/10.13140/RG.2.2.32245.54245`.

[21] G. Giuga, *Su una presumibile proprietà caratteristica dei numeri primi*, Ist. Lombardo Sci. Lett. Rend. Cl. Sci. Mat. Nat. (3) **14** (1950), 511–528.

[22] D. Borwein, J. M. Borwein, P. B. Borwein, and R. Girgensohn, *Giuga's conjecture on primality*, Amer. Math. Monthly **103** (1996), no. 1, 40–50.

[23] J. M. Borwein and E. Wong, *A survey of results relating to Giuga's conjecture on primality*, CRM Proc. Lecture Notes **11** (1997), 13–27.

[24] W. Butske, L. M. Jaje, and D. R. Mayernik, *On the equation* $\sum_{p|N} 1/p+1/N=1$*, pseudoperfect numbers, and perfectly weighted graphs*, Math. Comp. **69** (2000), no. 229, 407–420.

[25] J. M. Grau and A. M. Oller-Marcén, *Giuga numbers and the arithmetic derivative*, J. Integer Seq. **15** (2012), Article 12.4.1; arXiv:1103.2298.

[26] J. Sondow and K. MacMillan, *Primary pseudoperfect numbers, arithmetic progressions, and the Erdős–Moser equation*, Amer. Math. Monthly **124** (2017), no. 3, 232–240; arXiv:1812.06566.

[27] J. M. Grau, A. M. Oller-Marcén, and D. Sadornil, *On μ-Sondow numbers*, arXiv:2111.14211 (2021); Acta Math. Hungar. (2022).

[28] D. A. Cox, *Primes of the Form $x^2 + ny^2$: Fermat, Class Field Theory, and Complex Multiplication*, 2nd ed., Wiley, 2013.

[29] OEIS Foundation, *Sequence A054377 (primary pseudoperfect numbers)*, `https://oeis.org/A054377`.

[30] OEIS Foundation, *Sequence A007850 (Giuga numbers)*, `https://oeis.org/A007850`.

[31] OEIS Foundation, *Sequence A396915 (composite squarefree k with $k' + 2k$ a perfect square)*, `https://oeis.org/A396915`.

[32] J.-H. Evertse, *On sums of S-units and linear recurrences*, Compositio Math. **53** (1984), 225–244.

[33] J. Friedlander and H. Iwaniec, *The polynomial $X^2 + Y^4$ captures its primes*, Ann. of Math. **148** (1998), 945–1040.

[34] D. R. Heath-Brown, *Primes represented by $x^3 + 2y^3$*, Acta Math. **186** (2001), 1–84.

[35] C. L. Stewart, *On divisors of Lucas and Lehmer numbers*, Acta Math. **211** (2013), 291–314.

[36] Y. Bugeaud, P. Corvaja, and U. Zannier, *An upper bound for the G.C.D. of $a^n - 1$ and $b^n - 1$*, Math. Z. **243** (2003), 79–84.

[37] J. Merikoski, *On the largest prime factor of $n^2 + 1$*, J. Eur. Math. Soc. **25** (2023), 1253–1284.

[38] J. Bourgain, A. Gamburd, and P. Sarnak, *Affine linear sieve, expanders, and sundry applications*, Invent. Math. **179** (2010), 559–644.

[39] J.-H. Evertse, H. P. Schlickewei, and W. M. Schmidt, *Linear equations in variables which lie in a multiplicative group*, Ann. of Math. (2) **155** (2002), 807–836.

[40] P. Erdős, *On the sum $\sum_{k=1}^{x} d(f(k))$*, J. London Math. Soc. **27** (1952), 7–15.

[41] Y. Bilu, G. Hanrot, and P. M. Voutier, *Existence of primitive divisors of Lucas and Lehmer numbers* (with an appendix by M. Mignotte), J. Reine Angew. Math. **539** (2001), 75–122.